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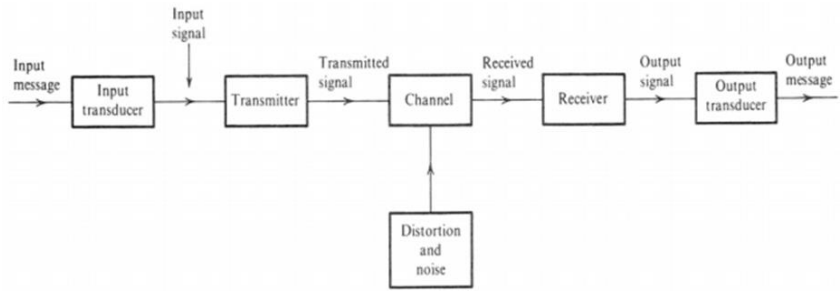
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1

Introduction And Basic Concept of Modern Communication and Technology: CDMA, WLL, GSM, VOIP, BLUETOOTH, WI-FI

1.1. Basic Communication System

A basic communication system, irrespective of the fact that it is wired or wireless, can be represented by the following block diagram:



The basic components and terminologies, which will be followed throughout the course, are:

- **Transducers**: Transducers are devices that convert a physical quantity into electrical signals, or vice versa. Here, the input transducer converts the input message, which could be spoken as it is in the case of a cellular network system, and converts it into electrical impulses, which is then sent. The received electrical impulses are converted back into speech at the output transducer.

- **Transmitter:** Transmitters are devices that convert the electrical signal into a form, say radio waves that can be used to transmit the message signal.

- **Channel**: The channel is the medium through which the transmitted signal passes. It is here that it faces the danger of being attenuated or being exposed to external distortive signals called noise signals.

- **Receiver**: Receivers converts the transmitted signal back to electrical impulses, after receiving it. It is here that certain signal processing processes are done on the received signal to either eliminate or at least reduce the effects of noise on the transmitted signal.

1.2. Wireless Communication

- Wireless communication is the process of transmission of data from the sender to the receiver without the need for any sort of physical medium or connections, such as wires and cables, for propagation.
- The channel for communication is generally considered to be air and the means of propagation are radio waves.
- The frequency of the radio waves varies based on application, lower frequencies to transmit over longer distances (like Deep-Space Communication) and higher frequencies over shorter distances (like Bluetooth).
- Apart from the conventionally used radio waves, there are other methods to achieve wireless communication, which include the use of light, magnetic and electric fields or even sound, though they are quite uncommon.
- Some examples of wireless communication include cellular phones, GPS devices, TV/AC remotes, walkie-talkies, and car keys.

Advantages of Wireless Communication:

Wireless communication has revolutionized the way how people communicate since its inception in 1896 when Marconi had invented the Radio. The advantages that wireless communication provide its wired counterparts are as follows:

- **Flexibility and Modularity**

Wireless communication is said to be flexible and modular. This means that the addition of more and more users into the network is a seamless and effortless process that doesn't require any additional infrastructure to be set up (other than getting the user a unique identification entity, such as SIM cards in cellular networks), unlike wired communication systems.

- **Cost-effectiveness**

Adding to the previous factor of flexibility, the absence of the need for additional infrastructure for every new member of the network implies a lower installation cost per person. Maintenance costs are also quite low as a majority of the components of the wireless systems are not physical entities that could fall victim to wear and tear.

- **Convenience**

A major advantage of wireless communication is its ability to provide user mobility, meaning that unlike wired communication, the destination address doesn't mean a fixed destination location. This effectively boils down to the fact that the user can be anywhere across the world (obviously somewhere where he's able to wirelessly access the network), and he'd still be able to communicate.

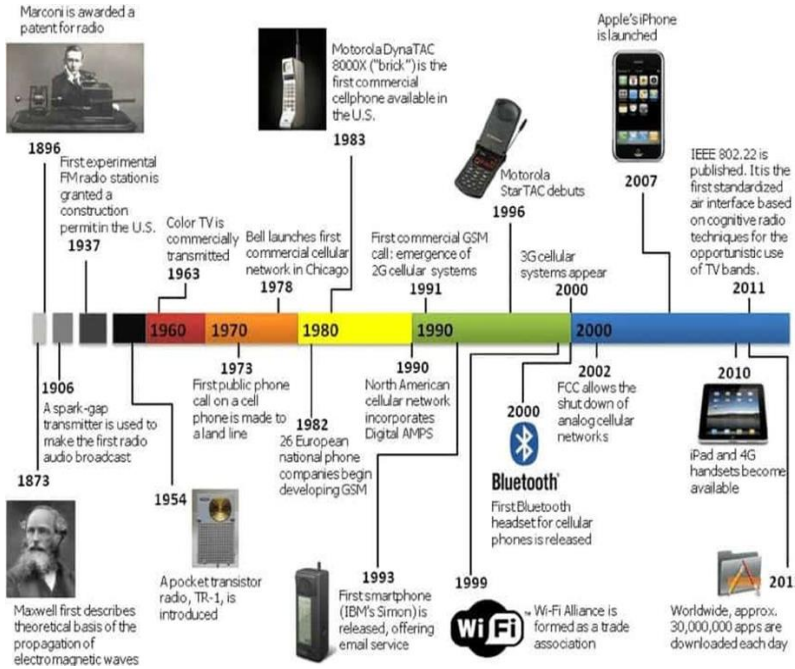
- **Constant Accessibility**

The wireless networks are up most of the time and can be accessed by any authorized person, from any place of his/her choice.

Wireless systems also ensure that people in areas where wired communication is not possible, such as rocky terrains where laying ground lines are tough, remain connected.

1.3. History Of Wireless Communication

The following image gives a brief overview of the evolution of wireless communication through the years:



Some major inventions/discoveries in wireless communication are as follows:

- In 1873, Maxwell proposed the idea of using electromagnetic waves to propagate the transmission of message signals, which is the underlying concept behind wireless communication.
- Guglielmo Marconi invented the first wireless device, the wireless Telegraph, in 1896 and was awarded the patent for his radio the same year. The very next year, the first

wireless message was sent, which read "Are you ready?", across a distance of 6km.

- Soon, advancements in radio broadcasting took place and in 1933, the first FM radio station was built in the United States.
- In 1946, the first commercial telephone service was set up by AT&T and Southern Bell in Saint Louis, Missouri, USA, and the service was made available to private customers as well.
- In 1947, a tiny device that revolutionized the computing industry was invented. The transistor, which replaced the much bigger vacuum tubes, made possible the marriage of computers and communications and also brought compactness into the picture.
- In 1954, the first among a line of transistor radios were built, which could fit in one's pocket comfortably.
- During the 1970s and 1980s, various companies such as Bell industries and Motorola came up with their idea of a commercial cellular phone for the masses, available for the public, which also initiated the first generation(1G) of mobile telephony.
- In the 1990s, GSM came into the picture, connecting many more people, and brought about the second generation of mobile telephony(2G).
- Wireless Fidelity(WiFi) came into existence in the late 1990s.
- Soon after, in the year 2000, Bluetooth was invented along with the release of the third- generation (3G) of mobile telephony. These two technologies revolutionized the way people shared messages, bringing more into the wireless world.
- In 2007, Apple released its first iPhone2 (the first "actual smartphone"), which lead to an exponential increase in the number of smartphones in the market.
- In 2010, the fourth generation(4G) of mobile telephony rolled out, which provide facilities such as Voice over LTE.

1.4. Code Division Multiple Access (CDMA)

CDMA stands for Code Division Multiple Access. It is one of the two second-generation systems for cellular phones available today. Even though the shift towards 5G cellular phone networks should, in theory, remove the split between the two types of networks, phones will still behave differently in areas where only 2G or3G networks are operating.

Code Division Multiple Access is a channel access method used by several radio communication *technologies*. It is a digital cellular technology and an example of multiple access. It is generally used for mobile communication.

Multiple accesses mean that several transmitters can send information simultaneously over a single communication channel. In this system, different CDMA codes are assigned to different users and the user can access the whole bandwidth for the entire duration. It optimizes the use of available bandwidth as it transmits over the entire frequency range and does not limit the user's frequency range.

Thus, CDMA allows several users to share a band of frequencies without undue interference between the users. It is used as a access method in many mobile phone standards.



CDMA technology was developed during World War II. It was developed by English allies to protect their wireless transmissions from jamming. When the war ended, Qualcomm patented this

technology and made it commercially available. The first CDMA system was launched in September 1995 in Hong Kong by Hutchison Telephone Co.

❖ Usage

- It is used in the Global Positioning System (GPS).
- It is used by several mobile phone companies (e.g. Qualcomm standard IS-2000 also known as CDMA2000)
- W-CDMA is used in UTMS 3G mobile phone standard.
- CDMA has been used in OmniTRACS satellite system for transportation.

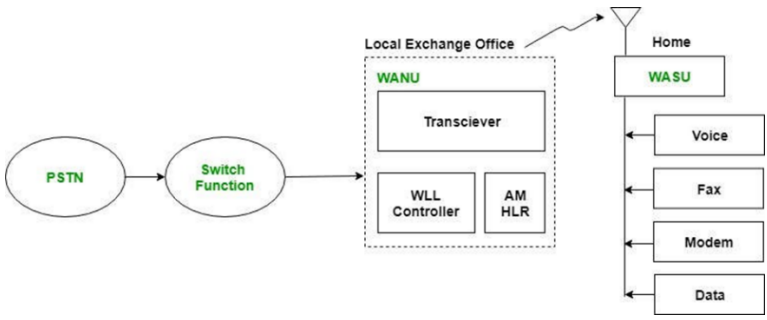
❖ **Categories of CDMA**

- Synchronous CDMA (orthogonal codes)
- Asynchronous CDMA (pseudorandom codes)

1.5. Wireless Local Loop (WLL)

Local loop is a circuit line from a subscriber's phone to the local central office (LCO). But the implementation of local loop of wires is risky for the operators, especially in rural and remote areas due to less number of users and increased cost of installation. Hence, the solution for it is the usage of wireless local loop (WLL) which uses wireless links rather than copper wires to connect subscribers to the local central office.

WLL Architecture:



WLL components:

1. **PSTN:** It is Public Switched Telephone Network which is a circuit switched network. It is a collection of world's interconnected circuit switched telephone networks.
2. **Switch Function:** Switch Function switches the PSTN among various WANUs.
3. **WANU:** It is short for Wireless Access Network Unit. It is present at the local exchange office. All local WASUs are connected to it. Its functions includes: Authentication, Operation & maintenance, Routing, Transceiving voice and data. It consists of following sub-components:
4. **Transceiver**: It transmits/receives data.
5. **WLL Controller:** It controls the wireless local loop component with WASU.
6. **AM:** It is short for Access Manager. It is responsible for authentication.
7. **HLR:** It is short for Home Location Register. It stores the details of all local WASUs.
8. **WASU (**Wireless Access Subscriber Units):

It is short for Wireless Access Subscriber Units. It is present at the house of the subscriber. It connects the subscriber to WANU and the power supply for it is provided locally.

Advantages of WLL:

- It eliminates the first mile or last mile construction of the network connection.
- Low cost due to no use of conventional copper wires.
- Much more secure due to digital encryption techniques used in wireless communication.
- Highly scalable as it doesn't require the installation of more wires for scaling it.

Features of WLL:

- Internet connection via modem

- Data service
- Voice service
- Fax service

1.6. GSM (Global System for Mobile Communication)

GSM stands for **G**lobal **S**ystem for Mobile Communication. It is a digital cellular technology used for transmitting mobile voice and data services. The concept of GSM emerged from a cell-based mobile radio system at Bell Laboratories in the early 1970s. GSM is the name of a standardization group established in 1982 to create a common European mobile telephone standard.

GSM is the most widely accepted standard in telecommunications and it is implemented globally. GSM is a circuit-switched system that divides each 200 kHz channel into eight 25 kHz time-slots. GSM operates on the mobile communication bands 900 MHz and 1800 MHz in most parts of the world. In the US, GSM operates in the bands 850 MHz and 1900 MHz.

GSM owns a market share of more than 70 percent of the world's digital cellular subscribers. GSM makes use of narrowband Time Division Multiple Access (TDMA) technique for transmitting signals. GSM was developed using digital technology. It has an ability to carry 64 kbps to 120 Mbps of data rates. Presently GSM supports more than one billion mobile subscribers in more than 210 countries throughout the world.

GSM provides basic to advanced voice and data services including roaming service. Roaming is the ability to use your GSM phone number in another GSM network.GSM digitizes and compresses data, then sends it down through a channel with two other streams of user data, each in its own timeslot.

Why GSM?
Listed below are the features of GSM that account for its popularity and wide acceptance.

- Improved spectrum efficiency
- International roaming
- Low-cost mobile sets and base stations (BSs)
- High-quality speech
- Compatibility with Integrated Services Digital Network (ISDN) and other telephone company services
- Support for new services

GSM History

The following table shows some of the important events in the rollout of the GSM system.

Years	Events
1982	Conference of European Posts and Telegraph (CEPT) establishes a GSM group to widen the standards for a pan-European cellular mobile system.
1985	A list of recommendations to be generated by the group is accepted.
1986	Executed field tests to check the different radio techniques recommended for the air interface.
1987	Time Division Multiple Access (TDMA) is chosen as the access method (with Frequency Division Multiple Access [FDMA]). The initial Memorandum of Understanding (MoU) is signed by telecommunication operators representing 12 countries.
1988	GSM system is validated.
1989	The European Telecommunications Standards Institute (ETSI) was given the responsibility of the GSM specifications.

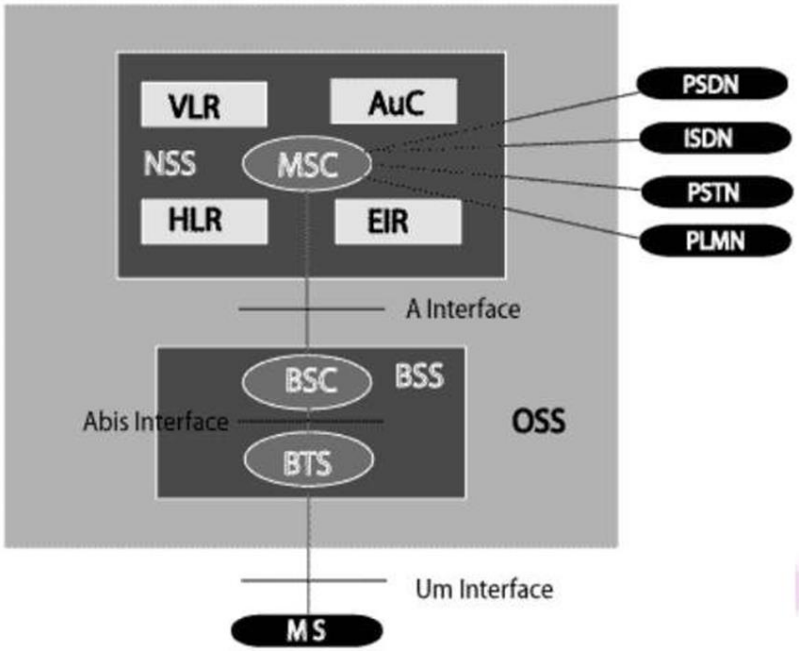
1990	Phase 1 of the GSM specifications is delivered
1991	Commercial launch of the GSM service occurs The DCS1800 specifications are finalized
1992	The addition of the countries that signed the GSM MoU takes place Coverage spreads to larger cities and airports.
1993	Coverage of main roads GSM services starts outside Europe.
1994	Data transmission capabilities launched. The number of networks rises to 69 in 43 countries by the end of 1994.
1995	Phase 2 of the GSM specifications occurs. Coverage is extended to rural areas.
1996	June: 133 network in 81 countries operational.
1997	July 200 network in 109 countries operational, around 44 million subscribers worldwide.
1999	Wireless Application Protocol (WAP) came into existence and became operational in 130 countries with 260 million subscribers.
2000	General Packet Radio Service(GPRS) came into existence.
2001	As of May 2001, over 550 million people were subscribers to mobile telecommunications.

A GSM network comprises of many functional units. These functions and interfaces are explained in this chapter. The GSM network can be broadly divided into:

- The Mobile Station (MS)
- The Base Station Subsystem (BSS)

- The Network Switching Subsystem (NSS)
- The Operation Support Subsystem (OSS)

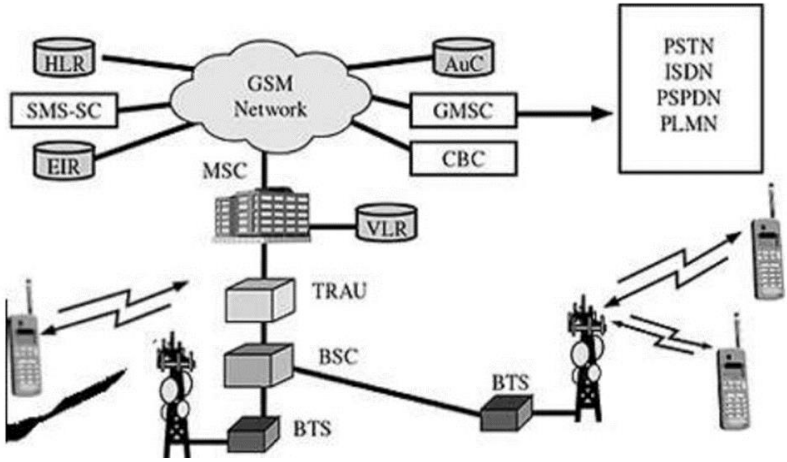
Given below is a simple pictorial view of the GSM architecture.



The additional components of the GSM architecture comprise of databases and messaging systems functions:

- Home Location Register (HLR)
- Visitor Location Register (VLR)
- Equipment Identity Register (EIR)
- Authentication Center (AuC)
- SMS Serving Center (SMS SC)
- Gateway MSC (GMSC)
- Chargeback Center (CBC)
- Transcoder and Adaptation Unit (TRAU)

The following diagram shows the GSM network along with the added elements:



The MS and the BSS communicate across the Um interface. It is also known as the air interface or the radio link. The BSS communicates with the Network Service Switching (NSS) center across the A interface.

GSM network areas

In a GSM network, the following areas are defined:

Cell : Cell is the basic service area; one BTS covers one cell. Each cell is given a Cell Global Identity (CGI), a number that uniquely identifies the cell.

Location Area : A group of cells form a Location Area (LA). This is the area that is paged when a subscriber gets an incoming call. Each LA is assigned a Location Area Identity (LAI). Each LA is served by one or more BSCs.

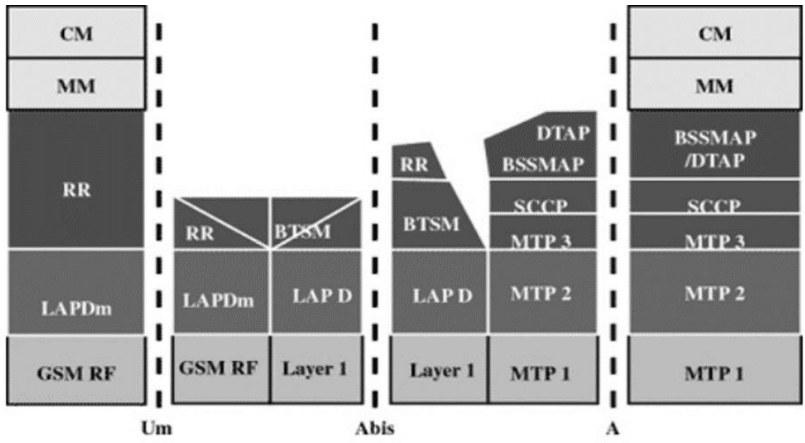
MSC/VLR Service Area : The area covered by one MSC is called the MSC/VLR service area.

PLMN : The area covered by one network operator is called the Public Land Mobile Network (PLMN). A PLMN can contain one or more MSCs.

GSM protocol stack

GSM architecture is a layered model that is designed to allow communications between two different systems. The lower layers assure the services of the upper-layer protocols. Each layer passes suitable notifications to ensure the transmitted data has been formatted, transmitted, and received accurately.

The GMS protocol stacks diagram is shown below:



MS Protocols

Based on the interface, the GSM signalling protocol is assembled into three general layers:

Layer 1 : The physical layer. It uses the channel structures over the air interface.

Layer 2 : The data-link layer. Across the Um interface, the data-link layer is a modified version of the Link access protocol for the D channel (LAP-D) protocol used in ISDN, called Link access protocol on the Dm channel (LAP-Dm). Across the A interface, the Message Transfer Part (MTP), Layer 2 of SS7 is used.

Layer 3 : GSM signalling protocol's third layer is divided into three sublayers: o Radio Resource Management (RR),

- Mobility Management (MM), and
- Connection Management (CM).

MS to BTS Protocols

The RR layer is the lower layer that manages a link, both radio and fixed, between the MS and the MSC. For this formation, the main components involved are the MS, BSS, and MSC. The responsibility of the RR layer is to manage the RR-session, the time when a mobile is in a dedicated mode, and the radio channels including the allocation of dedicated channels.

The MM layer is stacked above the RR layer. It handles the functions that arise from the mobility of the subscriber, as well as the authentication and security aspects. Location management is concerned with the procedures that enable the system to know the current location of a powered- on MS so that incoming call routing can be completed.

The CM layer is the topmost layer of the GSM protocol stack. This layer is responsible for Call Control, Supplementary Service Management, and Short Message Service Management. Each of these services are treated as individual layer within the CM layer. Other functions of the CC sub layer include call establishment, selection of the type of service (including alternating between services during a call), and call release.

BSC Protocols

The BSC uses a different set of protocols after receiving the data from the BTS. The Abis interface is used between the BTS and BSC. At this level, the radio resources at the lower portion of Layer 3 are changed from the RR to the Base Transceiver Station Management (BTSM). The BTS management layer is a relay function at the BTS to the BSC.

The RR protocols are responsible for the allocation and reallocation of traffic channels between the MS and the BTS. These services include controlling the initial access to the system, paging

for MT calls, the handover of calls between cell sites, power control, and call termination. The BSC still has some radio resource management in place for the frequency coordination, frequency allocation, and the management of the overall network layer for the Layer 2 interfaces.

To transit from the BSC to the MSC, the BSS mobile application part or the direct application part is used, and SS7 protocols is applied by the relay, so that the MTP 1-3 can be used as the prime architecture.

MSC Protocols

At the MSC, starting from the BSC, the information is mapped across the A interface to the MTP Layers 1 through 3. Here, Base Station System Management Application Part (BSS MAP) is said to be the equivalent set of radio resources. The relay process is finished by the layers that are stacked on top of Layer 3 protocols, they are BSS MAP/DTAP, MM, and CM. This completes the relay process.

To find and connect to the users across the network, MSCs interact using the control- signalling network. Location registers are included in the MSC databases to assist in the role of determining how and whether connections are to be made to roaming users. Each GSM MS user is given a HLR that in turn comprises of the user‘s location and subscribed services.

VLR is a separate register that is used to track the location of a user. When the users move out of the HLR covered area, the VLR is notified by the MS to find the location of the user. The VLR in turn, with the help of the control network, signals the HLR of the MS‘s new location. With the help of location information contained in the user‘s HLR, the MT calls can be routed to the user.

GSM addressing

GSM treats the users and the equipment in different ways. Phone

numbers, subscribers, and equipment identifiers are some of the known ones. There are many other identifiers that have been well-defined, which are required for the subscriber's mobility management and for addressing the remaining network elements. Vital addresses and identifiers that are used in GSM are addressed below.

International Mobile Station Equipment Identity (IMEI)

The International Mobile Station Equipment Identity (IMEI) looks more like a serial number which distinctively identifies a mobile station internationally. This is allocated by the equipment manufacturer and registered by the network operator, who stores it in the Entrepreneurs- in-Residence (EIR). By means of IMEI, one recognizes obsolete, stolen, or non-functional equipment.

Following are the parts of IMEI:

- **Type Approval Code (TAC)** : 6 decimal places, centrally assigned.
- **Final Assembly Code (FAC)** : 6 decimal places, assigned by the manufacturer.
- **Serial Number (SNR)** : 6 decimal places, assigned by the manufacturer.
- **Spare (SP)** : 1 decimal place.

Thus, IMEI = TAC + FAC + SNR + SP. It uniquely characterizes a mobile station and gives clues about the manufacturer and the date of manufacturing.

International Mobile Subscriber Identity (IMSI)

Every registered user has an original International Mobile Subscriber Identity (IMSI) with a valid IMEI stored in their Subscriber Identity Module (SIM).

IMSI comprises of the following parts:

- **Mobile Country Code (MCC)** : 3 decimal places,

internationally standardized.

- **Mobile Network Code (MNC)** : 2 decimal places, for unique identification of mobile network within the country.
- **Mobile Subscriber Identification Number (MSIN)** : Maximum 10 decimal places, identification number of the subscriber in the home mobile network.

Mobile Subscriber ISDN Number (MSISDN)

The authentic telephone number of a mobile station is the Mobile Subscriber ISDN Number (MSISDN). Based on the SIM, a mobile station can have many MSISDNs, as each subscriber is assigned with a separate MSISDN to their SIM respectively.

Listed below is the structure followed by MSISDN categories, as they are defined based on international ISDN number plan:

Country Code (CC) : Up to 3 decimal places.

- National Destination Code (NDC) : Typically, 2-3 decimal places.
- Subscriber Number (SN) : Maximum 10 decimal places.

Mobile Station Roaming Number (MSRN)

Mobile Station Roaming Number (MSRN) is an interim location dependent ISDN number, assigned to a mobile station by a regionally responsible Visitor Location Register (VLA). Using MSRN, the incoming calls are channelled to the MS.

The MSRN has the same structure as the MSISDN.

- **Country Code (CC)** : of the visited network.
- **National Destination Code (NDC)** : of the visited network.
- **Subscriber Number (SN)** : in the current mobile network.

Location Area Identity (LAI)

Within a PLMN, a Location Area identifies its own authentic Location Area Identity (LAI). The LAI hierarchy is based on international standard and structured in a unique format as mentioned below:

- **Country Code (CC)** : 3 decimal places.
- **Mobile Network Code (MNC)** : 2 decimal places.
- **Location Area Code (LAC)** : maximum 5 decimal places or maximum twice 8 bits coded in hexadecimal (LAC < FFFF).

Temporary Mobile Subscriber Identity (TMSI)

Temporary Mobile Subscriber Identity (TMSI) can be assigned by the VLR, which is responsible for the current location of a subscriber. The TMSI needs to have only local significance in the area handled by the VLR. This is stored on the network side only in the VLR and is not passed to the Home Location Register (HLR). Together with the current location area, the TMSI identifies a subscriber uniquely. It can contain up to 4 × 8 bits.

Local Mobile Subscriber Identity (LMSI)

Each mobile station can be assigned with a Local Mobile Subscriber Identity (LMSI), which is an original key, by the VLR. This key can be used as the auxiliary searching key for each mobile station within its region. It can also help accelerate the database access. An LMSI is assigned if the mobile station is registered with the VLR and sent to the HLR. LMSI comprises of four octets (4x8 bits).

Cell Identifier (CI)

Using a Cell Identifier (CI) (maximum 2 × 8) bits, the individual cells that are within an LA can be recognized. When the Global Cell Identity (LAI + CI) calls are combined, then it is uniquely defined.

GSM security

GSM is the most secured cellular telecommunications system available today. GSM has its security methods standardized. GSM maintains end-to-end security by retaining the confidentiality of calls and anonymity of the GSM subscriber.

Temporary identification numbers are assigned to the subscriber‘s number to maintain the privacy of the user. The privacy of the communication is maintained by applying encryption algorithms and frequency hopping that can be enabled using digital systems and signaling.

Mobile Station Authentication

The GSM network authenticates the identity of the subscriber through the use of a challenge-response mechanism. A 128-bit Random Number (RAND) is sent to the MS. The MS computes the 32-bit Signed Response (SRES) based on the encryption of the RAND with the authentication algorithm (A3) using the individual subscriber authentication key (Ki). Upon receiving the SRES from the subscriber, the GSM network repeats the calculation to verify the identity of the subscriber.

The individual subscriber authentication key (Ki) is never transmitted over the radio channel, as it is present in the subscriber's SIM, as well as the AUC, HLR, and VLR databases. If the received SRES agrees with the calculated value, the MS has been successfully authenticated and may continue. If the values do not match, the connection is terminated and an authentication failure is indicated to the MS.

The calculation of the signed response is processed within the SIM. It provides enhanced security, as confidential subscriber information such as the IMSI or the individual subscriber authentication key (Ki) is never released from the SIM during the authentication process.

Signaling and Data Confidentiality

The SIM contains the ciphering key generating algorithm (A8) that is used to produce the 64-bit ciphering key (Kc). This key is computed by applying the same random number (RAND) used in the authentication process to ciphering key generating algorithm (A8) with the individual subscriber authentication key (Ki).

GSM provides an additional level of security by having a way to change the ciphering key, making the system more resistant to eavesdropping. The ciphering key may be changed at regular intervals as required. As in case of the authentication process, the computation of the ciphering key (Kc) takes place internally within the SIM. Therefore, sensitive information such as the individual subscriber authentication key (Ki) is never revealed by the SIM. Encrypted voice and data communications between the MS and the network is accomplished by using the ciphering algorithm A5. Encrypted communication is initiated by a ciphering mode request command from the GSM network. Upon receipt of this command, the mobile station begins encryption and decryption of data using the ciphering algorithm (A5) and the ciphering key (Kc).

Subscriber Identity Confidentiality

To ensure subscriber identity confidentiality, the Temporary Mobile Subscriber Identity (TMSI) is used. Once the authentication and encryption procedures are done, the TMSI is sent to the mobile station. After the receipt, the mobile station responds. The TMSI is valid in the location area in which it was issued. For communications outside the location area, the Location Area Identification (LAI) is necessary in addition to the TMSI.

GSM Billing

GSM service providers are doing billing based on the services

they are providing to their customers. All the parameters are simple enough to charge a customer for the provided services.

Telephony Service

These services can be charged on per call basis. The call initiator has to pay the charges, and the incoming calls are nowadays free. A customer can be charged based on different parameters such as:

- International call or long distance call.
- Local call.
- Call made during peak hours.
- Call made during night time.
- Discounted call during weekends.
- Call per minute or per second.
- Many more other criteria can be designed by a service provider to charge their customers.

SMS Service

Most of the service providers charge their customer's SMS services based on the number of text messages sent. There are other prime SMS services available where service providers charge more than normal SMS charge. These services are being availed in collaboration of Television Networks or Radio Networks to demand SMS from the audiences.

Most of the time, the charges are paid by the SMS sender but for some services like stocks and share prices, mobile banking facilities, and leisure booking services, etc. the recipient of the SMS has to pay for the service.

GPRS Services

Using GPRS service, you can browse, play games on the Internet, and download movies. So a service provider will charge you based on the data uploaded as well as data downloaded on your mobile phone. These charges will be based on per Kilo Byte data downloaded/uploaded.

Additional parameter could be a QoS provided to you. If you want to watch a movie, then a low QoS may work because some data loss may be acceptable, but if you are downloading a zip file, then a single byte loss will corrupt your complete downloaded file.Another parameter could be peak and off peak time to download a data file or to browse the Internet.

Supplementary Services

Most of the supplementary services are being provided based on monthly rental or absolutely free. For example, call waiting, call forwarding, calling number identification, and call on hold are available at zero cost.

Call barring is a service, which service providers use just to recover their dues, etc., otherwise this service is not being used by any subscriber. Call conferencing service is a form of simple telephone call where the customers are charged for multiple calls made at a time. No service provider charges extra charge for this service.

Closed User Group (CUG) is very popular and is mainly being used to give special discounts to the users if they are making calls to a particular defined group of subscribers. Advice of Charge (AoC) can be charged based on number of queries made by a subscriber.

1.6.1. Difference between CDMA and GSM

The major difference between CDMA and GSM are given below.

Criteria	CDMA	GSM
Technology	CDMA is based on spread- spectrum technology which makes the optimum use of available bandwidth.	GSM operates on the wedge spectrum. it uses both time division multiple access (TDMA) and frequency division multiple access (fdma).

		TDMA provide multi-user access by cutting up the channel into different time slice and fdma provides the multi-user access by separating the used frequency.
Security	CDMA is more secure than GSM.	GSM is less secure than CDMA.
Global reach	CDMA is used in USA and some part of Canada and Japan. CDMA is used only by 24% of the users worldwide.	GSM is used over 80% of the world network in over 210 countries. GSM is used 76% of the users worldwide.
Data Transfer Rate	CDMA has faster data transfer as compared to GSM.	GSM has slower data transfer as compared to CDMA.
Radiation exposure	CDMA phones emits less radiation than GSM phones.	GSM phones emits continuous wave pulses and emits almost 28 times more radiation than CDMA phones.

1.6.2. What are GSM vs. CDMA vs. LTE differences?

The big difference among GSM, CDMA and LTE (long-term evolution) cellular-wireless communications is the technology behind them and the business objectives each is designed to meet. GSM is the oldest of the three. Developed and adopted as a standard in Europe, GSM used the processor/chip technologies available at the time to encode and decode data.

For a time, mobile operators deployed 2G GSM across many

countries worldwide except for the U.S. and several countries in South America. Incompatibility with existing analog AMPS systems largely drove these exceptions. To provide the necessary interim compatibility with GSM, they evaluated GSM's economies of scale for their networks. Carriers employed D-AMPS (Digital-Advanced Mobile Phone Service), a digital version of AMPS based on Interim Standard (IS)-136 for TDMA networking (itself an evolution of the original 2GL D-AMPS standard, IS-54) from the Electronics Industries Association/Telecommunication Industry Association. It eventually became clear that TDMA protocols weren't sufficiently spectrum efficient to support fast-growing cellular services, however. This led to the introduction of CDMA protocols.

ITU IS-95, also known as cdma One, became the CDMA digital cellular standard in 1993, gaining popularity in countries using older Analog AMPS systems. That said, IS-95 needed powerful processors because coding and decoding CDMA required significantly more compute power than decoding and coding TDMA. As a result, CDMA phones were more expensive than GSM models.

Cellular technology evolved from there. For data, GSM introduced GPRS, which led to EDGE, while cdma One led to ANSI-2000 1xRTT. That, in turn, led to EV-DO. Because of their superior efficiency, 3GPP adopted CDMA protocols under Wide-Band CDMA (W- CDMA) for implementation in 3G UMTS.

1.7. Voice Over Internet Protocol (VOIP)

- **Voice over Internet Protocol (VoIP),** is a technology that allowing you to make voice calls over a broadband Internet connection instead of an analog (regular) phone line. Some VoIP services allow you to call people using the same service, but others may allow you to call anyone. They can have a telephone number – including local, long- distance, mobile, and international numbers or not. Some VoIP services only work over your computer or a

special VoIP phone while other services allow you to use a traditional phone connected to a VoIP adapter.

- **How VoIP / Internet Voice Works –** Voice is converted into a digital signal by VoIP services that travel over the Internet. If the regular phone number is called, the signal is converted to a regular telephone signal i.e. an analog signal before it reaches the destination. VoIP can allow you to make a call directly from a computer having a special VoIP phone, or a traditional phone connected to a special adapter. Wireless hot spots in locations such as airports, hospitals, cafes, etc allow you to connect to the Internet and can enable you to use VoIP service wirelessly.
- **Equipments Required** – A high-speed Internet connection is required which can be through a cable modem or high-speed services such as a local area network. A computer, adaptor, or specialized phone is required. Some VoIP services only work over your computer or a special VoIP phone. Other services allow you to use a traditional phone connected to a VoIP adapter. If you use your computer some software and an inexpensive microphone are needed. VoIP phones plug directly into your broadband connection and operate largely like a traditional telephone. If you use a telephone with a VoIP adapter, you can dial just as you always have, and the service provider may also provide a dial tone.

Advantages of VoIP –

1. Some VoIP services offer features and services that are not available with a traditional phone, or are available but only for an additional fee.
2. Paying for both a broadband connection and a traditional telephone line can be avoided.
3. Smoother connection than an analog signal can be provided.

Disadvantages of VoIP –

1. Some VoIP services don't work during power outages and the service provider may not offer backup power.
2. Not all VoIP services connect directly to emergency services through emergency service numbers.
3. VoIP providers may or may not offer directory assistance.

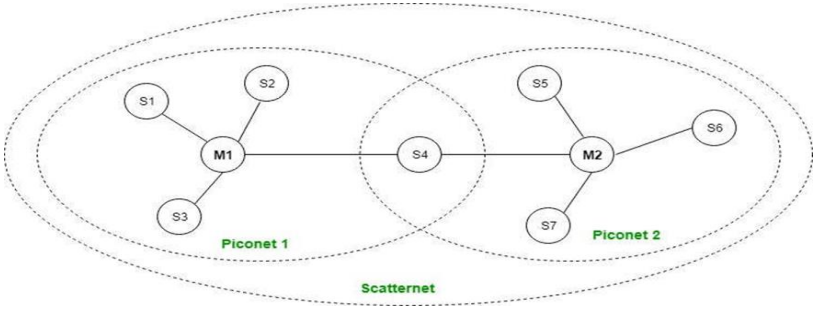
1.8. BLUETOOTH

It is a Wireless Personal Area Network (WPAN) technology and is used for exchanging data over smaller distances. This technology was invented by Ericson in 1994. It operates in the unlicensed, industrial, scientific and medical (ISM) band at 2.4 GHz to 2.485 GHz. Maximum devices that can be connected at the same time are 7. Bluetooth ranges upto 10 meters. It provides data rates upto 1 Mbps or 3 Mbps depending upon the version. The spreading technique which it uses is FHSS (Frequency hopping spread spectrum). A Bluetooth network is called a **piconet** and a collection of interconnected piconets is called **scatternet**.

Bluetooth Architecture:

The architecture of Bluetooth defines two types of networks:

1. Piconet
2. Scatternet



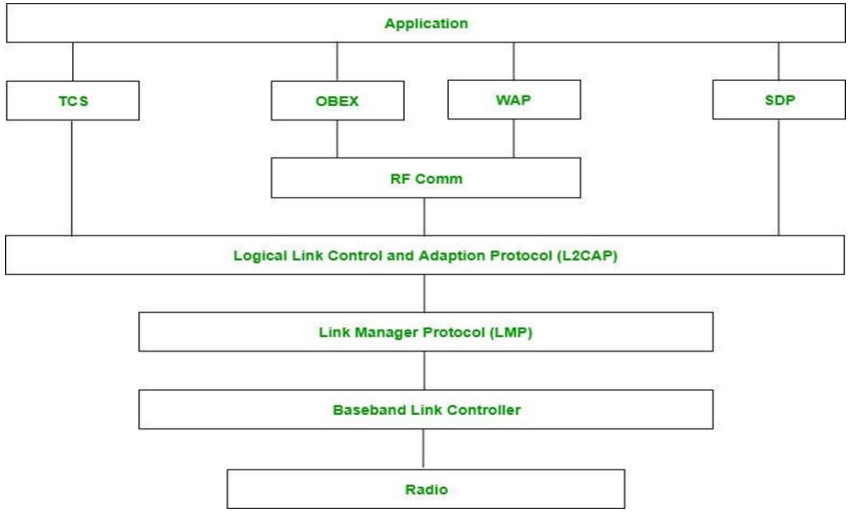
- **Piconet:**

Piconet is a type of Bluetooth network that contains one primary node called master node and seven active secondary nodes called slave nodes. Thus, we can say that there are total of 8 active nodes which are present at a distance of 10 meters. The communication between the primary and secondary node can be one-to-one or one-to-many. Possible communication is only between the master and slave; Slave-slave communication is not possible. It also have 255 parked nodes, these are secondary nodes and cannot take participation in communication unless it gets converted to the active state.

- **Scatternet:**

It is formed by using various piconets. A slave that is present in one piconet can act as master or we can say primary in another piconet. This kind of node can receive message from master in one piconet and deliver the message to its slave into the other piconet where it is acting as a slave. This type of node is refer as bridge node. A station cannot be master in two piconets.

Bluetooth protocol stack:



1. **Radio (RF) layer:**

It performs modulation/demodulation of the data into RF signals. It defines the physical characteristics of Bluetooth transceivers. It defines two types of physical link: connection-less and connection-oriented.

2. **Baseband Link layer:**

It performs the connection establishment within a piconet.

3. **Link Manager protocol layer:**

It performs the management of the already established links. It also includes authentication and encryption processes.

4. **Logical Link Control and Adaption protocol layer:**

It is also known as the heart of the Bluetooth protocol stack. It allows the communication between upper and lower layers of the Bluetooth protocol stack. It packages the data packets received from upper layers into the form expected by lower layers. It also performs the segmentation and multiplexing.

5. **SDP layer:**

It is short for Service Discovery Protocol. It allows to discover the services available on another Bluetooth enabled device.

6. **RF comm layer:**

It is short for Radio Frontend Component. It provides serial interface with WAP and OBEX.

7. **OBEX:**

It is short for Object Exchange. It is a communication protocol to exchange objects between 2 devices.

8. **WAP:**

It is short for Wireless Access Protocol. It is used for internet access.

9. **TCS:**

It is short for Telephony Control Protocol. It provides telephony service.

10. **Application layer:**

It enables the user to interact with the application.

Advantages:

- Low cost.
- Easy to use.
- It can also penetrate through walls.
- It creates an adhoc connection immediately without any wires.
- It is used for voice and data transfer.

Disadvantages:

- It can be hacked and hence, less secure.
- It has slow data transfer rate: 3 Mbps.
- It has small range: 10 meters.

1.9. WIFI (Wireless Fidelity)

Wi-Fi is a wireless networking technology that allows devices such as computers (laptops and desktops), mobile devices (smart phones and wearables), and other equipment (printers and video cameras) to interface with the Internet. It allows these devices--and many more--to exchange information with one another, creating a network.

Internet connectivity occurs through a wireless router. When you access Wi-Fi, you are connecting to a wireless router that allows your Wi-Fi-compatible devices to interface with the Internet.

Wi-Fi is a wireless networking technology that uses radio waves to provide wireless high- speed Internet access. A common misconception is that the term Wi-Fi is short for "wireless fidelity," however Wi-Fi is a trademarked phrase that refers to IEEE 802.11x standards.

Wi-Fi originated in Hawaii in 1971, where a wireless UHF packet network called ALOHAnet was used to connect the islands. Later

protocols developed in 1991 by NCR and AT&T called Wave LAN became the precursor to the IEEE 802.11 standards.

The Wi-Fi Alliance was formed in 1999 and currently owns the Wi-Fi registered trademark. It specifically defines Wi-Fi as any "wireless local area network (WLAN) products that are based on the Institute of Electrical and Electronics Engineers' (IEEE) 802.11 standards."

Initially, Wi-Fi was used in place of only the 2.4GHz 802.11b standard, however the Wi-Fi Alliance has expanded the generic use of the Wi-Fi term to include any type of network or WLAN product based on any of the 802.11 standards, including 802.11b, 802.11a, etc. in an attempt to stop confusion about wireless LAN interoperability.

What does Wi-Fi mean?

Wi-Fi is not an acronym; it is a brand name created by a marketing firm that's meant to serve as an interoperability seal for marketing efforts.

How does Wi-Fi work?

On the technical side, the IEEE 802.11 standard defines the protocols that enable communications with current Wi-Fi-enabled wireless devices, including wireless routers and wireless access points. Wireless access points support different IEEE standards.

Each standard is an amendment that was ratified over time. The standards operate on varying frequencies, deliver different bandwidth, and support different numbers of channels.

What is a wireless access point?

A wireless access point (AP) allows wireless devices to connect to the wireless network. Having a Cisco wireless network makes it easy to bring new devices online and provides flexible support to mobile workers.

What a wireless access point does for your network is similar to what an amplifier does for your home stereo. An access point takes the bandwidth coming from a router and stretches it so that many devices can go on the network from farther distances away. But a wireless access point does more than simply extend Wi-Fi. It can also give useful data about the devices on the network, provide proactive security, and serve many other practical purposes.

What is a wireless router?

Wireless routers are commonly found in homes. They're the hardware devices that Internet service providers use to connect you to their cable or xDSL Internet network.

A wireless router is sometimes referred to as a wireless local area network (WLAN) device. A wireless network is also called a Wi-Fi network.

A wireless router combines the networking functions of a wireless access point and a router. Read more about wireless routers.

What is a desktop Wi-Fi router?

The most common way for users to connect to the Internet wirelessly is with a desktop wireless (Wi-Fi) router. These routers look like small boxes with multiple short antennas to help broadcast the signal throughout a home or workplace. The farther a user is from the base Wi-Fi router, the weaker the signal. So multiple wireless routers, called range extenders, usually are placed throughout the workspace. Wi-Fi range extenders, placed in an array, boost or extend Internet coverage.

What is a mobile hotspot?

A mobile hotspot is a common feature on smartphones with both tethered and untethered connections. When you turn on your phone’s mobile hotspot, you share your wireless network connection with other devices that can then access the Internet.

What is portable Wi-Fi hotspot?

A portable Wi-Fi hotspot is a mobile hotspot obtained through a cell phone carrier. It's a small device that uses cellular towers that broadcast high-speed 3G or 4G broadband signals. Multiple devices, like iPads and laptops, can then connect wirelessly to the device, which in turn seamlessly connects to the Internet where ever you travel. Similar to a cell phone, the portable hotspot's monthly cost is based on the data usage plan you select. A portable Wi-Fi hotspot is a more reliable way to access the Internet than searching for static public Wi-Fi hotspots.

Difference between WiFi and WiMax:

WiFi	**WiMax**
Wifi is defined under IEEE 802.11x standards where x stands for various WiFi versions.	WiMax is defined under IEEE 802.16y standards where y stands for various WiMax versions.
WiFi is for LAN (Local Area Network) applications.	WiMax is for MAN (Metropolitan Area Network) applications.
WiFi does not guarantee any Quality of Service (Qos).	WiMax (Qos). guarantee quality of Service
WiFi network range is around 100 meters.	WiMax network can reach about 50-90 km.
WiFi MAC layer uses CSMA/CA protocol which is not connection oriented.	WiMax is connection oriented in nature.
WiFi is short range technology.	WiMax is long range technology.
WiFi connection can transmit upto 54 mbps.	WiMax connection can transmit upto 70 mbps.

1.10. Communication Technology (Generation) 1G, 2G, 3G, 4G & 5G

1G V Simply, the "**G**" stands for "**GENERATION**" . While you connected to internet, the speed of your internet is depends upon the signal strength that has been shown in alphabets like 2G, 3G, 4G etc. right next to the signal bar on your home screen. Each Generation is defined as a set of telephone **network standards** , which detail the technological implementation of a particular mobile phone system. The speed increases and the technology used to achieve that speed also changes. For eg, 1G offers 2.4 kbps, 2G offers 64 Kbps and is based on GSM, 3G offers 144 kbps- 2 mbps whereas 4G offers 100 Mbps - 1 Gbps and is based on **LTE technology** .

Features	**1G**	**2G**	**3G**	**4G**	**5G**
Start/Devlopment	1970/1984	1980/1999	1990/2002	2000/2010	2010/2015
Technology	AMPS, NMT, TACS	GSM	WCDMA	LTE, WiMax	MIMO, mm Waves
Frequency	30 KHz	1.8 Ghz	1.6 - 2 GHz	2 8 GHz	3 30 Ghz
Bandwidth	2 kbps	14.4 - 64 kbps	2 Mbps	2000 Mbps to 1 Gbps	1 Gbps and higher
Access System	FDMA	TDMA/CDMA	CDMA	CDMA	OFDM/BDMA
Core Network	PSTN	PSTN	Packet Network	Internet	Internet

The aim of wireless communication is to provide high quality,

reliable communication just like wired communication(optical fibre) and each **new generation** of services represents a big step(a leap rather) in that direction. This evolution journey was started in **1979** from 1G and it is still continuing to 5G. Each of the Generations has standards that must be met to officially use the G terminology. There are institutions in charge of standardizing each generation of mobile technology. Each generation has requirements that specify things like throughput, delay, etc. that need to be met to be considered part of that generation. Each generation built upon the research and development which happened since the last generation. 1G was not used to identify **wireless technology** until 2G, or the second generation, was released. That was a major jump in the technology when the wireless networks went from **analog to digital** .

- **1G - First Generation**

This was the first generation of **cell phone technology** . The very first generation of commercial cellular network was introduced in the late 70's with fully implemented standards being established throughout the 80's. It was introduced in 1987 by Telecom (known today as Telstra), Australia received its first cellular mobile phone network utilising a 1G analog system. 1G is an analog technology and the phones generally had poor battery life and voice quality was large without much security, and would sometimes experience **dropped calls** . These are the analog telecommunications standards that were introduced in the 1980s and continued until being replaced by 2G digital telecommunications. The maximum speed of 1G is **2.4 Kbps** .

- **2G - Second Generation**

Cell phones received their first major **upgrade** when they went from 1G to 2G. The main difference between the two mobile telephone systems (1G and 2G), is that the **radio signals** used by 1G network are analog, while 2G networks are **digital** . Main motive of this generation was to provide secure and reliable communication channel. It implemented the concept of **CDMA**

and **GSM** . Provided small data service like sms and mms. Second generation 2G cellular telecom networks were commercially launched on the GSM standard in Finland by Radiolinja (now part of Elisa Oyj) in 1991. 2G capabilities are achieved by allowing multiple users on a single channel via multiplexing. During 2G Cellular phones are used for data also along with voice. The advance in technology from 1G to 2G introduced many of the fundamental services that we still use today, such as SMS, **internal roaming** , conference calls, call hold and billing based on services e.g. charges based on long distance calls and real time billing. The max speed of 2G with General Packet Radio Service (**GPRS**) is 50 Kbps or 1 Mbps with Enhanced Data Rates for GSM Evolution (**EDGE**). Before making the major leap from 2G to 3G wireless networks, the lesser-known 2.5G and 2.75G was an interim standard that bridged the gap.

- **3G - Third Generation**

This generation set the standards for most of the wireless technology we have come to know and love. Web browsing, email, video downloading, picture sharing and other **Smartphone technology** were introduced in the third generation. Introduced commercially in 2001, the goals set out for third generation mobile communication were to facilitate greater voice and data capacity, support a wider range of applications, and increase data transmission at a **lower cost** .

The 3G standard utilises a new technology called **UMTS** as its core network architecture - Universal Mobile Telecommunications System. This network combines aspects of the 2G network with some new technology and protocols to deliver a significantly faster data rate. Based on a set of standards used for mobile devices and mobile telecommunications use services and networks that comply with the International Mobile Telecommunications-2000 (**IMT-2000**) specifications by the International Telecommunication Union. One of requirements set by IMT-2000 was that speed should be at least 200Kbps to call it as 3G service.

3G has Multimedia services support along with **streaming** are more popular. In 3G, Universal access and portability across different device types are made possible (Telephones, PDA's, etc.). 3G increased the efficiency of frequency spectrum by improving how audio is **compressed** during a call, so more simultaneous calls can happen in the same frequency range. The UN's International Telecommunications Union **IMT-2000** standard requires stationary speeds of 2Mbps and mobile speeds of 384kbps for a "true" 3G. The theoretical max speed for **HSPA+** is 21.6 Mbps.

Like 2G, 3G evolved into 3.5G and 3.75G as more features were introduced in order to bring about 4G. A 3G phone cannot communicate through a **4G network** , but newer generations of phones are practically always designed to be backward compatible, so a 4G phone can communicate through a 3G or even **2G network** .

- **4G - Fourth Generation**

4G is a very different technology as compared to **3G** and was made possible practically only because of the advancements in the technology in the last 10 years. Its purpose is to provide **high speed** , high quality and high capacity to users while improving security and lower the cost of voice and data services, multimedia and internet over IP. Potential and current applications include amended mobile web access, **IP telephony** , gaming services, high-definition mobile TV, video conferencing, 3D television, and cloud computing.

The key technologies that have made this possible are **MIMO** (Multiple Input Multiple Output) and **OFDM** (Orthogonal Frequency Division Multiplexing). The two important 4G standards are WiMAX (has now fizzled out) and **LTE** (has seen widespread deployment). LTE (Long Term Evolution) is a series of upgrades to existing UMTS technology and will be rolled out on Telstra's existing 1800MHz frequency band. The max speed of a 4G network when the device is moving is 100 Mbps or **1 Gbps**

for low mobility communication like when stationary or walking, latency reduced from around 300ms to less than 100ms, and significantly lower congestion. When 4G first became available, it was simply a little faster than 3G. 4G is not the same as **4G LTE** which is very close to meeting the criteria of the standards. To download a new game or stream a TV show in HD, you can do it **without buffering** .

Newer generations of phones are usually designed to be **backward-compatible** , so a 4G phone can communicate through a 3G or even 2G network. All carriers seem to agree that **OFDM** is one of the chief indicators that a service can be legitimately marketed as being 4G. OFDM is a type of digital modulation in which a signal is split into several narrowband channels at different frequencies. There are a significant amount of infrastructure changes needed to be implemented by service providers in order to supply because voice calls in **GSM** , **UMTS** and **CDMA2000** are circuit switched, so with the adoption of LTE, carriers will have to re-engineer their voice call network. And again, we have the fractional parts: **4.5G** and **4.9G** marking the transition of LTE (in the stage called LTE-Advanced Pro) getting us more MIMO, more D2D on the way to IMT-2020 and the requirements of **5G** .

- **5G - Fifth Generation**

5G is a generation currently **under development** , that's intended to improve on 4G. **5G** promises significantly faster data rates, higher connection density, much lower latency, among other improvements. Some of the plans for 5G include **device-to-device** communication, better battery consumption, and improved overall wireless coverage. The max speed of 5G is aimed at being as fast as **35.46 Gbps** , which is over 35 times faster than 4G.

Key technologies to look out for: **Massive MIMO** , Millimeter Wave Mobile Communications etc. Massive MIMO, milimetre wave, small cells, **Li-Fi** all the new technologies from the previous decade could be used to give 10Gb/s to a user, with an unseen low

latency, and allow connections for at least **100 billion devices** . Different estimations have been made for the date of commercial introduction of 5G networks. Next Generation Mobile Networks Alliance feel that 5G should be rolled out by **2020** to meet business and consumer demands.

- **6G - Sixth Generation**

6G is the sixth generation standard currently under development for wireless communications technologies supporting cellular data networks. It is the planned successor to 5G and will likely be significantly faster. Like its predecessors, 6G networks will probably be broadband cellular networks, in which the service area is divided into small geographical areas called cells, a 6G network works in combination of 4G and 5G networks. Several companies (Nokia, Ericsson, Huawei, Samsung, LG, Apple, Xiaomi), as well as several countries (India, China, Japan and Singapore), have shown interest in 6G networks.

6G networks are expected to exhibit even more heterogeneity (be even more diverse) than their predecessors and are likely to support applications beyond current mobile use scenarios, such as virtual and augmented reality (VR/AR), ubiquitous instant communications, pervasive intelligence and the Internet of Things (IoT). It is expected that mobile network operators will adopt flexible decentralized business models for 6G, with local spectrum licensing, spectrum sharing, infrastructure sharing, and intelligent automated management underpinned by mobile edge computing, artificial intelligence, short-packet communication and blockchain technologies.

Table 1.1 The evolution of 1G to 5G

Type	**Deployment**	**Technologies And Standards**	**Features**
1G	Analog telecommunicatio	Advanced Mobile Phone Service	Voice calls, NMT for

	n deployed in the 1980s	(AMPS) Nordic Mobile Telephone (NMT)	simple integrated data and messaging
2G	Digital cellular deployed in the 1990s	Code-division multiple access (CDMA) Global System for Mobile Communications (GSM)/ Enhanced Data rates for GSM Evolution (EDGE) Time-division multiple access (TDMA)	Voice, SMS text messages, low-rate data
3G	First broadband, deployed in 2000	CDMA2000 1X/Evolution-Data Optimized (EVDO) Universal Mobile Telecommunications Service (UMTS)/high-speed packet access (HSPA) Worldwide Interoperability for Microwave Access (WiMAX)	Offers speeds from 144 Kbps to 2 Mbps indoors, enabling rich content
4G	Deployed in 2010	LTE	100s of Mbps to 1 Gbps with video and streaming capabilities
5G	First deployed in 2018	International Telecommunication Union (ITU)/ International	3x higher spectral efficiency than 4G

		Mobile Communications (IMT)-2020 defined technical objectives 3rd Generation Partnership Project (3GPP) is developing 5G specifications	and peak downlink throughputs to peak 20 Gbps

4G LTE is a GSM technology and a major upgrade over 3G in terms of data transfer speeds. It offers no way of making phone calls in the traditional sense, however. To make regular phone calls, LTE uses specialized voice over Internet Protocol (VoIP) for what's referred to as VoLTE.

CDMA and GSM technologies eventually converged through Orthogonal Frequency Division Multiple Access (OFDMA), LTE's encoding protocol. OFDMA is also the encoding protocol used for WiMAX and Wi-Fi networks.

As 5G becomes more commonplace, there's an expectation that it will come with new encoding protocols. It's still too early to predict whether 5G will be a progressive evolution in telecommunications or mark a technological revolution in this market. Either way, most telecommunication industry watchers agree that its effects will be global in scale and dramatic.

Generation Network	**Technology Type**	**Typical Download Speed (Megabits Per Second)**	**Max Download Speed (Mbps)**
2G	GPRS	<0.1	0.1
	Edge	0.1	0.3
3G	3G (Basic)	0.1	0.3
	HSPA	1.5	7.2

	HSPA+	4	21
	DC-HSPA+	8	42
4G	LTE Category 4	15	150
4G+	LTE-Advanced Cat6	30	300
	LTE-Advanced Cat9	45	450
	LTE-Advanced Cat12	60	600
	LTE-Advanced Cat16	90	979
5G	5G	150-200	1,000-10,000 (1 to 10 gigabits per second)

1.11. Communication Over Radio

Spectrum Radio Frequency Spectrum is a key distinguishing factor used to compare alternative mobile radio systems. Radio spectrum for communications ranges from approximately 30 Hz (termed Extremely Low Frequency [ELF]) to above 100 GHz (termed Extremely High Frequency [EHF]). Because of its capability to provide very wide area coverage and penetrate sea water, ELF has been used for global systems for providing low-rate submarine communications. EHF, on the other hand, can be used for Line-of-Sight (LoS) microwave communications. Table shows the complete range of radio frequency spectrum used in communication systems and provides some examples of spectrum use.

Band Name	Frequency Range	Example Communication Use
Extremely Low Frequency	3-30 Hz	Submarine communications
Super Low Frequency	30-300 Hz	Submarine communications
Ultra Low Frequency	300-3,000 Hz	Underground communications
Very Low Frequency	3-30 kHz	Navigation
Low Frequency	30-300 kHz	AM broadcasting
Medium Frequency	300-3,000 kHz	AM broadcasting
High Frequency	3-30 MHz	Shortwave broadcast: amateur radio
Very High Frequency	30-300 MHz	Private mobile radio; FM and television broadcasting
Ultra High Frequency	300-3,000 MHz	Television broadcasting, cellular radio, and wireless LANs
Super High Frequency	3-30 GHz	Wireless LANs; point-to-point and point- to-multipoint microwave
Extremely High Frequency	30-300 GHz	Point-to-point microwave

Table 1-1 highlights how the characteristics of the different bands of the radio spectrum vary. In general, the lower the frequency, the better the range (for example, in the extreme case, a single ELF transmitter is able to cover the entire planet), but the bandwidths available are limited (for example, the same ELF systems typically provided a global system with total system capacity below 50 bps). Conversely, EHF systems can provide incredible capacity, but they incur significant attenuation by atmospheric effects due, for example, to extreme humidity, rain, or molecular absorption, and thus are prone to significant losses in non-Line-of-Sight (LoS) deployments.

- **Propagation**

Because of its relative scarcity, mobile systems are required to re-use the allocated radio spectrum across a particular network of cell sites. Radio frequency signals need to propagate between the cell site antenna and the mobile wireless terminal. As the signals propagate, they exhibit a path loss as the emitted energy is dispersed over an increasing area. Estimating the path loss is critical in determining both the coverage provided by a single cell site and the bandwidth available to the IP services offered in that cell coverage area. The benchmark of propagation loss is that of free space—in other words, the loss in a region that is free from all objects that might absorb or reflect the radio energy. Because the emitted energy from an isotropic antenna is dispersed over the surface of a sphere (with the transmitting antenna at the center of the sphere), the received energy is inversely proportional to the surface area of the sphere ($4\pi r^2$, where r is the radius of the sphere), as illustrated in Figure. Using this approach, you can see that the free space path loss follows an inverse square law with changing distance from the antenna, r.

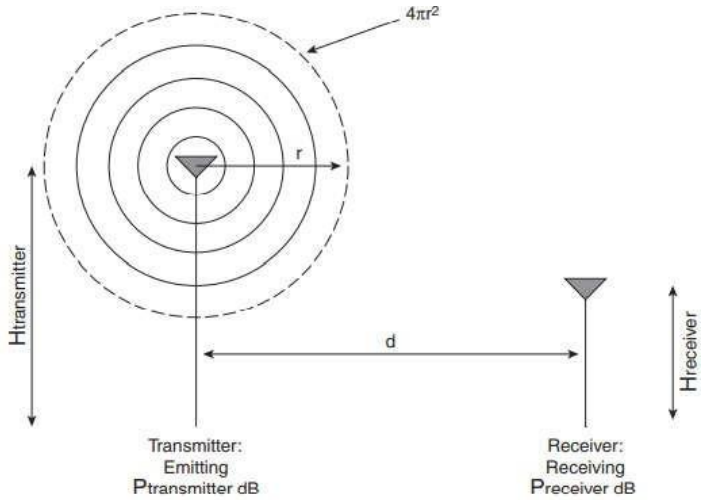


Figure 1-1 *Free Space Loss*

- **Fast Fading**

Whereas the previous analysis concentrated on propagation in ideal free space or with simple two- ray models, the reality is that mobile radio systems operate with a variety of obstacles and reflections both between and around the base station and mobile terminal, as shown in Figure 1-3. The combination of the disparate propagation paths is called multipath—where the transmitted signal arrives at the receiver from various directions over a multiplicity of paths, with each individual path having its own electrical path length and degree of attenuation.

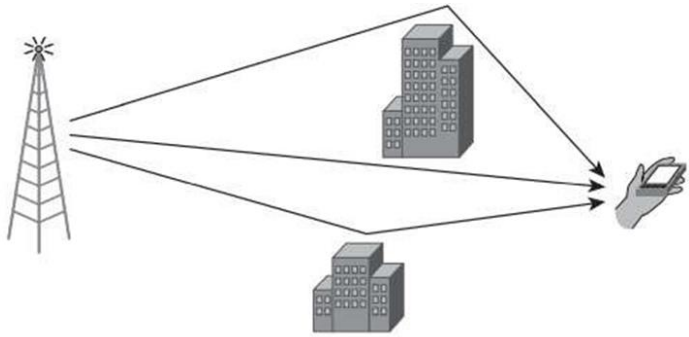


Figure 1-3 *Multipath Propagation*

1.12. Microwave Systems

Microwave signals are used to transmit data without the use of cables. The microwave signals are similar to **radio and television signals**. Microwaves are used for **long-distance communication**. In **microwave communication** parabolic antennas are mounted on towers to send data to other antennas.

- **Introduction to microwaves**

Microwaves – As the name implies, are very short waves .In general RF extends from dc upto Infrared region and these are forms of electromagnetic energy. A glance look at the various frequency ranges makes it clear that UHF(Ultra high frequency)

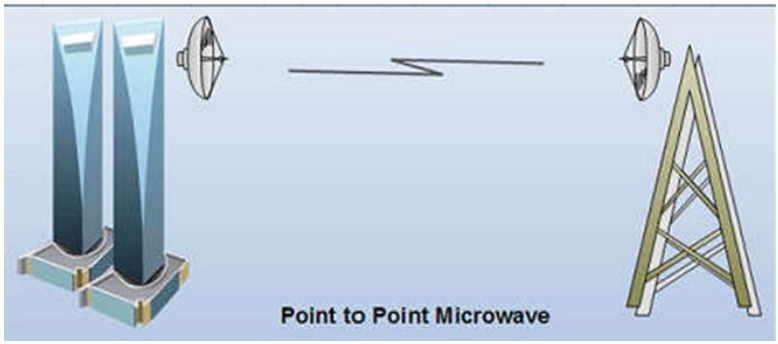
& SHF (super high frequencies) constitutes the Microwave frequency range with wave length (λ) extending from 1 to 100 cm The basic principle of low frequency radio waves and microwaves are the same .Here the phenomena are readily explained in terms of current flow in a closed electric circuit. At low frequencies , we talk in terms of lumped circuit elements such as C. L, R which can be easily identified and located in a circuit . On the other hand in Microwave circuitry , the inductance & capacitance are assumed to be distributed along a transmission line . Microwaves are electromagnetic waves whose frequencies range from 1 GHz to 1000 GHz (1 GHz =109). Microwaves so called since they are defined in terms of their wave length, micro in the sense tinny ness in wave length , period of cycle (CW wave) , λ is very short. Microwave is a signal that has a wave length of 1 foot or less $\lambda \leq$ 30.5 cm .= 1 foot. F= 984MHz approximately 1 GHz

Microwave radio, a form of radio transmission that use. Ultra-high frequencies developed out of experiments with radar (radio detecting and ranging) during the period preceding World War II. There are several frequency ranges assigned to microwave systems, all of which are in the Giga Hertz (GHz) range and the wavelength in the millimeter range. This very short wavelength gives rise to the term microwave. Such high frequency signals are especially susceptible to attenuation and, therefore must be amplified or repeated after a particular distance.

In order to maximize the strength of such a high frequency signal and, therefore, to increase the distance of transmission at acceptable levels, the radio beams are highly focused. The transmit antenna is centered in a concave, reflective metal dish which serves to focus the radio beam with maximum effect on the receiving antenna, as illustrated in Figure. The receiving antenna, similarly, is centered in a concave metal dish, which serves to collect the maximum amount of incoming signal.

It is a point-to-point, rather than a broadcast, transmission system. Additionally, each antenna must be within line of sight of the next antenna. Given the curvature of the earth, and the obvious

problems of transmitting through it, microwave hops generally are limited to 50 miles (80 km). If the frequencies are higher within the microwave band given in Table 2.1, this impact is more than lower frequencies in the same band.



Frequency Bands Maximum Antenna Separation Analog/Digital 4-6 GHz 32-48 km Analog 10-12 GHz 16-24 km Digital 18-23 GHz 8-11 km Digital.

General Properties of Microwave Transmission:

Configuration Microwave radio consists of antennae centered within reflective dishes that are attached to structures such as towers or buildings. Cables connect the antennae to the actual transmit (receive) equipment.

Bandwidth Microwave offers substantial bandwidth, often in excess of 6 Gbps.

Error Performance Microwave, especially digital microwave, performs well in this regard, assuming proper design. However, such high frequency radio is particularly susceptible to environmental interference, e.g. precipitation, haze, smog, and smoke. Generally speaking, however, microwave performs well in this regard.

Distance Microwave clearly is distance limited, especially at the higher frequencies. This limitation can be mitigated through

special and more complex arrays of antennae incorporating spatial diversity in order to collect more signals.

Security As is the case with all radio systems, microwave is inherently not secure. Security must be imposed through encryption (scrambling) of the signal.

Cost The acquisition, deployment and rearrangement cost of microwave can be high. However, it often compares very favorably with cabled systems, which require right-of-way, trenching, conduit, splicing, etc.

Applications Microwave originally was used for long haul voice and data communications. Competing long distance carriers, microwave was found the most attractive alternative to cabled systems, due to the speed and low cost qf deployment where feasible, however, fiber optic technology is currently used in this regard. Contemporary applications include private networks, interconnection of cellular radio switches, and as an alternative to cabled systems in consideration of difficult terrain.

Advantages of Microwave

1. It offers ease of communication over difficult terrain.
2. Microwaves have the ability to communicate over oceans.

Disadvantages of Microwave

1. It is an insecure communication.
2. Its propagation is susceptible to weather effects like rains, thunderstorms, etc.
3. Implementation and maintenance cost is high.
4. Bandwidth is limited.

APPLICATION AREAS OF MICROWAVES

- RADAR
- Surveillance (air traffic control)

- Navigation (direction finding)
- Meteorology 2-MEDICINE
- Treatment of Diseases
- Microwave Imaging 3-SURVEYING
- LAND HEATING INDUSTRIAL QUALITY CONTROL
- RADIO ASTRONOMY NAVIGATION VIA GLOBAL POSITIONING SYSTEMS
- REMOTE SENSING POWER TRANSMISSION

1.13. Communication Satellite

A satellite is an object that revolves around another object. For example, earth is a satellite of The Sun, and moon is a satellite of earth.

A **communication satellite** is a **microwave repeater station** in a space that is used for telecommunication, radio and television signals. A communication satellite processes the data coming from one earth station and it converts the data into another form and send it to the second earth station.

Communication satellites may be owned by government or private organizations. Presently, there are more than 2000 communication satellites in the sky. Some of its uses are −

- Internet
- Military operations
- Television
- Telephone
- Radio

Communication satellites are broadly categorized into three types depending upon the orbit in which they are placed.

- **Geostationary Satellite (GEO)** − They are at 36,000 km from the earth's surface. They have same orbital period as earth's rotation. So they appear to be still in the sky. At least 3 GEOs are needed for global coverage.

- **Medium Earth Orbit Satellite (MEO)** − They are placed between the two Van Allen belts, at a distance between 2,000 km to 36,000 km from the earth's surface. At least 10 MEOs are needed for global coverage.
- **Low Earth Orbit Satellite (LEO)** − They are situated below the Lower Van Allen belt. Their orbital altitude is 160 km to 2000 km. For global coverage, as high as 50 LEOs are required.

How a Satellite Works:

Two stations on earth want to communicate through radio broadcast but are too far away to use conventional means. The two stations can use a relay station for their communication. One earth station transmits the signal to the satellite.

Uplink frequency is the frequency at which ground station is communicating with satellite. The satellite transponder converts the signal and sends it down to the second earth station, and this is called **Downlink frequency**. The second earth station also communicates with the first one in the same way.

Advantages of Satellite

The advantages of Satellite Communications are as follows −

- The Coverage area is very high than that of terrestrial systems.
- The transmission cost is independent of the coverage area.
- Higher bandwidths are possible.

Disadvantages of Satellite

- The disadvantages of Satellite Communications are as follows −
- Launching satellites into orbits is a costly process.
- The bandwidths are gradually used up.
- High propagation delay for satellite systems than the conventional terrestrial systems.

1.14. Satellite Communication Basics Details

- The process of satellite communication begins at an **earth station**. Here an installation is designed to transmit and receive signals from a satellite in orbit around the earth. Earth stations send information to satellites in the form of high powered, high frequency (GHz range) signals.
- The satellites **receive** and **retransmit** the signals back to earth where they are received by other earth stations in the coverage area of the satellite. **Satellite's footprint** is the area which receives a signal of useful strength from the satellite.
- The transmission system from the earth station to the satellite through a channel is called the **uplink**. The system from the satellite to the earth station through the channel is called the **downlink**.

Satellite Frequency Bands:

The satellite frequency bands which are commonly used for communication are the **Cband, Ku- band,** and **Ka-band**. C-band and Ku-band are the commonly used frequency spectrums by today's satellites.

It is important to note that there is an inverse relationship between frequency and wavelength i.e., when frequency increases, wavelength decreases this helps to understand the relationship between **antenna diameter** and **transmission frequency**. Larger antennas (satellite dishes) are necessary to gather the signal with increasing wavelength.

Earth Orbits

A satellite when launched into space, needs to be placed in certain orbit to provide a particular way for its revolution, so as to maintain accessibility and serve its purpose whether scientific, military or commercial. Such orbits which are assigned to satellites, with respect to earth are called as **Earth Orbits**. The satellites in these orbits are Earth Orbit Satellites.

The important kinds of Earth Orbits are −

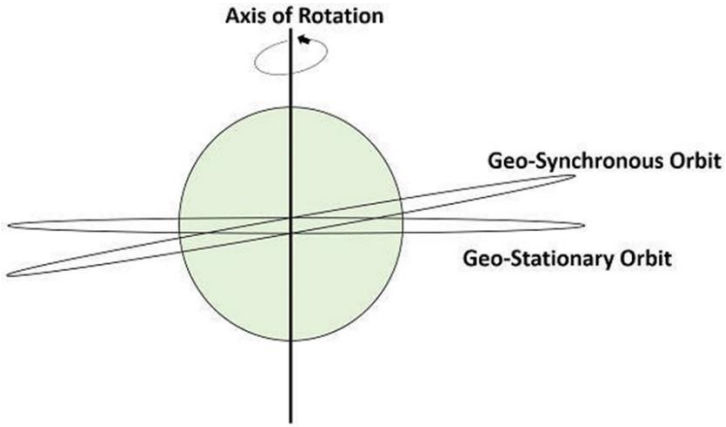
- Geo-synchronous Earth Orbit
- Geo-stationary Earth Orbit
- Medium Earth Orbit
- Low Earth Orbit

Geo-synchronous Earth Orbit (GEO) Satellites

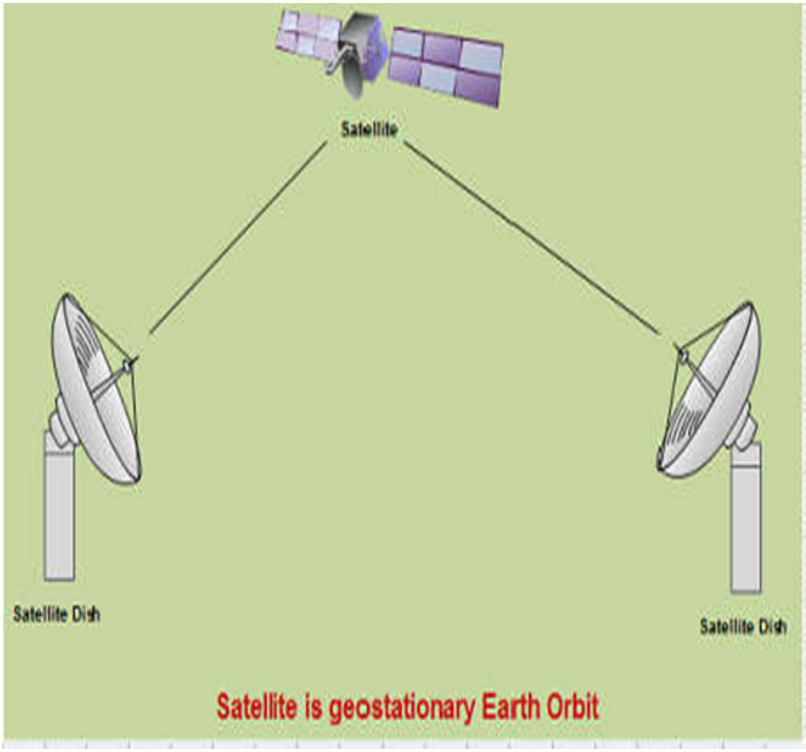
A Geo-synchronous Earth orbit Satellite is one which is placed at an altitude of 22,300 miles above the Earth. This orbit is synchronized with a **side real day** (i.e., 23hours 56minutes). This orbit can **have inclination and eccentricity**. It may not be circular. This orbit can be tilted at the poles of the earth. But it appears stationary when observed from the Earth.

The same geo-synchronous orbit, if it is **circular** and in the plane of equator, it is called as geo- stationary orbit. These Satellites are placed at 35,900kms (same as geosynchronous) above the Earth's Equator and they keep on rotating with respect to earth's direction (west to east). These satellites are considered **stationary** with respect to earth and hence the name implies.

Geo-Stationary Earth Orbit Satellites are used for weather forecasting, satellite TV, satellite radio and other types of global communications.



The above figure shows the difference between Geo-synchronous and Geo- Stationary orbits. The Axis of rotation indicates the movement of Earth.



The main point to note here is that every Geo-Stationary orbit is a Geo-Synchronous orbit. But every Geo-Synchronous orbit is NOT a Geo-stationary orbit.

Medium Earth Orbit (MEO) Satellites

Medium earth orbit (MEO) satellite networks will orbit at distances of about 8000 miles from earth's surface. Signals transmitted from a MEO satellite travel a shorter distance. This translates to improved signal strength at the receiving end. This shows that smaller, more lightweight receiving terminals can be used at the receiving end.

Since the signal is travelling a shorter distance to and from the satellite, there is less transmission delay. **Transmission delay** can be defined as the time it takes for a signal to travel up to a satellite and back down to a receiving station.

For real-time communications, the shorter the transmission delay, the better will be the communication system. As an example, if a GEO satellite requires 0.25 seconds for a round trip, then MEO satellite requires less than 0.1 seconds to complete the same trip. MEOs operates in the frequency range of 2 GHz and above.

Low Earth Orbit (LEO) Satellites

The LEO satellites are mainly classified into three categories namely, little LEOs, big LEOs, and Mega-LEOs. LEOs will orbit at a distance of 500 to 1000 miles above the earth's surface.

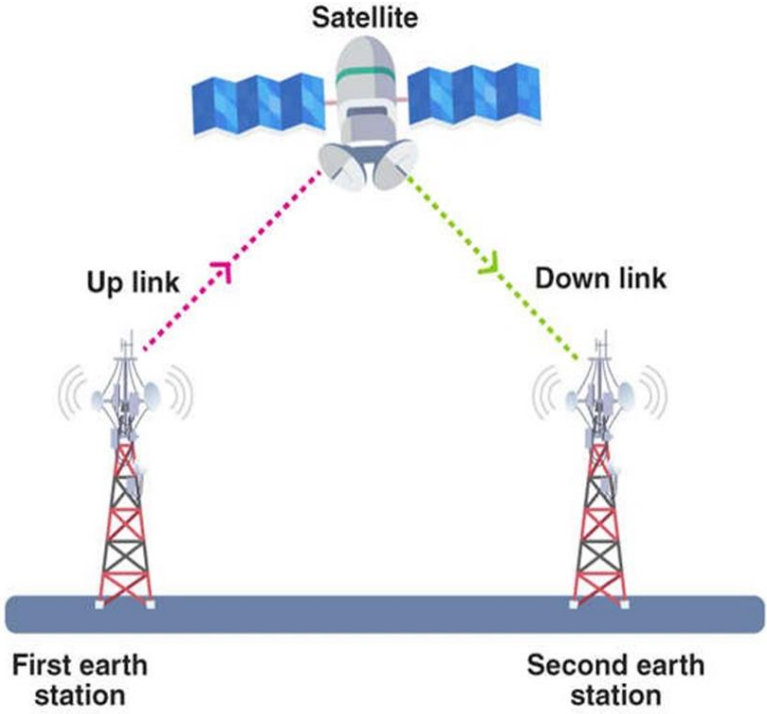
This relatively short distance reduces transmission delay to only 0.05 seconds. This further reduces the need for sensitive and bulky receiving equipment. Little LEOs will operate in the 800 MHz (0.8 GHz) range. Big LEOs will operate in the 2 GHz or above range, and Mega-LEOs operates in the 20-30 GHz range.

The higher frequencies associated with **Mega-LEOs** translates into more information carrying capacity and yields to the capability of real-time, low delay video transmission scheme.

High Altitude Long Endurance (HALE) Platforms

Experimental HALE platforms are basically highly efficient and lightweight airplanes carrying communications equipment. This will act as **very low earth orbit geosynchronous satellites**.

These crafts will be powered by a combination of battery and solar power or high efficiency turbine engines. HALE platforms will offer **transmission delays of less than 0.001 seconds** at an altitude of only 70,000 feet, and even **better signal strength** for very lightweight hand-held receiving devices.



1.15. Orbital Slots

Here there may arise a question that with more than **200 satellites** up there in geosynchronous orbit, how do we keep them from running into each other or from attempting to use the same location in space? To answer this problem, international regulatory bodies like the International Telecommunications Union (**ITU**) and national government organizations like the Federal Communications Commission (**FCC**) designate the locations on the geosynchronous orbit where the communications satellites can be located.

These locations are specified in degrees of longitude and are called as **orbital slots**. The FCC and ITU have progressively reduced the required spacing down to only 2 degrees for C-band and Ku- band satellites due to the huge demand for orbital slots.

Applications of Satellite Communication

- Telephone
- Television
- Digital cinema
- Radio broadcasting
- Amateur radio
- Internet access
- Military
- Disaster Management

Satellite communication has a number of advantages:

Advantages :

- Through satellite transmission, coverage over geographical area is quite large mainly for sparsely populated areas.
- High bandwidth.
- Wireless and mobile communication applications can be easily established by satellite communication independent of location.
- It is used in wide variety of applications such as global mobile communication, private business networks, Long distance telephone transmission, weather forecasting, radio/TV signal broadcasting, gathering intelligence in military, navigation of ships and air crafts, connecting remote areas, television distribution etc.
- Security in satellite transmission is usually provided by the coding and decoding equipment.
- Service from one single provider is easy to obtain and uniform service is available.
- Over long distances, it can be cheaper.
- The laying and maintenance is easy and cheap in satellite communication therefore it is best alterative.
- During critical condition, each Earth Station may be removed relatively quickly from a location and reinstalled somewhere else.

- Ground station sites are easy to install and maintain.

Disadvantages :

- Design, development, investment and insurance of satellite requires higher cost.
- To reach the satellite from Earth, time can vary between 270 milliseconds and return again to 320 milliseconds. This propagation delay can cause an echo over telephone connections
- Satellites are not easy to repair and maintain.
- Some circumstances like weather or sunspots affect the satellite's signal and can cause interference and make proper operation of the satellite very difficult.
- It requires to be monitored and controlled on regular periods so that it remains in the orbit, once it has been launched.

1.16. Radar Communication

We can observe different objects around the world. Similarly, radar-like radio detection and ranging is used to assist the pilots while flying through fog because the pilot cannot notice that where they are traveling. The radar used in the airplanes is similar to a torchlight that works with radio waves in place of light. The airplane transmits a blinking radar signal and listens out for any indications of that signal from nearby objects. Once the indications are noticed, then the airplane identifies something is near & it uses the time taken for the indications to reach for discovering how distant away it is. This article discusses an overview of Radar and its working.

Who Invented Radar?

Similar to several inventions, the radar system is not easy to give credit to an individual because it was the outcome of earlier work on the properties of electromagnetic radiation for the accessibility of numerous electronic devices. The question of main concern is

more complicated by the hide of military privacy under which radio location techniques were examined in different countries in the early days of World War-II.

This review writer finally concluded that when radar system is a clear case of direct creation, Robert Watson-Watt's note on Aircraft's Detection & Location of by Radio Methods published immediately 50 years ago. So it was the most significant solitary publication in this field. British achievement in the fight of Britain allocated much to the expansion of a radar system that included technical growth with operational feasibility.

What is a Radar System?

RADAR stands for Radio Detection and Ranging System. It is basically an electromagnetic system used to detect the location and distance of an object from the point where the RADAR is placed. It works by radiating energy into space and monitoring the echo or reflected signal from the objects. It operates in the UHF and microwave range.

A radar is an electromagnetic sensor, used to notice, track, locate, and identify different objects which are at certain distances. The working of radar is, it transmits electromagnetic energy in the direction of targets to observe the echoes and returns from them. Here the targets are nothing but ships, aircraft, astronomical bodies, automotive vehicles, spacecraft, rain, birds, insects, etc. Instead of noticing the target's location and velocity, it also obtains their shape and size sometimes.

The main objective of radar as compared with infrared and optical sensing devices is to discover faraway targets under difficult climate conditions & determines their distance, range, through precision. Radar has its own transmitter which is known as a source of illumination for placing targets. Generally, it works in the microwave area of the electromagnetic spectrum that is calculated in hertz when frequencies extend from 400 MHz to 40 GHz. The essential components which are used in the radar

Radar undergoes quick development during the years 1930-the 40s to reach the requirements of the military. It is still broadly used through the armed forces, wherever several technological advances have created. Simultaneously, radar is also utilized in civilian applications particularly in controlling air traffic, observation of weather, navigation of ship, environment, sensing from remote areas, observation of planetary, measurement of speed in industrial applications, space surveillance, law enforcement, etc.

Working Principle

The **radar working principle** is very simple because it transmits electromagnetic power as well as examines the energy returned back to the target. If the returned signals are received again at the position of their source, then an obstacle is in the transmission way. This is the working principle of radar.

Fundamentals of Radar

The RADAR system generally consists of a transmitter that produces an electromagnetic signal which is radiated into space by an antenna. When this signal strikes an object, it gets reflected or reradiated in many directions. This reflected or echo signal is received by the radar antenna which delivers it to the receiver, where it is processed to determine the geographical statistics of the object.

The range is determined by calculating the time taken by the signal to travel from the RADAR to the target and back. The target's location is measured in angle, from the direction of the maximum amplitude echo signal, the antenna points to. To measure the range and location of moving objects, the Doppler Effect is used.

The essential parts of this system include the following.

- **A Transmitter:** It can be a power amplifier like a Klystron, Travelling Wave Tube, or a power Oscillator like a Magnetron. The signal is first generated using a

waveform generator and then amplified in the power amplifier.

- **Waveguides:** The waveguides are transmission lines for transmission of the RADAR signals.
- **Antenna:** The antenna used can be a parabolic reflector, planar arrays, or electronically steered phased arrays.
- **Duplexer:** A duplexer allows the antenna to be used as a transmitter or a receiver. It can be a gaseous device that would produce a short circuit at the input to the receiver when the transmitter is working.
- **Receiver:** It can be a superheterodyne receiver or any other receiver which consists of a processor to process the signal and detect it.
- **Threshold Decision:** The output of the receiver is compared with a threshold to detect the presence of any object. If the output is below any threshold, the presence of noise is assumed.

How Does Radar use Radio?

Once the radar is placed on a ship or plane, then it requires a similar essential set of components to produce radio signals, transmit them into space and receive them by something, and finally display the information to understand it. A magnetron is one kind of device, used to generate radio signals which are used through radio. These signals are similar to light signals because they travel at the same speed but their signals are much longer with fewer frequencies.

The light signals wavelength is 500 nanometers, whereas the radio signals used by radar normally range from centimeters to meters. In an electromagnetic spectrum, both the signals like radio and light are made with variable designs of magnetic and electrical energy throughout the air. The magnetron in radar generates microwaves the same as a microwave oven. The main disparity is that the magnetron within radar has to transmit the signals several miles, rather than just small distances, so it is more powerful as well as much larger.

Whenever the radio signals have been transmitted, then an antenna functions as a transmitter to transmit them into the air. Generally, the antenna shape is bent so it mainly focuses the signals into an exact and narrow signal; however radar antennas also normally revolve so they can notice actions over a huge area.

The radio signals travel outside from the antenna with 300,000 km per second speed until they strike something and some of them return back to the antenna. In a radar system, there is an essential device namely a duplexer. This device is used to make the antenna change from side to side in between a transmitter & a receiver.

Types of Radar

There are different types of radars which include the following.

- **Bistatic Radar**

This type of radar system includes a Tx-transmitter & an Rx-receiver that is divided through a distance that is equivalent to the distance of the estimated object. The transmitter & the receiver are situated at a similar position is called a monastic radar whereas the very long-range surface to air & air to air military hardware uses the bistatic radar.

- **Doppler Radar**

It is a special type of radar that uses the Doppler Effect to generate data velocity regarding a target at a particular distance. This can be obtained by transmitting electromagnetic signals in the direction of an object so that it analyzes how the action of the object has affected the returned signal's frequency.

This change will give very precise measurements for the radial component of an object's velocity within relation toward the radar. The applications of these radars involve different industries like meteorology, aviation, healthcare, etc.

- **Monopulse Radar**

This kind of radar system compares the obtained signal using a

particular radar pulse next to it by contrasting the signal as observed in numerous directions otherwise polarizations. The most frequent type of monopulse radar is the conical scanning radar. This kind of radar evaluates the return from two ways to measure the position of the object directly. It is significant to note that the radars which are developed in the year 1960 are monopulse radars.

- **Passive Radar**

This kind of radar is mainly designed to notice as well as follow the targets through processing indications from illumination within the surroundings. These sources comprise communication signals as well as commercial broadcasts. The categorization of this radar can be done in the same category of bistatic radar.

- **Instrumentation Radar**

These radars are designed for testing aircraft, missiles, rockets, etc. They give different information including space, position, and time both in the analysis of post-processing & real-time.

- **Weather Radars**

These are used to detect the direction and weather by using radio signals through circular or horizontal polarization. The frequency choice of weather radar mainly depends on a compromise of performance among attenuation as well as precipitation refection as an outcome of atmospheric water steam. Some types of radars are mainly designed to employ Doppler shifts to calculate the wind speed as well as dual- polarization to recognize the types of rainfall.

- **Mapping Radar**

These radars are mainly used to examine a large geographical area for the applications of remote sensing & geography. As a result of synthetic aperture radar, these are restricted to quite

stationary targets. There are some particular radar systems used to detect humans after walls that are more different as compared with the ones found within construction materials.

- **Navigational Radars**

Generally, these are the same to search radars but, they available with small wavelengths that are capable of replicating from the ground & from stones. These are commonly used on commercial ships as well as long-distance airplanes. There are different navigational radars like marine radars which are placed commonly on ships to avoid a collision as well as navigational purposes.

- **Pulsed RADAR**

Pulsed RADAR sends high power and high-frequency pulses towards the target object. It then waits for the echo signal from the object before another pulse is sent. The range and resolution of the RADAR depend on the pulse repetition frequency. It uses the Doppler shift method.

The principle of RADAR detecting moving objects using the Doppler shift works on the fact that echo signals from stationary objects are in the same phase and hence get canceled while echo signals from moving objects will have some changes in phase. These radars are classified into two types.

- **Pulse-Doppler**

It transmits high pulse repetition frequency to avoid Doppler ambiguities. The transmitted signal and the received echo signal are mixed in a detector to get the Doppler shift and the difference signal is filtered using a Doppler filter where the unwanted noise signals are rejected.

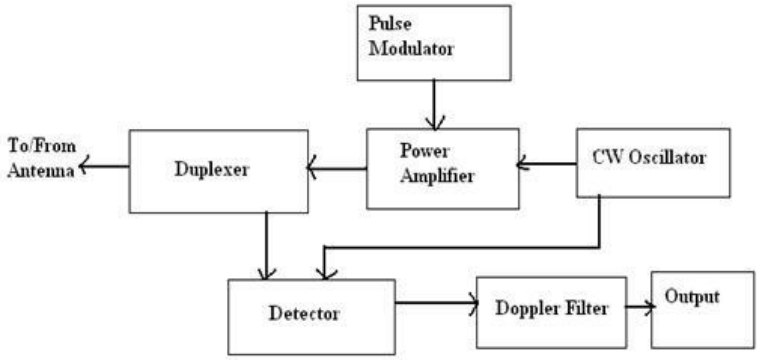


Figure 1.1 Block Diagram of Pulsed Doppler RADAR

Moving Target Indicator

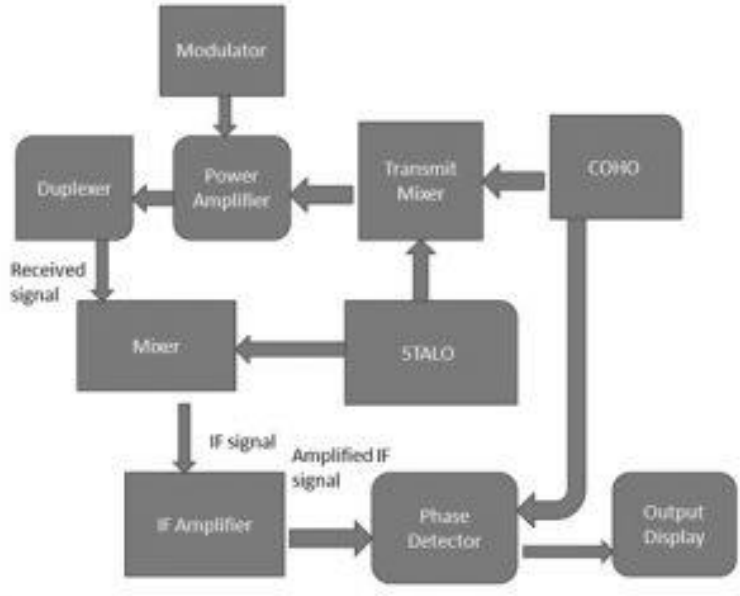


Figure 1.2 MTI Radar

It transmits low pulse repetition frequency to avoid range ambiguities. In an MTI RADAR system, the received echo signals

from the object are directed towards the mixer, where they are mixed with the signal from a stable local oscillator (STALO) to produce the IF signal.

This IF signal is amplified and then given to the phase detector where its phase is compared with the phase of the signal from the Coherent Oscillator (COHO) and the difference signal is produced. The Coherent signal has the same phase as the transmitter signal. The coherent signal and the STALO signal are mixed and given to the power amplifier which is switched on and off using the pulse modulator.

Continuous Wave

The continuous wave RADAR doesn't measure the range of the target but rather the rate of change of range by measuring the Doppler shift of the return signal. In a CW RADAR electromagnetic radiation is emitted instead of pulses. It is basically used for speed measurement.

The RF signal and the IF signal are mixed in the mixer stage to generate the local oscillator frequency. The RF signal is then transmitted signal and the received signal by the RADAR antenna consists of the RF frequency plus the Doppler shift frequency. The received signal is mixed with the local oscillator frequency in the second mixture stage to generate the IF frequency signal.

This signal is amplified and given to the third mixture stage where it is mixed with the IF signal to get the signal with Doppler frequency. This Doppler frequency or Doppler shift gives the rate of change of range of the target and thus the velocity of the target is measured.

Radar Range Equation

There are different kinds of versions available for the radar range equations. Here, the following equation is one of the fundamental types for an only antenna system.

Applications

The **applications of radar** include the following.

- ✓ **Military Applications**

It has 3 major applications in the Military:

- In air defense, it is used for target detection, target recognition, and weapon control (directing the weapon to the tracked targets).
- In a missile system to guide the weapon.
- Identifying enemy locations on the map.

- ✓ **Air Traffic Control**

It has 3 major applications in Air Traffic control:

- To control air traffic near airports. The Air Surveillance RADAR is used to detect and display the aircraft's position in the airport terminals.
- To guide the aircraft to land in bad weather using Precision Approach RADAR.
- To scan the airport surface for aircraft and ground vehicle positions

- ✓ **Remote Sensing**

It can be used for observing whether or observing planetary positions and monitoring sea ice to ensure a smooth route for ships.

- ✓ **Ground Traffic Control**

It can also be used by traffic police to determine the speed of the vehicle, controlling the movement of vehicles by giving warnings about the presence of other vehicles or any other obstacles behind them.

✓ **Space**

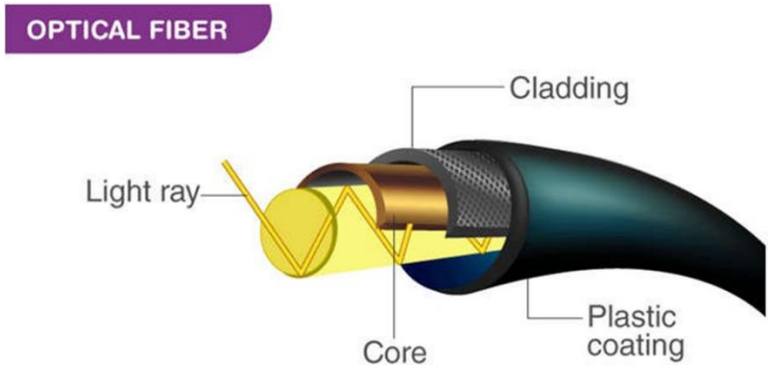
It has 3 major applications

- To guide the space vehicle for a safe landing on the moon
- To observe the planetary systems
- To detect and track satellites
- To monitor the meteors

1.17. Fiber Optics Cable

What is an Optical Fiber?

Optical fiber is the technology associated with data transmission using light pulses travelling along with a long fiber which is usually made of plastic or glass. Metal wires are preferred for transmission in optical fiber communication as signals travel with fewer damages. Optical fibers are also unaffected by electromagnetic interference. The fiber optical cable uses the application of total internal reflection of light. The fibers are designed such that they facilitate the propagation of light along with the optical fiber depending on the requirement of power and distance of transmission. Single-mode fiber is used for long-distance transmission, while multimode fiber is used for shorter distances. The outer cladding of these fibers needs better protection than metal wires.



Types of Optical Fibers

The types of optical fibers depend on the refractive index, materials used, and mode of propagation of light.

The classification based on the refractive index is as follows:

- **Step Index Fibers:** It consists of a core surrounded by the cladding, which has a single uniform index of refraction.
- **Graded Index Fibers:** The refractive index of the optical fiber decreases as the radial distance from the fiber axis increases.

The classification based on the materials used is as follows:

- **Plastic Optical Fibers:** The poly methyl methacrylate is used as a core material for the transmission of the light.
- **Glass Fibers:** It consists of extremely fine glass fibers.

The classification based on the mode of propagation of light is as follows:

- **Single-Mode Fibers:** These fibers are used for long-distance transmission of signals.
- **Multimode Fibers:** These fibers are used for short-distance transmission of signals.

The mode of propagation and refractive index of the core is used to form four combination types of optic fibers as follows:

- Step index-single mode fibers
- Graded index-Single mode fibers
- Step index-Multimode fibers
- Graded index-Multimode fibers

How Does an Optical Fibre Work?

The optical fiber works on the principle of total internal reflection. Light rays can be used to transmit a huge amount of data, but there is a problem here – the light rays travel in straight lines. So

unless we have a long straight wire without any bends at all, harnessing this advantage will be very tedious. Instead, the optical cables are designed such that they bend all the light rays' inwards (using TIR). Light rays travel continuously, bouncing off the optical fiber walls and transmitting end to end data. Although light signals do degrade over progressing distances, depending on the purity of the material used, the loss is much less compared to using metal cables. A Fibre Optic Relay System consists of the following components:

- The Transmitter – It produces the light signals and encodes them to fit to transmit.
- The Optical Fibre – The medium for transmitting the light pulse (signal).
- The Optical Receiver – It receives the transmitted light pulse (signal) and decodes them to be fit to use.
- The Optical Regenerator – Necessary for long-distance data transmission.

Advantages of Optical Fibre Communication

- Economical and cost-effective
- Thin and non-flammable
- Less power consumption
- Less signal degradation
- Flexible and lightweight

1.18. ISDN-their properties

ISDN or Integrated Services Digital Network is a circuit-switched telephone network system that transmits both data and voice over a digital line. You can also think of it as a set of communication standards to transmit data, voice, and signaling.

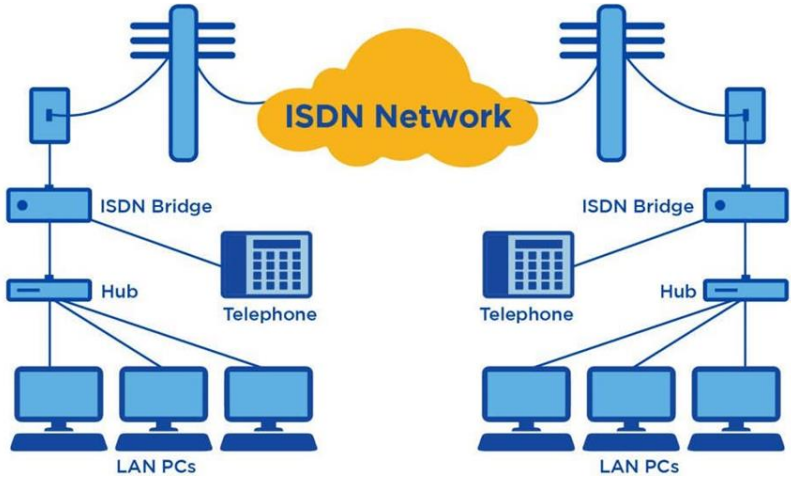
These digital lines could be copper lines. It was designed to move outdated landline technology to digital.

ISDN connections have a reputation for providing better speeds

and higher quality than traditional connections. Faster speeds and better connections allow data transmissions to travel more reliably.

The modern upgrade to an ISDN would be using a **SIP trunk provider** — they use the data for business phone service to a PBX.

How ISDN works



It's easy enough to define ISDN, but do you know how it works?

Most people use ISDN for high-speed internet when options like DSL or cable modem connections are not available.

Setting up ISDN is something you'll want to work on with your Internet Service Provider (ISP). A lot of the steps can easily be done from your home.

Your ISDN will be plugged in through a traditional **POTS (Plain Old Telephone Service**) line that can access both phone numbers at once.

You'll have to make sure you have a working POTS line and assigned phone numbers to begin.

After that, you can follow the steps below to get your voice and data communications up and running.

What are the advantages of ISDN?

- So why do people use ISDN?
- It first started as an alternative to your dial-up connection that provided higher internet speeds.
- To access the internet with ISDN, users had to connect through a digital modem.
- People still use ISDN for internet access in areas where broadband internet isn't an option. For the most part, ISDN for internet access is being phased out.
- There have been many attempts to improve the ISDN service.
- Broadband ISDN, also known as B-ISDN, transmitted data over fiber optic cable. Another attempt was ISDN BRI which attempted to improve voice services.

Some of the reasons people choose ISDN are:

- It offers multiple digital services that operate through the same copper wire
- Digital signals broadcast through telephone lines.
- ISDN provides a higher data transfer rate.
- Can connect devices and allow them to operate over a single line. This includes credit card readers, fax machines, and other manifold devices.
- It is up and running faster than other modems.

1.18.1. Geographic Information System (GIS)

Introduction

Geographical Information System (GIS) is a technology that provides the means to collect and use geographic data to assist in the development of Agriculture. A digital map is generally of much greater value than the same map printed on a paper as the digital version can be combined with other sources of data for

analyzing information with a graphical presentation. The GIS software makes it possible to synthesize large amounts of different data, combining different layers of information to manage and retrieve the data in a more useful manner. GIS provides a powerful means for agricultural scientists to better service to the farmers and farming community in answering their query and helping in a better decision making to implement planning activities for the development of agriculture.

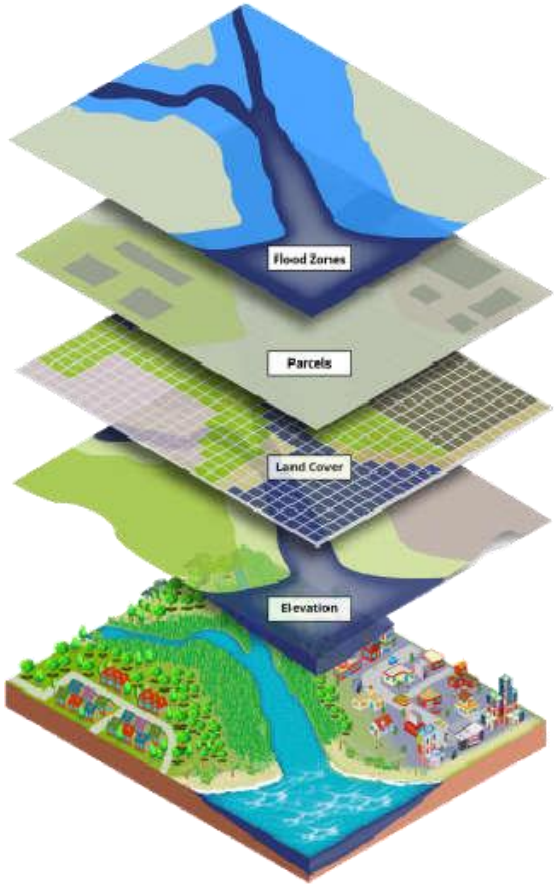


Overview of GIS

A Geographical Information System (GIS) is a system for capturing, storing, analyzing and managing data and associated attributes, which are spatially referenced to the Earth. The geographical information system is also called as a geographic information system or geospatial information system. It is an information system capable of integrating, storing, editing, analyzing, sharing, and displaying geographically referenced information. In a more generic sense, GIS is a software tool that allows users to create interactive queries, analyze the spatial information, edit data, maps, and present the results of all these operations. GIS technology is becoming essential tool to combine various maps and remote sensing information to generate are used in real time environment. Geographical information system is the various models, which science utilizing the geographic concepts, applications and systems.

Geographical Information System can be used for scientific investigations, resource management, asset management, environmental impact assessment, urban planning, cartography, criminology, history, sales, marketing, and logistics. For example, agricultural planners might use geographical data to decide on the best locations for a location specific crop planning, by combining data on soils, topography, and rainfall to determine the size and location of biologically suitable areas. The final output could include overlays with land ownership, transport, infrastructure, labour availability, and distance to market centers.

What is GIS mapping?



GIS mapping produces visualizations of geospatial information. The 4 main ideas of Geographic Information Systems (GIS) are:

- Create geographic data.
- Manage it in a database.
- Analyze and find patterns.
- Visualize it on a map.

Because viewing and analyzing data on maps impacts our understanding of data, we can make better decisions using GIS.

What are applications of GIS?

Every day, GIS powers millions of decisions around world such as these 1000 GIS applications. It makes a big impact in our life and you might not even realize. For example, we use GIS for:

- Pinpointing new store locations
- Reporting power outages
- Analyzing crime patterns
- Routing in car navigation
- Forecasting and predicting weather

Components of GIS

GIS enables the user to input, manage, manipulate, analyze, and display geographically referenced data using a computerized system. To perform various operations with GIS, the components of GIS such as software, hardware, data, people and methods are essential.

Software

GIS software provides the functions and tools needed to store, analyze, and display geographic information. Key software components are (a) a database management system (DBMS) (b) tools for the input and manipulation of geographic information (c) tools that support geographic query, analysis, and visualization (d) a graphical user interface (GUI) for easy access to tools. GIS

software are either commercial software or software developed on Open Source domain, which are available for free. However, the commercial software is copyright protected, can be expensive and is available in terms number of licensees. Currently available commercial GIS software includes Arc/Info, Intergraph, MapInfo, Gram++ etc. Out of these Arc/Info is the most popular software package. And, the open source software are AMS/MARS etc.

Hardware

Hardware is the computer on which a GIS operates. Today, GIS runs on a wide range of hardware types, from centralized computer servers to desktop computers used in stand-alone or networked configurations.

Data

The most important component of a GIS is the data. Geographic data or Spatial data and related tabular data can be collected in-house or bought from a commercial data provider. Spatial data can be in the form of a map/remotely-sensed data such as satellite imagery and aerial photography.

These data forms must be properly georeferenced (latitude/longitude). Tabular data can be in the form attribute data that is in some way related to spatial data. Most GIS software comes with inbuilt Database Management Systems (DBMS) to create and maintain a database to help organize and manage data.

GIS stores location data as thematic layers. Each data set has an attribute table that stores information about the feature. The two main types of GIS data are raster and vector:

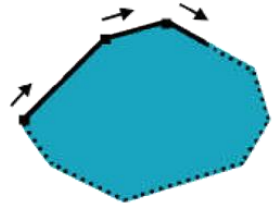
Raster

Raster look like grids because they store data in rows and columns. They can be discrete or continuous. For example, we often represent land cover, temperature data and imagery as raster data.



Vector

Vectors are points, lines and polygons with vertices. For example, fire hydrants, contours and administrative boundaries are often vectors.



Users

GIS technology is of limited value without the users who manage the system and to develop plans for applying it. GIS users range from technical specialists who design and maintain the system to those who use it to help them do their everyday work. 6 These users are largely interested in the results of the analyses and may have no interest or knowledge of the methods of analysis. The user-friendly interface of the GIS software allows the nontechnical users to ha e easy access to GIS analytical capabilities without needing to know detailed software commands. A simple User Interface (UI) can consist of menus and pull-down graphic windows so that the user can perform required analysis with a few key presses without needing to learn specific commands in detail.

Functions of GIS

General-purpose GIS software performs six major tasks such as input, manipulation, management, query and analysis, Visualization.

2

Information Security: Introduction

2.1. Information Security

The protection of information and information systems from unauthorised access, use, or disruption.

It is important for employees and all stakeholders to understand information security policies and guidelines. It is also necessary to have good work practices that comply with security policies so that the effects of possible breaches can be minimised.

Information Security is not only about securing information from unauthorized access. Information Security is basically the practice of preventing unauthorized access, use, disclosure, disruption, modification, inspection, recording or destruction of information. Information can be physical or electronic one. Information can be anything like Your details or we can say your profile on social media, your data in mobile phone, your biometrics etc. Thus Information Security spans so many research areas like Cryptography, Mobile Computing, Cyber Forensics, Online Social Media etc.

During First World War, Multi-tier Classification System was developed keeping in mind sensitivity of information. With the beginning of Second World War formal alignment of Classification System was done. Alan Turing was the one who successfully decrypted Enigma Machine which was used by Germans to encrypt warfare data.

Why is it important to keep the University network secure?

- **Data theft:** Hackers accessing student or employee personal or confidential details;
- **Data loss:** Unauthorized entities manipulating or deleting important data;
- **Law suits** against the University;
- Loss of **Reputation**; and
- **Financial loss** including recovery expenses.

Types of Information Security

Some of the types of Information Security are:

- **Application Security:** This covers a wide area of vulnerabilities in software, web, and application programming interfaces. The vulnerabilities can exist in user authentication and other areas which can create entry points for potential threats and information security breaches.
- **Information Security:** This refers to the security of the entire infrastructure related to this area.
- **Cloud Security:** This refers to the shared cloud environment and information security for the building and hosting a secure infrastructure for client applications.
- **Cryptography:** Encryption and Cryptography play a big part in information security in many fields. A beginner can easily relate to the encrypted messaging in WhatsApp to prevent information breaches when transmitted between two individuals.
- **Vulnerability Management:** This is the process of screening all system areas to find out any potential weak zones. This is critical in an era where organizations are continually upgrading their hardware and adding new applications where any weak and vulnerable points need to be plugged at the earliest.

2.2. Ensuring Security

The key areas to consider when safeguarding the University's information and assets are listed below:

1. **Update your software.**

    - New versions of software are released to address security problems that have been found. Updating your software ensures you take full advantage of all the security upgrades.
    - If you do not update the software you can put your computer at risk of viruses and other problems because the software is no longer supported.

2. **Use anti-virus software.**

    - Anti-virus companies spend their time ensuring their software helps stops known viruses. If you have a current and up-to-date version, you can be assured that the software is looking out for problems and blocking them.

3. **Be suspicious of unsolicited phone calls or emails.**

    - Unsolicited emails and phone calls are trying to get you to do something that will benefit someone else. It might be just spam trying to get you to buy things, or it might be trying to get you to access something that will put a virus on your computer or give others access to your information**.**

4. **Back up your data.**

    - If you have a problem with your computer and it needs to be reset or even replaced, you will still have access to your information if you have backed it up.

5. **Use legitimate software.**

You should always use legitimate software that you have

purchased from a vendor or downloaded from the company's website. Why?

- Using pirated copies of software on university devices will mean that the University could be prosecuted if discovered.
- If you use pirated copies you open up your computer to viruses that you may not know about. The software itself may contain a virus, or it won't be supported by the vendor, meaning you won't receive regular security updates.

6. Set strong passwords and use different passwords for different accounts.

- A password that is strong and changed regularly makes it harder for people to access your information.
- If you use the same password for all your accounts and one account is compromised, the person accessing your account is more likely to be able to guess all your other passwords and access those accounts too.
- If you store an unencrypted list of your passwords on your computer and someone gains access to the computer, they then have all your passwords.

8. Do not lose your device.

- If someone gets your device, they may have access to all your information and plenty of time to access it.

2.3. Malicious Programs

Malware definition

Malware, short for malicious software, is a blanket term for viruses, worms, trojans and other harmful computer programs hackers use to wreak destruction and gain access to sensitive information. As Microsoft puts it, "[malware] is a catch-all term to refer to any software designed to cause damage to a single

computer, server, or computer network." In other words, software is identified as malware based on its intended use, rather than a particular technique or technology used to build it.

This means that the question of, say, what the difference is between malware and a virus misses the point a bit: a virus is a type of malware, so all viruses are malware (but not every piece of malware is a virus).

Types of malwares

- **Virus**

Viruses are a subgroup of malware. A virus is malicious software attached to a document or file that supports macros to execute its code and spread from host to host. Once downloaded, the virus will lay dormant until the file is opened and in use. Viruses are designed to disrupt a system's ability to operate. As a result, viruses can cause significant operational issues and data loss.

- **Worms**

Worms are a malicious software that rapidly replicates and spreads to any device within the network. Unlike viruses, worms do not need host programs to disseminate. A worm infects a device via a downloaded file or a network connection before it multiplies and disperses at an exponential rate. Like viruses, worms can severely disrupt the operations of a device and cause data loss.

- **Trojan virus**

Trojan viruses are disguised as helpful software programs. But once the user downloads it, the Trojan virus can gain access to sensitive data and then modify, block, or delete the data. This can be extremely harmful to the performance of the device. Unlike normal viruses and worms, Trojan viruses are not designed to self-replicate.

- **Spyware**

Spyware is malicious software that runs secretly on a computer and reports back to a remote user. Rather than simply disrupting a device's operations, spyware targets sensitive information and can grant remote access to predators. Spyware is often used to steal financial or personal information. A specific type of spyware is a keylogger, which records your keystrokes to reveal passwords and personal information.

- **Adware**

Adware is malicious software used to collect data on your computer usage and provide appropriate advertisements to you. While adware is not always dangerous, in some cases adware can cause issues for your system. Adware can redirect your browser to unsafe sites, and it can even contain Trojan horses and spyware. Additionally, significant levels of adware can slow down your system noticeably. Because not all adware is malicious, it is important to have protection that constantly and intelligently scans these programs.

- **Ransomware**

Ransomware is malicious software that gains access to sensitive information within a system, encrypts that information so that the user cannot access it, and then demands a financial payout for the data to be released. Ransomware is commonly part of a phishing scam. By clicking a disguised link, the user downloads the ransomware. The attacker proceeds to encrypt specific information that can only be opened by a mathematical key they know. When the attacker receives payment, the data is unlocked.

- **Fileless malware**

Fileless malware is a type of memory-resident malware. As the term suggests, it is malware that operates from a victim's computer's memory, not from files on the hard drive. Because there are no files to scan, it is harder to detect than traditional

malware. It also makes forensics more difficult because the malware disappears when the victim computer is rebooted. In late 2017, the Cisco Talos threat intelligence team posted an example of fileless malware that they called DNS Messenger.

How to prevent malware

With spam and phishing email being the primary vector by which malware infects computers, the best way to prevent malware is make sure your email systems are locked down tight—and your users know how to spot danger. We recommend a combination of carefully checking attached documents and restricting potentially dangerous user behavior—as well as just familiarizing your users with common phishing scams so that their common sense can kick in.

When it comes to more technical preventative measures, there are a number of steps you can take, including keeping all your systems patched and updated, keeping an inventory of hardware so you know what you need to protect, and performing continuous vulnerability assessments on your infrastructure. When it comes to ransomware attacks in particular, one way to be prepared is to always make backups of your files, ensuring that you'll never need to pay a ransom to get them back if your hard drive is encrypted.

Malware protection

Antivirus software is the most widely known product in the category of malware protection products; despite "virus" being in the name, most offerings take on all forms of malware. While high-end security pros dismiss it as obsolete, it's still the backbone of basic anti-malware defense. Today's best antivirus software is from vendors Kaspersky Lab, Symantec and Trend Micro, according to recent tests by AV-TEST.

When it comes to more advanced corporate networks, endpoint security offerings provide defense in depth against

malware. They provide not only the signature-based malware detection that you expect from antivirus, but anti-spyware, personal firewall, application control and other styles of host intrusion prevention. Gartner offers a list of its top picks in this space, which include products from Cylance, CrowdStrike, and Carbon Black.

How to detect malware

It's fully possible—and perhaps even likely—that your system will be infected by malware at some point despite your best efforts. How can you tell for sure? *CSO* columnist Roger Grimes has written a deep dive into how to diagnose your PC for potential malware that you might find helpful.

When you get to the level of corporate IT, there are also more advanced visibility tools you can use to see what's going on in your networks and detect malware infections. Most forms of malware use the network to either spread or send information back to their controllers, so network traffic contains signals of malware infection that you might otherwise miss; there are a wide range of network monitoring tools out there, with prices ranging from a few dollars to a few thousand. There are also SIEM tools, which evolved from log management programs; these tools analyze logs from various computers and appliances across your infrastructure looking for signs of problems, including malware infection. SIEM vendors range from industry stalwarts like IBM and HP Enterprise to smaller specialists like Splunk and Alien Vault.

Malware removal

How to remove malware once you're infected is in fact the million dollar question. Malware removal is a tricky business, and the method can vary depending on the type you're dealing with. *CSO* has information on how to remove or otherwise recover from rootkits, ransomware, and crypto jacking. We also have a guide to auditing your Windows registry to figure out how to move forward.

If you're looking for tools for cleansing your system, Tech Radar has a good roundup of free offerings, which contains some familiar names from the antivirus world along with newcomers like Malwarebytes.

Malware examples

We've already discussed some of the current malware threats looming large today. But there is a long, storied history of malware, dating back to infected floppy disks swapped by Apple II hobbyists in the 1980s and the Morris Worm spreading across Unix machines in 1988. Some of the other high-profile malware attacks have included:

- **ILOVEYOU**, a worm that spread like wildfire in 2000 and did more than $15 billion in damage
- **SQL Slammer**, which ground internet traffic to a halt within minutes of its first rapid spread in 2003
- **Conficker**, a worm that exploited unpatched flaws in Windows and leveraged a variety of attack vectors – from injecting malicious code to phishing emails – to ultimately crack passwords and hijack Windows devices into a botnet.
- **Zeus**, a late '00s keylogger Trojan that targeted banking information
- **CryptoLocker**, the first widespread ransomware attack, whose code keeps getting repurposed in similar malware projects
- **Stuxnet**, an extremely sophisticated worm that infected computers worldwide but only did real damage in one place: the Iranian nuclear facility at Natanz, where it destroyed uranium-enriching centrifuges, the mission it was built for by U.S. and Israeli intelligence agencies

Malware trends

You can count on cyber criminals to follow the money. They will target victims depending on likelihood of delivering their

malware successfully and size of potential payout. If you look at malware trends over the past few years, you will see some fluctuation in terms of the popularity of certain types of malware and who the most common victims are—all driven by what the criminals believe will have the biggest ROI.

Recent research reports indicate some interesting shifts in malware tactics and targets. Cryptominers, which had surpassed ransomware as the most common type of malware, are falling out of favor due to the decline in cryptocurrency values. Ransomware is becoming more targeted, moving away from a shotgun approach.

Malware attacks on businesses spike

Businesses saw a 79 percent increase in the amount of malware they dealt with in 2018 over 2017, according to the Malwarebytes Labs State of Malware Report 2019. "What we usually see year-end or quarterly end is that there has been some sort of increase or large amounts of detections on the consumer side," says Adam Kujawa, director of Malwarebytes Labs. "On the business side it might slowly grow, but certainly nothing like we've seen this last six months." By comparison, consumer detections decreased by 3 percent over the same period.

"We've observed that there is a significant push by cyber criminals to move away from consumers and put their really heavy stuff against businesses instead," Kujawa adds.

That "really heavy stuff" comes largely in the form of older consumer-focused malware that's "been weaponized" to become a bigger, more versatile threat for business. Kujawa cites Emotet as one of the most significant. "It's a nasty little information stealing Trojan that also installs additional malware, spreads laterally, and acts as its own spam sender. Once it infects a system, it starts sending email and tries to infect other people."

Emotet has been around since 2014 and targeted mainly consumers. Originally, it infected a computer looking for an

individual's financial or credit card information to steal. Since then, it's picked up new capabilities inspired by or borrowed from other successful malware like Wannacry or EternalBlue. "Now it's become much more modular and we see it able to use these exploits to traverse through a corporate network whereas before they were limited to a single endpoint," says Kujawa. "Even if it's a small network in a small business, it's more juicy than infecting Grandma."

Lateral movement of malware is increasing, according to the Global Threat Report: The Year of the Next-Gen Cyberattack from Carbon Black. Nearly 60 percent of malware attacks on business are now designed to move laterally across a network.

One reason for the spike in malware attacks on business might be the EU's General Data Privacy Regulation (GDPR). Kujawa believes it's possible that attackers stepped up business attacks thinking that it would be harder to steal personal and other data after the regulation went into effect. That combined with the decline of cryptocurrency values and stepped up defenses against ransomware turned attackers to what worked in the past. "They always [go back to what works]," he says. "Cyber crime is cyclical. It always comes back around."

Cryptomining attacks decline

The Malwarebyte Labs report has seen a shift away from cryptomining starting in the second quarter of 2018, due largely to the decline in cryptocurrency values. Still, the number of cryptomining detections increased for the year by 7 percent.

Instead, cyber criminals are turning to information stealing malware like Emotet to turn a profit. "Overall, it seems as though criminals have reached the consensus that sometimes stealing is better than mining," the report stated.

Ransomware becoming more targeted

Kujawa notes that small and medium-sized businesses (SMBs) are

becoming more popular targets. He attributes this to the likelihood of being paid for ransomware attacks—SMBs often can't afford the downtime and see paying ransom as the fastest way to recover. They also often softer targets than larger businesses.

Ransomware detections actually declined by 26 percent worldwide in 2018, according to the Malwarebytes report. However, ransomware detections at businesses rose by 28 percent. Industries most often targeted were consulting, education, manufacturing and retail. Kujawa believes criminals focus on these industries because of opportunity and likelihood of ransoms being paid.

2.4. Cryptography

Basic Concepts

- **Cryptography** The art or science encompassing the principles and methods of transforming an intelligible message into one that is unintelligible, and then retransforming that message back to its original form
- **Plaintext** The original intelligible message
- **Cipher text** The transformed message
- **Cipher** An algorithm for transforming an intelligible message into one that is unintelligible by transposition and/or substitution methods
- **Key** Some critical information used by the cipher, known only to the sender& receiver **Encipher (encode)** The process of converting plaintext to cipher text using a cipher and a key **Decipher (decode)** the process of converting cipher text back into plaintext using a cipher and a key
- **Cryptanalysis** The study of principles and methods of transforming an unintelligible message back into an intelligible message without knowledge of the key. Also called code breaking
- **Cryptology** Both cryptography and cryptanalysis
- **Code** An algorithm for transforming an intelligible

message into an unintelligible one using a code-book Cryptograph

Cryptography

- Cryptographic systems are generally classified along 3 independent dimensions:

Type of operations used for transforming plain text to cipher text

- All the encryption algorithms are based on two general principles: substitution, in which each element in the plaintext is mapped into another element, and transposition, in which elements in the plaintext are rearranged.

The number of keys used

- If the sender and receiver uses same key then it is said to be symmetric key (or)

single key (or) conventional encryption.

- If the sender and receiver use different keys then it is said to be public key encryption.

The way in which the plain text is processed

- A block cipher processes the input and block of elements at a time, producing output block for each input block.
- A **stream cipher** processes the input elements continuously, producing output element one at a time, as it goes along.



2.5. Cryptanalysis

The process of attempting to discover X or K or both is known as cryptanalysis. The strategy used by the cryptanalysis depends on the nature of the encryption scheme and the information available to the cryptanalyst.

There are various types of cryptanalytic attacks based on the amount of information known to the cryptanalyst.

- **Cipher text only –** A copy of cipher text alone is known to the cryptanalyst.
- **Known plaintext –** The cryptanalyst has a copy of the cipher text and the corresponding plaintext.
- **Chosen plaintext –** The cryptanalysts gains temporary access to the encryption machine. They cannot open it to find the key, however; they can encrypt a large number of suitably chosen plaintexts and try to use the resulting cipher texts to deduce the key.
- **Chosen cipher text –** The cryptanalyst obtains temporary access to the decryption machine, uses it to decrypt several string of symbols, and tries to use the results to deduce the key.

Cryptographic algorithms

Cryptosystems use a set of procedures known as cryptographic algorithms, or ciphers, to encrypt and decrypt messages to secure communications among computer systems, devices and applications.

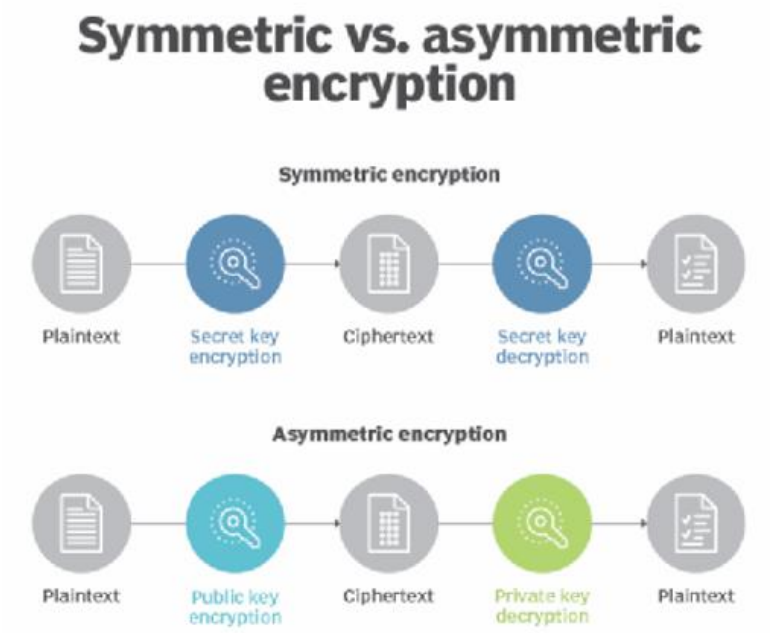
A cipher suite uses one algorithm for encryption, another algorithm for message authentication and another for key exchange. This process, embedded in protocols and written in software that runs on operating systems (OSes) and networked computer systems, involves:

- Public and private key generation for data

encryption/decryption
- Digital signing and verification for message authentication
- Key exchange

Types of cryptography

Single-key or symmetric-key encryption algorithms create a fixed length of bits known as a block cipher with a secret key that the creator/sender uses to encipher data (encryption) and the receiver uses to decipher it. One example of symmetric-key cryptography is the Advanced Encryption Standard (AES). AES is a specification established in November 2001 by the National Institute of Standards and Technology (NIST) as a Federal Information Processing Standard (FIPS 197) to protect sensitive information. The standard is mandated by the U.S. government and widely used in the private sector.



In June 2003, AES was approved by the U.S. government for

classified information. It is a royalty- free specification implemented in software and hardware worldwide. AES is the successor to the Data Encryption Standard (DES) and DES3. It uses longer key lengths -- 128-bit, 192-bit, 256-bit -- to prevent brute force and other attacks.

Symmetric cryptography uses a single key while asymmetric cryptography uses a key pair to encrypt and decrypt data.

Public-key or asymmetric-key encryption algorithms use a pair of keys, a public key associated with the creator/sender for encrypting messages and a private key that only the originator knows (unless it is exposed or they decide to share it) for decrypting that information.

Examples of public-key cryptography include:

- RSA, used widely on the internet
- Elliptic Curve Digital Signature Algorithm (ECDSA) used by Bitcoin
- Digital Signature Algorithm (DSA) adopted as a Federal Information Processing Standard for digital signatures by NIST in FIPS 186-4
- Diffie-Hellman key exchange

To maintain data integrity in cryptography, hash functions, which return a deterministic output from an input value, are used to map data to a fixed data size. Types of cryptographic hash functions include SHA-1 (Secure Hash Algorithm 1), SHA-2 and SHA-3.

2.6. DIGITAL SIGNATURES

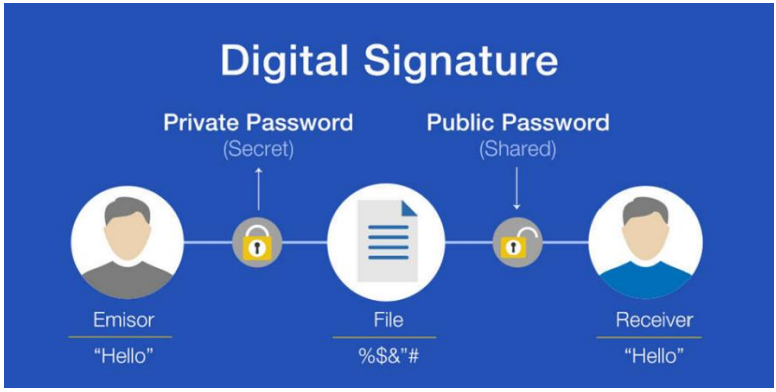
Digital signatures are like electronic "fingerprints." In the form of a coded message, the digital signature securely associates a signer with a document in a recorded transaction. Digital signatures use a standard, accepted format, called Public Key Infrastructure (PKI), to provide the highest levels of security and universal

acceptance. They are a specific signature technology implementation of electronic signature (eSignature).

What's the difference between a digital signature and an electronic signature?

The broad category of electronic signatures (eSignatures) encompasses many types of electronic signatures. The category includes digital signatures, which are a specific technology implementation of electronic signatures. Both digital signatures and other eSignature solutions allow you to sign documents and authenticate the signer. However, there are differences in purpose, technical implementation, geographical use, and legal and cultural acceptance of digital signatures versus other types of eSignatures.

In particular, the use of digital signature technology for eSignatures varies significantly between countries that follow open, technology-neutral eSignature laws, including the United States, United Kingdom, Canada, and Australia, and those that follow tiered eSignature models that prefer locally defined standards that are based on digital signature technology, including many countries in the European Union, South America, and Asia. In addition, some industries also support specific standards that are based on digital signature technology.

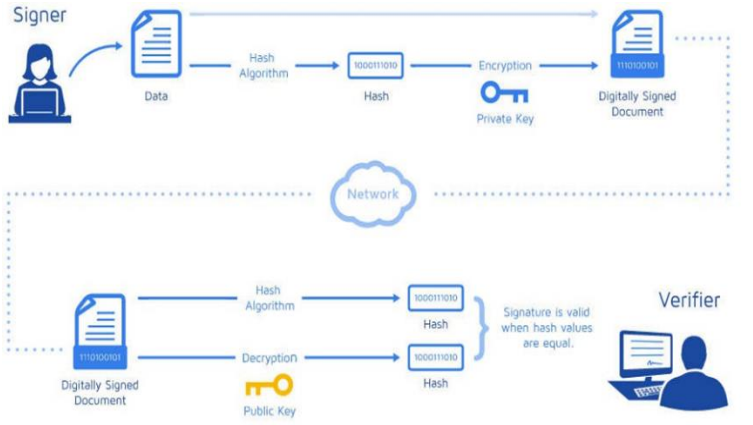


How do digital signatures work?

Digital signatures, like handwritten signatures, are unique to each signer. Digital signature solution providers, such as DocuSign, follow a specific protocol, called PKI. PKI requires the provider to use a mathematical algorithm t one key is private.

When a signer electronically signs a document, the signature is created using the signer's private key, which is always securely kept by the signer. The mathematical algorithm acts like a cipher, creating data matching the signed document, called a hash, and encrypting that data. The resulting encrypted data is the digital signature. The signature is also marked with the time that the document was signed. If the document changes after signing, the digital signature is invalidated.

As an example, Jane signs an agreement to sell a timeshare using her private key. The buyer receives the document. The buyer who receives the document also receives a copy of Jane's public key. If the public key can't decrypt the signature (via the cipher from which the keys were created), it means the signature isn't Jane's, or has been changed since it was signed. The signature is then considered invalid.



To protect the integrity of the signature, PKI requires that the keys

be created, conducted, and saved in a secure manner, and often requires the services of a reliable Certificate Authority (CA). Digital signature providers, like DocuSign, meet PKI requirements for safe digital signing.

Create a digital signature?

eSignature providers, such as DocuSign, that offer solutions based on digital signature technology, make it easy to digitally sign documents. They provide an interface for sending and signing documents online and work with the appropriate Certificate Authorities to provide trusted digital certificates.

Depending upon the Certificate Authority you are using, you may be required to supply specific information. There also may be restrictions and limitations on whom you send documents to for signing and the order in which you send them. DocuSign's interface walks you through the process and ensures that you meet all of these requirements. When you receive a document for signing via email, you must authenticate as per the Certificate Authority's requirements and then "sign" the document by filling out a form online.

2.7. Firewall

What is a Firewall?

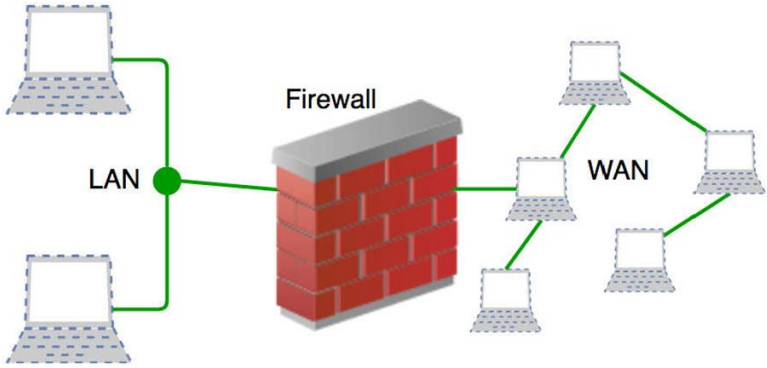
A Firewall is a security system to protect an internal network from unauthorized servers and networks based on predefined rules. It acts as a barrier and only allows the secured network to send or receive data.

How does a Firewall work?

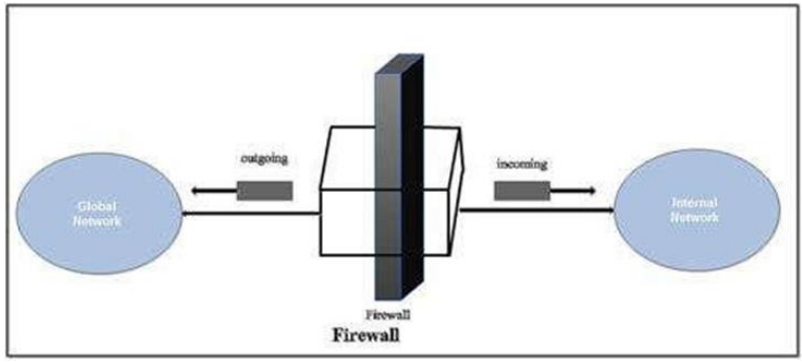
A Firewall analyses the network traffic and filters it so that the unsecured and suspicious networks cannot attack the system. The point where information is exchanged with an external network is called a port.

How is Firewall different from an Antivirus?

A firewall is a security network designed to protect computer system's and networks from malicious attacks. Whereas, Antivirus is a software utility program designed to protect a system from internal attacks from viruses.



A software firewall prevents unwanted access to the computer over a network connection by identifying and preventing communication over risky ports. Computers communicate over many different recognized ports, and the firewall will tend to permit these without prompting or alerting the user.



History and Development of Firewall

Discussed below is a brief account of the evolution and

development of Firewall and why it had become an important part of network security.

The term 'Firewall' actually meant a wall which intended to confine a fire within a line of adjacent buildings. It was only in the late 1980s when this was acknowledged as a computer terminology.

It was during this time that the Internet has started to emerge as a new tool for global use. Thus, having a means which could secure the transmission and flow of data was required by many.

Until the Firewall was introduced, routers performed the same function as it restricted the number of people who could use a particular network.

Types of Firewall

There are various types of Firewalls. Described below are each of them in detail for a better and simplistic understanding:

1. Packet Filtering Firewall

- One of the oldest types of Firewall
- This type of Firewall creates a checkpoint at the traffic router. Only the secure and verified IP address or networks are allowed for the further flow of data
- The data packets are not verified, i.e. the information or data is not opened at the Firewall stage
- They are easy to use and do not overload the device and do not affect its processing or functioning speed

2. Application Level Gateway Firewall

- It is also known as Proxy Firewall
- When the user connects with the destination server, it forms a connection with the application gateway
- The proxy then connects with the destination server and takes up the decision of forwarding the data packets

- It is a bit more secure in comparison to Packet Filtering Firewall
- Strong Memory and processors are required for using this Firewall

3. Circuit Level Gateway Firewall

- This works as the Sessions layer of the OSI Model
- Using this, two Transmission Control Protocol (TCP) connections can be set up together
- It can easily let the flow of data packets continue without consuming major computer resources
- These Firewalls are not much efficient as they do not check the data packets and incase a data packet comprises malware, it will allow it to pass if the TCP connections are successfully done

4. Stateful Inspection Firewall

- It is a combination of data packet inspection and TCP connection. Until both the fields are verified, the information cannot be approved
- They are less straining for the computer resources
- However, they are a bit slow in comparison to other Firewalls

5. Next-Generation Firewall

- The recently launched Firewall systems are known as the Next-Gen Firewalls
- Under this, the data packets are also thoroughly checked before being passed on to the destination address
- These are still on the platform of improving and evolving and intend to use modern technology for automatic detection of errors and network safety

6. Software Firewall

- Any firewall which is installed in a local device or a cloud

server is called a Software Firewall

- They can be the most beneficial in terms of restricting the number of networks being connected to a single device and control the in-flow and out-flow of data packets
- Software Firewall also time-consuming

7. Hardware Firewall

- They are also known as Physical-appliance based firewalls
- It ensures that the malicious data is stopped before it reaches the endpoint of the network at risk

Functions of Firewall

Following are the function of a Firewall. Candidates can refer to these to apprehend the basic functions of a Firewall:

- Any data which enters or exits a computer network has to pass through the Firewall
- All the valuable information stays intact if the data packets are securely passed through the Firewall
- Every time a data packets passed through a Firewall, it records it which allows the user to record the network activity
- No data can be modified as it is held securely within the data packets

Precisely, a Firewall ensures that all the data is secure and any malicious data trying to enter the internal network is not allowed to pass through.

A firewall can be simple or complex, depending on how specifically you want to control your Internet traffic. A simple firewall might require only that you configure the software in the router that connects your intranet to your ISP. A more complex firewall might be a computer running UNIX and specialized software.

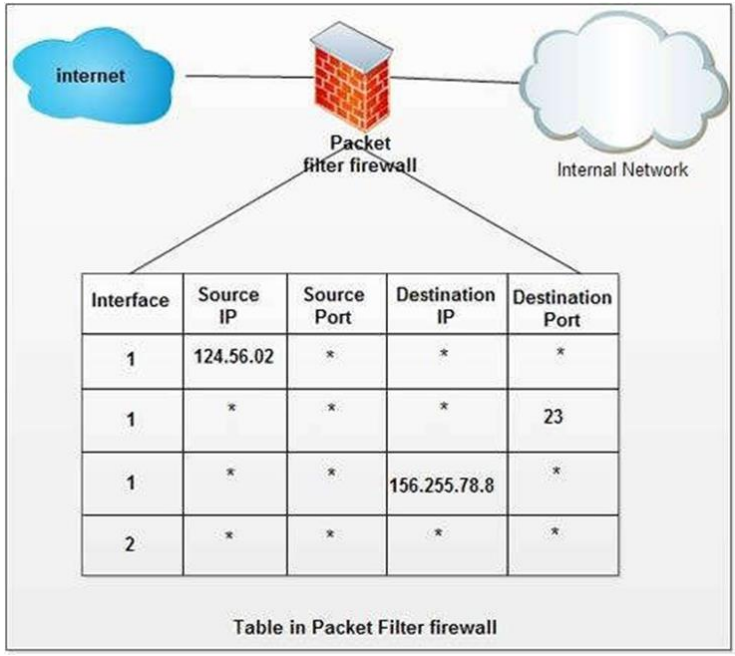
Firewall systems fall into two categories

- Network-Level
- Application-Level.

Network-Level Firewalls:

It can be used as packet filter. These firewalls examine only the headers of each packet of information passing to or from the Internet. The firewall accepts or rejects packets based on the packet's sender, receiver, and port. For example, the firewall might allow e-mail and Web packets to and from any computer on the intranet, but allow telnet (remote login) packets to and from only selected computers.

Packet filter firewall maintains a filtering table that decides which packets are to be forwarded or discarded. A packet filter firewall filters at the network or transport layer.



Interface	Source IP	Source Port	Destination IP	Destination Port
1	124.56.02	*	*	*
1	*	*	*	23
1	*	*	156.255.78.8	*
2	*	*	*	*

Table in Packet Filter firewall

As shown in fig. the packets are filtered according to following specifications:

1. Incoming packets from network 124.56.0.2 are block (* means any).
2. Incoming packets destined for any internal TELNET server (port 23) are blocked.
3. Incoming packets for internal host 156.255.7.8.8 are blocked.
4. Outgoing packets destined for an HTTP server (port 80) are blocked i.e. employees of organization are not allowed to browse the internet and cannot send any HTTP request.

Application-Level Firewalls:

These firewalls handle packets for each Internet service separately, usually by running a program called a *proxy server*, which accepts e-mail, Web, chat, newsgroup, and other packets from computers on the intranet, strips off the information that identifies the source of the packet, and passes it along to the Internet.

When the replies return, the proxy server passes the replies back to the computer that sent the original message. A proxy server can also log all the packets that pass by, so that you have a record of who has access to your intranet from the Internet, and vice versa.

2.8. Users Identification and Authentication

User authentication verifies the identity of a user attempting to gain access to a network or computing resource by authorizing a human-to-machine transfer of credentials during interactions on a network to confirm a user's authenticity. The term contrasts with machine authentication, which is an automated authentication method that does not require user input.

Authentication helps ensure only authorized users can gain access to a system by preventing unauthorized users from gaining access and potentially damaging systems, stealing information or causing other problems. Almost all human-to-computer interactions -- other than guest and automatically logged-in accounts -- perform a user authentication. It authorizes access on both wired and wireless networks to enable access to networked and internet-connected systems and resources.

There is a growing demand for different types of user authentication technologies for both online and in physical systems. The motivation to authenticate users ranges from access control reasons to business development purposes like adding e-commerce elements.

Organizations need to understand that passwords are not the only way to authenticate users. There is a wide variety of authentication technologies and an even greater range of activities that require authentication methods.

What Is Authentication?

Authentication is the process of identifying users that request access to a system, network, or device. Access control often determines user identity according to credentials like username and password. Other authentication technologies like biometrics and authentication apps are also used to authenticate user identity.

Why Is User Authentication Important?

User authentication is a method that keeps unauthorized users from accessing sensitive information. For example, User A only has access to relevant information and cannot see the sensitive information of User B.

Cybercriminals can gain access to a system and steal information when user authentication is not secure. The data breaches companies like Adobe, Equifax, and Yahoo faced are examples of

what happens when organizations fail to secure their user authentication.

Hackers gained access to Yahoo user accounts to steal contacts, calendars and private emails between 2012 and 2016. The Equifax data breach in 2017 exposed credit card data of more than 147 million consumers. Without a secure authentication process, any organization could be at risk.

Common Authentication Types

Cybercriminals always improve their attacks. As a result, security teams are facing plenty of authentication-related challenges. This is why companies are starting to implement more sophisticated incident response strategies, including authentication as part of the process. The list below reviews some common authentication methods used to secure modern systems.

1. Password-based authentication

Passwords are the most common methods of authentication. Passwords can be in the form of a string of letters, numbers, or special characters. To protect yourself you need to create strong passwords that include a combination of all possible options.

However, passwords are prone to phishing attacks and bad hygiene that weakens effectiveness. An average person has about 25 different online accounts, but only 54% of users use different passwords across their accounts.

The truth is that there are a lot of passwords to remember. As a result, many people choose convenience over security. Most people use simple passwords instead of creating reliable passwords because they are easier to remember.

The bottom line is that passwords have a lot of weaknesses and are not sufficient in protecting online information. Hackers can easily guess user credentials by running through all possible combinations until they find a match.

2. Multi-factor authentication



Multi-Factor Authentication (MFA) is an authentication method that requires two or more independent ways to identify a user. Examples include codes generated from the user's smartphone, Captcha tests, fingerprints, voice biometrics or facial recognition.

MFA authentication methods and technologies increase the confidence of users by adding multiple layers of security. MFA may be a good defense against most account hacks, but it has its own pitfalls. People may lose their phones or SIM cards and not be able to generate an authentication code.

3. Certificate-based authentication

Certificate-based authentication technologies identify users, machines or devices by using digital certificates. A digital certificate is an electronic document based on the idea of a driver's license or a passport.

The certificate contains the digital identity of a user including a public key, and the digital signature of a certification authority. Digital certificates prove the ownership of a public key and issued only by a certification authority.

Users provide their digital certificates when they sign in to a server. The server verifies the credibility of the digital signature and the certificate authority. The server then uses cryptography to confirm that the user has a correct private key associated with the certificate.

4. Biometric authentication

Biometrics authentication is a security process that relies on the unique biological characteristics of an individual. Here are key advantages of using biometric authentication technologies:

- Biological characteristics can be easily compared to authorized features saved in a database.
- Biometric authentication can control physical access when installed on gates and doors.
- You can add biometrics into your multi-factor authentication process.

Biometric authentication technologies are used by consumers, governments and private corporations including airports, military bases, and national borders. The technology is increasingly adopted due to the ability to achieve a high level of security without creating friction for the user. Common biometric authentication methods include:

- **Facial recognition**—matches the different face characteristics of an individual trying to gain access to an approved face stored in a database. Face recognition can be inconsistent when comparing faces at different angles or comparing people who look similar, like close relatives. Facial liveness like ID R&D's passive facial liveness prevents spoofing.
- **Fingerprint scanners**—match the unique patterns on an individual's fingerprints. Some new versions of fingerprint scanners can even assess the vascular patterns in people's fingers. Fingerprint scanners are currently the most popular biometric technology for everyday

consumers, despite their frequent inaccuracies. This popularity can be attributed to iPhones.

- **Speaker Recognition** —also known as voice biometrics, examines a speaker's speech patterns for the formation of specific shapes and sound qualities. A voice-protected device usually relies on standardized words to identify users, just like a password.
- **Eye scanners**—include technologies like iris recognition and retina scanners. Iris scanners project a bright light towards the eye and search for unique patterns in the colored ring around the pupil of the eye. The patterns are then compared to approved information stored in a database. Eye-based authentication may suffer inaccuracies if a person wears glasses or contact lenses.

5. Token-based authentication

Token-based authentication technologies enable users to enter their credentials once and receive a unique encrypted string of random characters in exchange. You can then use the token to access protected systems instead of entering your credentials all over again. The digital token proves that you already have access permission. Use cases of token-based authentication include RESTful APIs that are used by multiple frameworks and clients.

Authentication technology is always changing. Businesses have to move beyond passwords and think of authentication as a means of enhancing user experience. Authentication methods like biometrics eliminate the need to remember long and complex passwords. As a result of enhanced authentication methods and technologies, attackers will not be able to exploit passwords, and a data breach will be prevented.

2.9. Security Awareness and Policies, Application Areas Requiring Security

Information security notes means leadership must protect corporate information, and the information of customers,

associates, and employees, placed in their custody. Leadership's needs to ensure that every employee understands the corporate concern with the protection of information with a Information Security Policy.

An effective information protection program cannot be solely defined in terms of trust. Rather, it must be based upon the same prudent business practices that applied to earlier manual systems; careful definition of individual responsibilities, separation of controls, maintenance of audit trails, protection of vital records, and access to information limited, based on "need to know".

Information security notes include policies, procedures, controls, reviews, and especially, separation of duties. They must have real and continued management backing and involvement.

Management must be pro-active regarding information security policy. They must let their people know the practices are important through example. They must take pride in their program. They must help bring about attitudinal changes in their people through strongly administered awareness programs, bringing about a clear understanding of the reasoning behind information security, instilling real concern about information protection, and instilling dissatisfaction with anything less than great execution. Executive leadership must ensure and apprise newly appointed account management details of the information security policy.

Make each manager responsible not only for the protection and integrity of assets under their control, but also for compliance with Corporate Information Protection Policies and Procedures. Each manager must make the necessary risk management decisions that consider the unique environment in which their resources are used, and must be prepared to justify their decisions during an audit.

Policy Prologue (Policy)

Our company information is one of most important assets, second

only to its employees. Our company information includes all proprietary information, in any form, related to the business of our company that has been created using the resources of our company.

As our company competes in the global marketplace, it is very important that each employee understands the competitive value of our company information security notes and their responsibility to protect it. We must be able to demonstrate that it has exercised due diligence in protecting our company information. In this way we can successfully assert its rights to our company information through legal channels.

This information security policy identifies the high level guidance for Information Protection for all business entities of our company. It is based on the three sides of an Information Protection triangle where the three triangle legs consist of: Information Security, Business Continuity, and IT Compliance.

Information Protection Triangle

The **Information Security** leg consists of confidentiality and integrity. Confidentiality ensures that our company information is not disclosed to anyone who is not authorized to access it. Linked to this concept is the idea of "need-to-know", authorizing access only to those who can demonstrate a legitimate need for our information. Integrity ensures that information cannot be accidentally or intentionally modified or destroyed.

The **Business Continuity** leg consists of mitigation, crisis management, and contingency management. Mitigation deals with reducing or eliminating risks. Crisis management deals with the planning and training of people for the survival of the our company team and the business entity following a disaster.

Contingency management deals with planning for the recovery and continuation of critical internal and customer business functions following a service interruption, and the testing of

business recovery plans. This segment has also been known as availability in the past, but the scope is much broader today.

The **IT Compliance** leg consists of practices that do not fall within the scope of the other two legs. information security notes cannot be solely defined in terms of trust. Rather, it must be based upon the same prudent business practices that applied to earlier manual systems; careful definition of individual responsibilities, separation of controls, maintenance of audit trails, protection of vital records, and access to information limited, based on "need-to-know".

Included in the scope of the IT Compliance leg is adherence to the laws and ethics that govern us, i.e., copyright infringement, software licensing, export compliance, etc. These are controls, laws, or ethics principles, and are exactly what auditors look for, which is why the leg is called IT Compliance (being able to pass a stringent audit because the business is controlled, information is adequately protected, and laws are not being violated).

Information Security Notes and Policy

The following identifies Information Protection requirements for our company

1. Information, in any form, relating to the business of our company, and created using the resources of our company, is an asset owned by our company. Our company information should be protected from unauthorized disclosure, modification, and destruction, whether intentional or unintentional.
2. All our company employees are collectively and individually responsible for protecting information. They must comply with the policy set forth in this document and any other information protection documents derived from this policy. These include employee codes of conduct and other documents required as part of the terms of employment.

3. We mandate ongoing awareness and training activities to inform our employees of information protection issues are mandated as part of the ongoing support of information protection.
4. Our company information, when created, must be assessed according to its value and sensitivity to disclosure, and be managed according to "need-to-know". We identify controls that define our information security notes handling, retention, and destruction requirements.
5. Our company information must bear the legal and business markings as pertinent to communicate ownership, rights, and handling instructions. Some markings include, but are not limited to, the following: Copyright, Trademark, Patent, and the classification level.
6. Our company information cannot be released to the public through media interviews, publications, seminars, conversations, or in any other manner without a review procedure and management approval.
7. The computing and communication environment in which we create, process, store, and transmit information must be assessed as to the criticality of its existence and function relative to the business. Contingency management measures based on this assessment must be defined, implemented, maintained, and tested.
8. Systems and applications which process or communicate our company information must be created and maintained using a system development methodology that addresses the three sides of the Information Protection triangle, where the three sides consist of: Information Security, Business Continuity, and IT Compliance.
9. Our company and its employees complies with all legal requirements and all contractual agreements

requiring information protection, e.g., copyright infringement, software licensing, patents, etc.

10. Our company and its employees should complies with the laws and regulations within each country where we conduct our business. This includes, but is not limited to import and export regulations and the cross-border transfer of information and technology.
11. Our company and its employees should comply with our information retention requirements that support the business and support legal requirements.
12. Our company reserves the right to monitor and audit any and all activity and information security notes within the computer and communication resources utilized in support of our business. We may authorize a third party to exercise this right on behalf of our company.
13. Our company entities shall conduct an annual self-assessment regarding information protection controls, implementing corrective action where needed.
14. Suspicion or occurrence of any fraudulent activity, unauthorized disclosure, modification, or destruction of our company information security notes and intrusions to the computing and communications environment must be reported to our company Security

2.10. Physical Security

Environment

Data centers and computer rooms must have good physical security and strong protection from disaster and security threats, whether natural or caused by other reasons, in order to minimize the extent of loss and disruption.

Backup media containing business essential and/or mission critical information must be housed off-site at a safe distance from

the main site in order to avoid damage arising from a disaster at the main site.

Equipment Security

All Information Systems must be placed in a secure environment or attended by staff to prevent unauthorized access.

Staff in possession of laptop, portable computer, personal digital assistant, or mobile computing devices for business purposes must safeguard the equipment in his/her possession, and must not leave the equipment unattended without proper security measures.

IT equipment must not be taken off the property without proper control.

Physical Access Control

- A list of persons who are authorized to gain access to data centers, computer rooms or other areas supporting critical activities, where computer equipment and data are located or stored, must be kept up-to-date and be reviewed periodically.
- All access keys, cards, passwords, etc. for entry to any of the computer systems and networks must be physically secured or subject to well-defined and strictly enforced security procedures.
- All visitors to data centers or computer rooms must be monitored at all times by an authorized person.
- Automatic protection features (e.g. password protected screen saver, keyboard lock) in servers, computer terminals, workstations or microcomputers should be activated if there has been no activity for a predefined period of time to prevent an illegal system access attempt. Alternatively, the logon session and connection should be terminated. Also, user workstations should be switched off, if appropriate, before leaving work for the day or

before a prolonged period of inactivity.

- All staff with separate personal offices that can be directly accessed from a public area and contain Information System(s) should lock the doors when these offices are not in use.

The display screen of an Information System on which classified information can be viewed must be carefully positioned so that unauthorized persons cannot readily shoulder- surf.

2.11. Access Control Security

Data Access Control

- Access to information must not be allowed unless authorized by the relevant information owner.
- Data access rights must be granted to users based on a need-to-know basis.
- Data access rights must be clearly defined and reviewed periodically.
- Access to an Information Security Notes System containing confidential or classified information must be restricted by means of logical access control.

Authentication

- Access to classified information security notes without appropriate authentication must not be allowed.
- Authentication must be performed in a manner commensurate with the sensitivity of the information to be accessed.
- Consecutive unsuccessful log-in attempts must be controlled.

Privacy

Management reserves the right to examine all information stored in or transmitted by company- owned computer systems.

User Identification

- Each user identity (user-ID) must uniquely identify only one user. Shared or group user-IDs are not permitted unless explicitly approved by the IT Security Officer.
- Users are responsible for all activities performed with their user-IDs.

User Privileges Management

- All accounts must be revoked after a pre-defined period of inactivity.
- User privileges must be reviewed periodically.
- At the time that a member of the staff is transferred or ceases to provide services to the company, all related Information Systems privileges must be promptly terminated.
- The use of special privileges must be restricted and controlled.

Password Management

- Business entities must define a strict password policy that details at least, minimum password length, initial assignment, restricted words and format, password life cycle, and include guidelines on suitable system and user password selection.
- Passwords must not be shared or divulged unless necessary (e.g., helpdesk assistance, shared PC and shared files). The risk of sharing passwords is that it increases the probability of security being compromised. If passwords must be shared, explicit approval from the IT Security Officer must be obtained. In addition, the shared passwords should be changed promptly when the need no longer exists and should be changed frequently if sharing is required on a regular basis.
- Passwords must always be well protected when held in storage. Passwords must be encrypted when transmitted

over an un-trusted communication network. Compensating controls must be applied to reduce the risk exposure of Information Systems to an acceptable level if encryption is not available.

- Staff are prohibited from capturing or otherwise obtaining passwords, decryption keys, or any other access control mechanism, which could permit unauthorized access.
- All vendor-supplied default passwords must be changed before any Information System is put into operation.
- All passwords must be promptly changed if they are suspected of being compromised, or disclosed to vendors for maintenance and support.

Network Access Control

Prior approval from the IT Security Officer is required to connect an Information System with another Information System under the control of another entity. The security level of the Information Security Notes System being connected must not be downgraded.

Logging

- Business entities must define policies relating to the logging of activities of Information Security Notes Systems under their control according to the business needs and data classification.
- Any log kept must provide sufficient information to support comprehensive audits of the effectiveness of, and compliance of information security notes measures.
- Logs must be retained for a period commensurate with their usefulness as an audit tool. During this period, such logs must be secured such that they cannot be modified, and can only be read by authorized persons.
- Logs must not be used to profile the activity of a particular user unless it relates to a necessary audit activity supported by the IT Security Officer.
- Regular checking on log records, especially on

system/application where classified information is processed/stored, must be performed, not only on the completeness but also the integrity of the log records. All system and application errors which are suspected to be triggered as a result of security breaches must be reported and logged.

- Clock synchronization should be configured to keep the clocks of Information Security Notes Systems in sync.

2.12. Data Security

Overall Data Confidentiality

- Information security notes that may compromise the security of those systems must not be disclosed to users, or any other third parties, except on a need-to-know basis and only if authorized by the IT Security Officer.
- Staff must not disclose information about the individuals, business entities or specific information security notes systems that have suffered from damages caused by computer crimes and computer abuses, or the specific methods used to exploit certain system vulnerabilities, to any people other than those who are handling the incident and responsible for the security of such systems, or authorized investigators involved in the investigation of the crime or abuse.
- Staff must not disclose to any unauthorized persons the nature and location of the Information Systems, and the information security notes controls that are in use or the way in which they are implemented.
- All stored information classified as confidential or above must be encrypted.
- Business entities must comply with handling in relation to Information Security notes security including, but not limited to, storage, transmission, processing, and destruction of classified information.

Information Backup

- Backup and recovery procedures must be well documented, properly implemented, and tested periodically.
- Backups must be carried out at regular intervals.
- Backup activities must be reviewed regularly.
- Backups must be stored off-site at a remote distance from the main site, and be protected. Backup media should also be protected against unauthorized access, misuse, or corruption during transportation.

Application Security

Application Development & Maintenance

- Application development staff must include information security notes planning and implement the appropriate information security notes measures and controls for systems under development according to the systems' security requirements.
- Documentation and listings of applications must be properly maintained and restricted on a need-to-know basis.
- Formal testing and review on the information security notes controls must be performed prior to implementation.
- The integrity of an application must be maintained with appropriate security controls such as version control mechanisms and separation of environments for development, system testing, acceptance testing, and live operation.
- Application development staff must not be permitted to access production information unless necessary.

Configuration Management & Control

- Change control procedures for requesting and approving

program/system changes must be documented.

- Changes affecting existing information security notes protection mechanisms must be carefully considered.
- Installation of all computer equipment and software must be done under control and audit.
- Business entities must ensure that staffs are formally advised of the impact of security changes and usage on Information Systems.

Network & Communication Security

General Network Protection

- Internal network addresses, configurations and related system or network information must not be publicly released without the approval of the concerned entity.
- All internal networks with connections to other networks or publicly accessible computer networks must be properly protected.
- Security measures must be in place to prevent unauthorized remote access to the systems and data.
- Staff are prohibited from connecting workstations to an external network by means of any communication device, such as dial-up modem, wireless interface, or broadband link, if the workstations are simultaneously connected to a local area network (LAN) or another internal communication network, without the approval of the concerned entity.
- Staff must not connect any unauthorized Information System device to an Information System without prior approval as designated by the entity.
- Proper configuration and administration of information / communication systems is required and must be reviewed regularly.
- Connections and links made to other networks must not compromise the information security notes processed at another, and vice versa.
- Confidential/Restricted information security notes must

be encrypted when transmitted over an un-trusted communication network.

- Top Secret/Secret information must be transmitted only under encryption and inside an isolated LAN approved by the IT Security Officer.

Internet Security

- All Internet access must be either through centrally arranged Internet gateways or the entities own Internet gateway conforming to internal security standards. In circumstances where this is not feasible or having regard to the mode of use, i.e., such modes of use may include, for example, Internet surfing, email exchange, and the use of official, portable computers while on business. The relevant standalone machines must still be protected by any applicable security mechanisms.
- Business entities may consider allowing Internet access through stand-alone machines, provided that there is an approval and control mechanism at an appropriate level within the business entity.
- Business entities should consider the value versus inconvenience of implementing technologies to blocking non-business web sites. The ability to connect with a specific web site does not in itself imply that users of systems are permitted to visit that site.
- Each entity must clearly define and communicate to users its internet policy in relation to acceptable Internet usage.
- All software and files downloaded from the Internet must be screened and verified with anti-virus software.
- Staff should not execute mobile code or software downloaded from the Internet unless the code is from a known and trusted source.

Email Security

- Each entity must clearly define and communicate to users its email policy in relation to acceptable email usage.

- Systems administrators must establish and maintain a systematic information security notes process for the recording, retention, and destruction of electronic mail messages and accompanying logs.
- Incoming/outgoing email must be screened for computer viruses and malicious codes.
- Internal email address lists containing entries for authorized users must be properly maintained and protected from unauthorized access and modification.
- Email transmission of classified information must be transmitted only on an Information System approved by the IT Security Officer.
- Emails from suspicious sources should not be opened or forwarded.

Protection Against Computer Virus and Malicious Code

- Anti-virus software must always be enabled on all local area network servers and personal computers, and computers connecting to the internal network via remote access.
- Business entities must protect their Information Security Notes Systems from computer viruses and malicious codes. Virus signatures, malicious code definitions as well as their detection and repair engines must be updated regularly and whenever necessary.
- Storage media and files from unknown source or origin must not be used unless the storage media and files have been checked and cleaned for computer viruses and malicious codes.
- Users must not intentionally write, generate, copy, propagate, execute or be involved in introducing computer viruses or malicious codes.
- Business entities must implement proper measures to protect their wireless or mobile computing devices against computer viruses and malicious codes.

Software and Patch Management

- Computers and networks must only run software that comes from trustworthy sources.
- No unauthorized application software must be loaded onto an Information System without prior approval from IT Security Officer as designated by the entity.
- Business entities must protect their Information Security Notes Systems from known vulnerabilities by applying the latest security patches recommended by the product vendors or implementing other compensating security measures.
- Before security patches are applied, proper risk evaluation and testing should be conducted to minimize undesirable effects to Information Systems.

Wireless Security

- Business entities must document, monitor, and control wireless networks with connection to internal networks.
- Proper authentication and encryption security controls must be employed to protect data communication over wireless networks with connection to internal networks.

Security Risk Assessments & Audits

Security Risk Assessment

- Information security notes risk assessments for information systems and production applications must be performed at least once every two years. A security risk assessment must also be performed prior to major enhancements and changes associated with these systems or applications.
- Use of software and programs for security risk assessment analysis must be restricted and controlled.

Security Auditing

- Information Systems must be periodically evaluated by auditors of an independent and trusted party to determine the minimum set of controls required to reduce risk to an acceptable level.
- Auditing of compliance of computer and network security policies must be performed periodically.
- Use of software and programs for security audit analysis must be restricted and controlled.

Security Incident Management

Security Incident Monitoring

- Business entities must establish an incident detection and monitoring mechanism to detect, contain and ultimately prevent security incidents.
- Business entities must ensure that system logs and other supporting information are retained for the proof and tracing of security incidents.

Security Incident Response

- Business entities must establish, document and maintain a security incident handling/reporting procedure for their Information Systems.
- Staff must be made aware of the security incident handling/reporting procedure that is in place and must observe and follow it accordingly.
- All network or systems software malfunctions, information security alerts, warnings, suspected vulnerabilities, and the like, and suspected network security problems, must be reported immediately only to the responsible party according to the incident handling procedure.
- Immediate follow-up actions are required on suspected system intrusion according to security incident handling/reporting procedures.

2.13. Mobile Commerce: Introduction, Growth, Success Stories of Mobile Commerce, Technologies for Mobile Commerce, M- Commerce in India, Digital Marketing.

M-Commerce, short for Mobile Commerce, is a branch of commerce that deals with the digital transactions made from the smartphone. It doesn't require physical contact between two people for sending or receiving the money.



In other words, we can say that **M-commerce** (mobile commerce) is the buying and selling of goods and services through wireless handheld devices such as smartphones and tablets. As a form of e-commerce, m-commerce enables users to access online shopping platforms without needing to use a desktop computer.

Examples of M-commerce –

- In-app purchasing
- Mobile banking
- Virtual marketplace apps like the Amazon mobile app or a digital wallet such as **Apple Pay, Android Pay, and Samsung Pay.**
- Mobile ticketing

According to Wikipedia, *the term* **mobile commerce** *was originally coined in 1997 by Kevin Duffey at the launch of the Global Mobile Commerce Forum, to mean "the delivery of electronic commerce*

capabilities directly into the consumer's hand, anywhere, via wireless technology."

2.14. Types of M-Commerce

M-commerce can be categorized by function as either mobile shopping, mobile banking or mobile payments. Mobile shopping allows for a customer to purchase a product from a mobile device, using an application such as Amazon, or over a web app.

A subcategory of mobile shopping is app commerce, which is a transaction that takes place over a native app. Mobile banking includes any handheld technology that enables customers to conduct transactions.

This is typically done through a secure, dedicated app provided by the banking institution. Mobile payments enable users to buy products using a mobile device. Digital wallets, such as Apple Pay, allow a customer to buy a product without needing to swipe a card or pay with physical cash.

M-Commerce is a vast field which doesn't end at buying a commodity online- it includes all the behaviors related to checking any stuff online from your smartphone.

The different types of Mobile Commerce are-

1. **Browsing for stuff online on your mobile** – This type deals with surfing for stuff online like looking for Groceries, Daily essentials, and electronics. We're talking about dedicated apps, optimized websites, or even social media platforms like Facebook or Instagram that allow in-app purchases or linking to online stores.
2. **Purchasing app-delivery stuff** (Food delivery or e-commerce apps) – This type deals with the services related to ordering things like food, cab pickup, and others.
3. **Mobile banking** – Mobile banking is the method that

involves accessing the features of the bank using online methods. The transactions are made from specific apps designed by apps, though some financial services companies are now experimenting with chatbots or messaging apps to deliver customer service.

4. **Mobile App payments(Google Pay or PayPal)** – This type primarily focuses on making payment transactions through apps. The user is required to register their credit or debit card and then use it whenever they want.
5. **Purchasing or renting digital content (Netflix, Spotify, etc) on a mobile device** – This involves buying or renting online services like Music Streaming, Video Streaming, or commodity- based on a rent basis.
6. **Mobile person-to-person payments(Venmo or Xoom)** – This type involves sending money from one person to another, be it for buying stuff or sending it to someone in need. This type of transaction can be made by sending money to someone's number or to their bank account directly.

2.15. How Mobile Commerce Works?



With most **m-commerce** enabled platforms, the mobile device is

connected to a wireless network that can be used to make online product purchases. For those in charge of developing an M-commerce application, important ***KPIs*** to monitor include the *total mobile traffic, the total amount of traffic on the application, average order value, and the value of orders over time.*

Similarly, tracking the mobile add to cart rate will help developers examine, if visitors are becoming customers. M-commerce developers may also be interested in logging *average page loading times, mobile cart conversion rates and SMS subscriptions.*

In terms of mobile payment products, they operate through a form of peer-to-peer (P2P) sharing. Once a mobile device is paired with a bank card's information, the phone can be waved over a payment terminal to pay for a product. This contact less payment via a mobile device is feasible due to the use of Near Field Communication(NFC).

Applications of Mobile Commerce-

Mobile commerce has entered into all spheres of our daily lives like- finance, retail, telecommunication, healthcare, information technology, sales and services. Need for M-commerce has increased multi-fold in recent times because of the ease of functioning and accessibility they offer.

Some of the applications of mobile commerce are-

1. M-Commerce for finance
2. M-Commerce for Retail and After sale Services
3. M-commerce & Mobile ticketing
4. M-commerce & Mobile marketing
5. M-commerce for mobile entertainment
6. M-commerce for hotel reservations
7. M-commerce for Healthcare & Medicine
8. M-commerce for Information
9. M-commerce for Gaming
10. M-commerce for Intra-office communication

Advantages of M-Commerce



Advantages of M-Commerce are-

1. Increases the amount of customer retention by being easily accessible to them
2. Wider variety of commodities and services
3. More options for selecting the product
4. Convenient for the consumer to compare the pricing, product reviews, and making purchases without the use of a computer.
5. Multiple options for payments like the credit card and debit card payments,
6. Better user experience

Disadvantages of M-Commerce are-

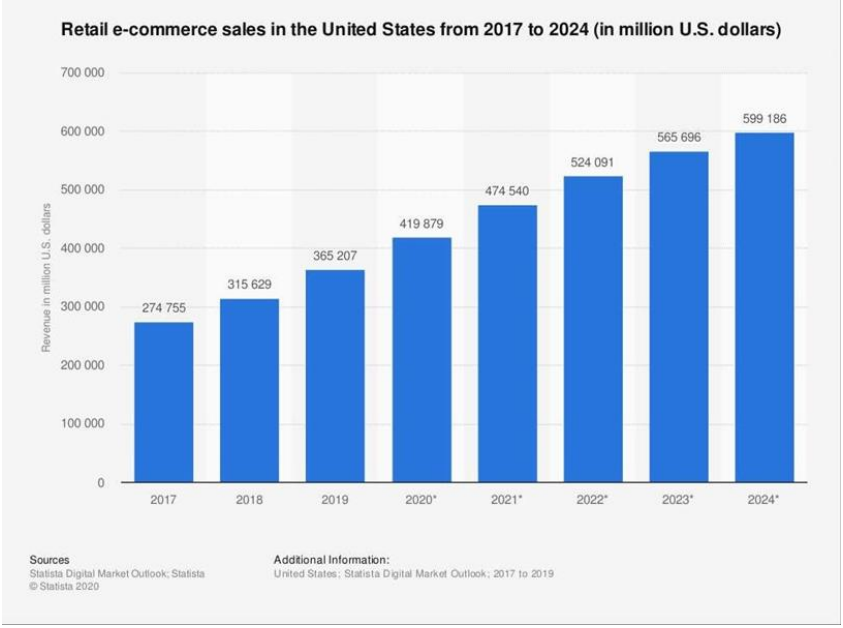
1. The technology required to set up M-commerce at the moment is very expensive. The initial cost to **begin a business** is very high.
2. In developing countries, the network signal is not very good and the providers are not capable of providing good speed. Due to this, it is not suitable for data transfer.
3. There is a risk of security- Although it is quite safe these

days, there is always a chance of data leaking and intrusions.

4. A poorly designed app can lead to lesser number of purchases because of customer dissatisfaction.
5. Mobile payment is still not available across many locations in the world.

Future scope of M-Commerce

M-Commerce is rapidly expanding in today's time. Everyone is busy and they don't have the time or energy to go to the market and buy stuff for them. Instead, they prefer ordering from their home itself. The biggest trend for M-Commerce is its market share/ growth. **M-Commerce** spending in 2018 was calculated at $501 Billion and it is expected to grow more to $740 Billion by 2023.



Major Trends for M-Commerce indicating Great Future Scope-

1. **The mobile wallet market will grow to $5.3 trillion by**

the end of 2020 – Mobile wallets have made our life easier by reducing the hassle of carrying a wallet with us everywhere. It offers the convenience of making as many transactions we want by just storing our credit or debit card details just once. A proof of this trend is the fact that the market size of mobile wallets has grown from *$1.65 Trillion to $5.3 Trillion* between 2013 and 2020.

2. **M-Commerce market will see new contenders between 2020 and 2025** – Currently, the payment markets are captured by old players like PayPal, Google, Apple, and SAP. But in the coming years, you can see the increase in the number of new contenders capturing the majority amount of market share. They will bring more opportunities for customers and share the market share more uniformly.
3. **The mobile commerce revenue is expected to hit $3.56 Trillion by 2021** – The revenue generated has been increasing with each passing year and will touch $3.56 trillion by 2021. This is proof of the increasing demand for mobile-based shopping.

M-Commerce Key Statistics & Insights-

To help you understand the potential of mobile commerce, check these insights and key statistics:

- 96% of Americans own a cellphone.
- 8 out of 10 Americans shop on their Mobile devices.
- Mobile commerce is expected to outgrow non-mobile commerce in 2021.
- Mobile digital advertising spend is double that of desktop advertising spend—$71 billion was spent on mobile advertising in 2018, while only $37 billion was spent on desktop.
- On Black Friday of 2018, 66% of shopping was done on mobile devices.
- In 2021, mobile commerce sales will account for more than half of total e-commerce sales.

- Today, one-third of purchasing decisions is influenced by researching products on mobile devices.
- More than half of retailers already have a mobile app and over 50% of the rest is going to build one soon.
- In mid-2019, mobile e-commerce spending in the United States reached a smashing $41.2 billion.
- The total number of US-based shoppers who make mobile purchases will reach 168.7 million.
- The volume of mobile commerce sales will be more than double between 2017 and 2020, reaching $ 336 billion this year.

What is the Scope of Mobile Commerce in India?

In India, the majority of the people, irrespective of their ages, are using a smartphone. Especially during the coronavirus pandemic, the few people who didn't use a smartphone probably started using one. From children in their nappies to grannies in their chairs are using a smartphone for multiple reasons. On average, the age group between 25 and 34 are using these smartphones to shop online. The m-commerce niche is to be completely explored and utilized in its full potential and India is actively working in that regard. Following are a few initiatives taken by the Government of India in order to encourage m-commerce:

1. Unified Payment Interface
2. GST Implementation
3. Mobile Wallets
4. The Digital India Makeover
5. Startup India
6. Skill India

3

Artificial Intelligence

3.1. What is AI?

- **A**rtificial Intelligence (AI) is a branch of *Science* which deals with helping machines find solutions to complex problems in a more human-like fashion.
- This generally involves borrowing characteristics from human intelligence, and applying them as algorithms in a computer friendly way.
- A more or less flexible or efficient approach can be taken depending on the requirements established, which influences how artificial the intelligent behavior appears
- Artificial intelligence can be viewed from a variety of perspectives.
    - ✓ From the perspective of **intelligence** artificial intelligence is making machines "intelligent" -- acting as we would expect people to act.
        - The inability to distinguish computer responses from human responses is called the Turing test.
        - Intelligence requires knowledge
        - Expert problem solving - restricting domain to allow including significant relevant knowledge
    - ✓ From a **business** perspective AI is a set of very powerful tools, and methodologies for using those tools to solve business problems.
    - ✓ From a **programming** perspective, AI includes the

study of symbolic programming, problem solving, and search.

- Typically AI programs focus on symbols rather than numeric processing.
- Problem solving - achieve goals.
- Search - seldom access a solution directly. Search may include a variety of techniques.

AI programming languages include:

- LISP, developed in the 1950s, is the early programming language strongly associated with AI. LISP is a functional programming language with procedural extensions. LISP (LISt Processor) was specifically designed for processing heterogeneous lists -- typically a list of symbols. Features of LISP are run- time type checking, higher order functions (functions that have other functions as parameters), automatic memory management (garbage collection) and an interactive environment.
- The second language strongly associated with AI is PROLOG. PROLOG was developed in the 1970s. PROLOG is based on first order logic. PROLOG is declarative in nature and has facilities for explicitly limiting the search space.
- Object-oriented languages are a class of languages more recently used for AI programming. Important features of object-oriented languages include: concepts of objects and messages, objects bundle data and methods for manipulating the data, sender specifies what is to be done receiver decides how to do it, inheritance (object hierarchy where objects inherit the attributes of the more general class of objects). Examples of object-oriented languages are Smalltalk, Objective C, C++. Object oriented extensions to LISP (CLOS - Common LISP Object System) and PROLOG (L&O - Logic & Objects) are also used.

- Artificial Intelligence is a new electronic machine that stores large amount of information and process it at very high speed
- The computer is interrogated by a human via a teletype It

passes if the human cannot tell if there is a computer or human at the other end

- The ability to solve problems
- It is the science and engineering of making intelligent machines, especially intelligent computer programs. It is related to the similar task of using computers to understand human intelligence

Importance of AI

- ***Game Playing***

You can buy machines that can play master level chess for a few hundred dollars. There is some AI in them, but they play well against people mainly through brute force computation--looking at hundreds of thousands of positions. To beat a world champion by brute force and known reliable heuristics requires being able to look at 200 million positions per second.

- ***Speech Recognition***

In the 1990s, computer speech recognition reached a practical level for limited purposes. Thus United Airlines has replaced its keyboard tree for flight information by a system using speech recognition of flight numbers and city names. It is quite convenient. On the other hand, while it is possible to instruct some computers using speech, most users have gone back to the keyboard and the mouse as still more convenient.

- ***Understanding Natural Language***

Just getting a sequence of words into a computer is not enough. Parsing sentences is not enough either. The computer has to be provided with an understanding of the domain the text is about, and this is presently possible only for very limited domains.

- ***Computer Vision***

The world is composed of three-dimensional objects, but the

inputs to the human eye and computers' TV cameras are two dimensional. Some useful programs can work solely in two dimensions, but full computer vision requires partial three-dimensional information that is not just a set of two-dimensional views. At present there are only limited ways of representing three-dimensional information directly, and they are not as good as what humans evidently use.

- ***Expert Systems***

A ``knowledge engineer'' interviews experts in a certain domain and tries to embody their knowledge in a computer program for carrying out some task. How well this works depends on whether the intellectual mechanisms required for the task are within the present state of AI. When this turned out not to be so, there were many disappointing results. One of the first expert systems was MYCIN in 1974, which diagnosed bacterial infections of the blood and suggested treatments. It did better than medical students or practicing doctors, provided its limitations were observed. Namely, its ontology included bacteria, symptoms, and treatments and did not include patients, doctors, hospitals, death, recovery, and events occurring in time. Its interactions depended on a single patient being considered. Since the experts consulted by the knowledge engineers knew about patients, doctors, death, recovery, etc., it is clear that the knowledge engineers forced what the experts told them into a predetermined framework. The usefulness of current expert systems depends on their users having common sense.

- ***Heuristic Classification***

One of the most feasible kinds of expert system given the present knowledge of AI is to put some information in one of a fixed set of categories using several sources of information. An example is advising whether to accept a proposed credit card purchase. Information is available about the owner of the credit card, his record of payment and also about the item he is buying and about the establishment from which he is buying it (e.g., about whether

there have been previous credit card frauds at this establishment).

- *The applications of AI are shown in Fig:*
  - ✓ Consumer Marketing
    - Have you ever used any kind of credit/ATM/store card while shopping?
    - if so, you have very likely been "input" to an AI algorithm
    - All of this information is recorded digitally
    - Companies like Nielsen gather this information weekly and search for patterns
      - general changes in consumer behavior
      - tracking responses to new products
      - identifying customer segments: targeted marketing, e.g., they find out that consumers with sports cars who buy textbooks respond well to offers of new credit cards.
    - Algorithms ("data mining") search data for patterns based on mathematical theories of learning
  - ✓ Identification Technologies
    - ID cards e.g., ATM cards
    - can be a nuisance and security risk: cards can be lost, stolen, passwords forgotten, etc
    - Biometric Identification, walk up to a locked door
      - Camera
      - Fingerprint device
      - Microphone

- Computer uses biometric signature for identification
- Face, eyes, fingerprints, voice pattern
- This works by comparing data from person at door with stored library
- Learning algorithms can learn the matching process by analyzing a large library database off-line, can improve its performance.

✓ Intrusion Detection
- Computer security - we each have specific patterns of computer use times of day, lengths of sessions, command used, sequence of commands, etc
  - would like to learn the "signature" of each authorized user
  - can identify non-authorized users
- How can the program automatically identify users?
  - record user's commands and time intervals
  - characterize the patterns for each user
  - model the variability in these patterns
  - classify (online) any new user by similarity to stored patterns

✓ Machine Translation
- Language problems in international business
  - e.g., at a meeting of Japanese,

Korean, Vietnamese and Swedish investors, no common language
  - If you are shipping your software manuals to 127 countries, the solution is ; hire translators to translate
  - would be much cheaper if a machine could do this!
- How hard is automated translation
  - very difficult!
  - e.g., English to Russian
  - not only must the words be translated, but their meaning also!

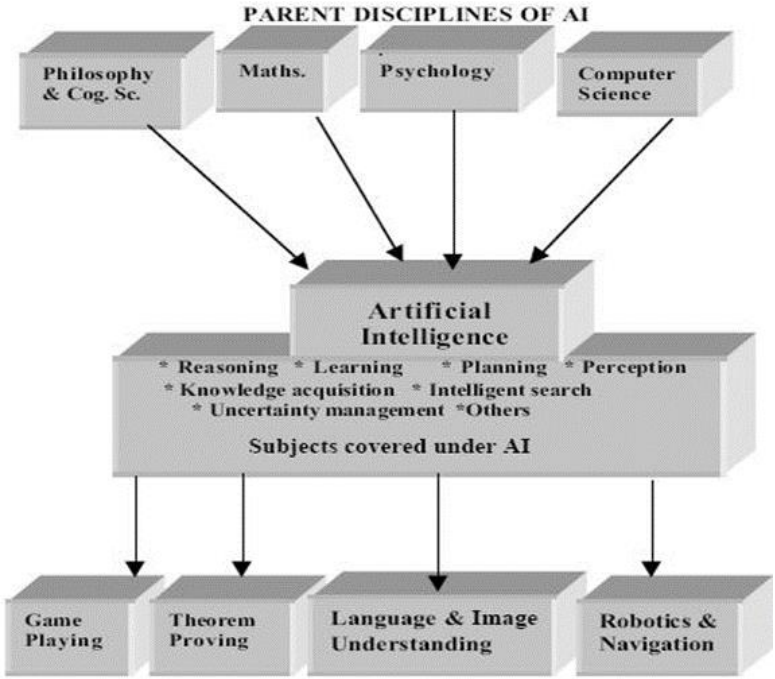


Figure 3.1 Application areas of AI

Early work in AI

- "Artificial Intelligence (AI) is the part of computer science concerned with designing intelligent computer systems, that is, systems that exhibit characteristics we associate with intelligence in human behaviour – understanding language, learning, reasoning, solving problems, and so on."
- Scientific Goal To determine which ideas about knowledge representation, learning, rule systems, search, and so on, explain various sorts of real intelligence.
- Engineering Goal To solve real world problems using AI techniques such as knowledge representation, learning, rule systems, search, and so on.
- Traditionally, computer scientists and engineers have been more interested in the engineering goal, while psychologists, philosophers and cognitive scientists have been more interested in the scientific goal.
- The Roots - Artificial Intelligence has identifiable roots in a number of older disciplines, particularly:
  - ✓ Philosophy
  - ✓ Logic/Mathematics
  - ✓ Computation
  - ✓ Psychology/Cognitive Science
  - ✓ Biology/Neuroscience
  - ✓ Evolution
- There is inevitably much overlap, e.g. between philosophy and logic, or between mathematics and computation. By looking at each of these in turn, we can gain a better understanding of their role in AI, and how these underlying disciplines have developed to play that role.

- **Philosophy**

  - ✓ ~400 BC Socrates asks for an algorithm to distinguish piety from non-piety.
  - ✓ ~350 BC Aristotle formulated different styles of

deductive reasoning, which could mechanically generate conclusions from initial premises, e.g. Modus Ponens If A ? B and A then B If A implies B and A is true then B is true when it's raining you get wet and it's raining then you get wet

- ✓ 1596 – 1650 Rene Descartes idea of mind-body dualism – part of the mind is exempt from physical laws.
- ✓ 1646 – 1716 Wilhelm Leibnitz was one of the first to take the materialist position which holds that the mind operates by ordinary physical processes – this has the implication that mental processes can potentially be carried out by machines.

□ **Logic/Mathematics**

- ✓ Earl Stanhope's Logic Demonstrator was a machine that was able to solve syllogisms, numerical problems in a logical form, and elementary questions of probability.
- ✓ 1815 – 1864 George Boole introduced his formal language for making logical inference in 1847 – Boolean algebra.
- ✓ 1848 – 1925 Gottlob Frege produced a logic that is essentially the first-order logic that today forms the most basic knowledge representation system.
- ✓ 1906 – 1978 Kurt Gödel showed in 1931 that there are limits to what logic can do. His Incompleteness Theorem showed that in any formal logic powerful enough to describe the properties of natural numbers, there are true statements whose truth cannot be established by any algorithm.
- ✓ 1995 Roger Penrose tries to prove the human mind has non-computable capabilities.

□ **Computation**

- ✓ 1869 William Jevon's Logic Machine could handle

Boolean Algebra and Venn Diagrams, and was able to solve logical problems faster than human beings.

- ✓ 1912 – 1954 Alan Turing tried to characterise exactly which functions are capable of being computed. Unfortunately it is difficult to give the notion of computation a formal definition. However, the Church-Turing thesis, which states that a Turing machine is capable of computing any computable function, is generally accepted as providing a sufficient definition. Turing also showed that there were some functions which no Turing machine can compute (e.g. Halting Problem).
- ✓ 1903 – 1957 John von Neumann proposed the von Neuman architecture which allows a description of computation that is independent of the particular realisation of the computer.
- ✓ 1960s Two important concepts emerged: Intractability (when solution time grows at least exponentially) and Reduction (to 'easier' problems).

□ **Psychology / Cognitive Science**

- ✓ Modern Psychology / Cognitive Psychology / Cognitive Science is the science which studies how the mind operates, how we behave, and how our brains process information.
- ✓ Language is an important part of human intelligence. Much of the early work on knowledge representation was tied to language and informed by research into linguistics.
- ✓ It is natural for us to try to use our understanding of how human (and other animal) brains lead to intelligent behavior in our quest to build artificial intelligent systems. Conversely, it makes sense to explore the properties of artificial systems (computer models/simulations) to test our hypotheses concerning human systems.

- ✓ Many sub-fields of AI are simultaneously building models of how the human system operates, and artificial systems for solving real world problems, and are allowing useful ideas to transfer between them.

□ **Biology / Neuroscience**

- ✓ Our brains (which give rise to our intelligence) are made up of tens of billions of neurons, each connected to hundreds or thousands of other neurons.
- ✓ Each neuron is a simple processing device (e.g. just firing or not firing depending on the total amount of activity feeding into it). However, large networks of neurons are extremely powerful computational devices that can learn how best to operate.
- ✓ The field of Connectionism or Neural Networks attempts to build artificial systems based on simplified networks of simplified artificial neurons.
- ✓ The aim is to build powerful AI systems, as well as models of various human abilities.
- ✓ Neural networks work at a sub-symbolic level, whereas much of conscious human reasoning appears to operate at a symbolic level.
- ✓ Artificial neural networks perform well at many simple tasks, and provide good models of many human abilities. However, there are many tasks that they are not so good at, and other approaches seem more promising in those areas.

□ **Evolution**

- ✓ One advantage humans have over current machines/computers is that they have a long evolutionary history.
- ✓ Charles Darwin (1809 – 1882) is famous for his work on evolution by natural selection. The idea is that

fitter individuals will naturally tend to live longer and produce more children, and hence after many generations a population will automatically emerge with good innate properties.

- ✓ This has resulted in brains that have much structure, or even knowledge, built in at birth.
- ✓ This gives them at the advantage over simple artificial neural network systems that have to learn everything.
- ✓ Computers are finally becoming powerful enough that we can simulate evolution and evolve good AI systems.
- ✓ We can now even evolve systems (e.g. neural networks) so that they are good at learning.
- ✓ A related field called genetic programming has had some success in evolving programs, rather than programming them by hand.

□ **Sub-fields of Artificial Intelligence**

- ✓ Neural Networks – e.g. brain modelling, time series prediction, classification
- ✓ Evolutionary Computation – e.g. genetic algorithms, genetic programming
- ✓ Vision – e.g. object recognition, image understanding
- ✓ Robotics – e.g. intelligent control, autonomous exploration
- ✓ Expert Systems – e.g. decision support systems, teaching systems
- ✓ Speech Processing– e.g. speech recognition and production
- ✓ Natural Language Processing – e.g. machine translation
- ✓ Planning – e.g. scheduling, game playing
- ✓ Machine Learning – e.g. decision tree learning, version space learning

□ **Speech Processing**

- ✓ As well as trying to understand human systems, there are also numerous real world applications: speech recognition for dictation systems and voice activated control; speech production for automated announcements and computerinterfaces.
- ✓ How do we get from sound waves to text streams and vice-versa?



Cen tre fo r Spee ch and Lan gua ge

□ **Natural Language Processing**

- ✓ For example, machine understanding and translation of simple sentences:

□ **Planning**

- ✓ Planning refers to the process of choosing/computing the correct sequence of steps to solve a given problem.
- ✓ To do this we need some convenient representation of the problem domain. We can define states in some formal language, such as a subset of predicate logic, or a series of rules.
- ✓ A plan can then be seen as a sequence of operations that transform the initial state into the goal state, i.e. the problem solution. Typically we will use some kind of search algorithm to find a good plan.

□ **Common Techniques**

- ✓ Even apparently radically different AI systems (such as rule based expert systems and neural networks) have many common techniques.

- ✓ Four important ones are:
    - Knowledge Representation: Knowledge needs to be represented somehow – perhaps as a series of if-then rules, as a frame based system, as a semantic network, or in the connection weights of an artificial neural network.
    - Learning: Automatically building up knowledge from the environment – such as acquiring the rules for a rule based expert system, or determining the appropriate connection weights in an artificial neural network.
    - Rule Systems: These could be explicitly built into an expert system by a knowledge engineer, or implicit in the connection weights learnt by a neural network.
    - Search: This can take many forms – perhaps searching for a sequence of states that leads quickly to a problem solution, or searching for a good set of connection weights for a neural network by minimizing a fitness function.

AI and related fields

- □ **Logical AI**

What a program knows about the world in general the facts of the specific situation in which it must act, and its goals are all represented by sentences of some mathematical logical language. The program decides what to do by inferring that certain actions are appropriate for achieving its goals.

- □ **Search**

AI programs often examine large numbers of possibilities, e.g. moves in a chess game or inferences by a theorem proving program. Discoveries are continually made about how to do this more efficiently in various domains.

- **Pattern Recognition**

When a program makes observations of some kind, it is often programmed to compare what it sees with a pattern. For example, a vision program may try to match a pattern of eyes and a nose in a scene in order to find a face. More complex patterns, e.g. in a natural language text, in a chess position, or in the history of some event are also studied.

- **Representation**

Facts about the world have to be represented in some way. Usually languages of mathematical logic are used.

- **Inference**

From some facts, others can be inferred. Mathematical logical deduction is adequate for some purposes, but new methods of *non-monotonic* inference have been added to logic since the 1970s. The simplest kind of non-monotonic reasoning is default reasoning in which a conclusion is to be inferred by default, but the conclusion can be withdrawn if there is evidence to the contrary. For example, when we hear of a bird, we can infer that it can fly, but this conclusion can be reversed when we hear that it is a penguin.

It is the possibility that a conclusion may have to be withdrawn that constitutes the non- monotonic character of the reasoning. Ordinary logical reasoning is monotonic in that the set of conclusions that can the drawn from a set of premises is a monotonic increasing function of the premises.

- **Common sense knowledge and reasoning**

This is the area in which AI is farthest from human-level, in spite of the fact that it has been an active research area since the 1950s. While there has been considerable progress, e.g. in developing systems of *non-monotonic reasoning* and theories of action, yet more new ideas are needed.

- **Learning from experience**

Programs do that. The approaches to AI based on *connectionism* and *neural nets* specialize in that. There is also learning of laws expressed in logic. Programs can only learn what facts or behaviors their formalisms can represent, and unfortunately learning systems are almost all based on very limited abilities to represent information.

- **Planning**

Planning programs start with general facts about the world (especially facts about the effects of actions), facts about the particular situation and a statement of a goal. From these, they generate a strategy for achieving the goal. In the most common cases, the strategy is just a sequence of actions.

- **Epistemology**

This is a study of the kinds of knowledge that are required for solving problems in the world.

- **Ontology**

Ontology is the study of the kinds of things that exist. In AI, the programs and sentences deal with various kinds of objects, and we study what these kinds are and what their basic properties are. Emphasis on ontology begins in the 1990s.

- **Heuristics**

A heuristic is a way of trying to discover something or an idea imbedded in a program. The term is used variously in AI. *Heuristic functions* are used in some approaches to search to measure how far a node in a search tree seems to be from a goal. *Heuristic predicates* that compare two nodes in a search tree to see if one is better than the other, i.e. constitutes an advance toward the goal, may be more useful.

- **Genetic Programming**

Genetic programming is a technique for getting programs to solve a task by mating random Lisp programs and selecting fittest in millions of generations.

3.2. Search and Control Strategies

Problem solving is an important aspect of Artificial Intelligence. A problem can be considered to consist of a goal and a set of actions that can be taken to lead to the goal. At any given time, we consider the state of the search space to represent where we have reached as a result of the actions we have applied so far. For example, consider the problem of looking for a contact lens on a football field.

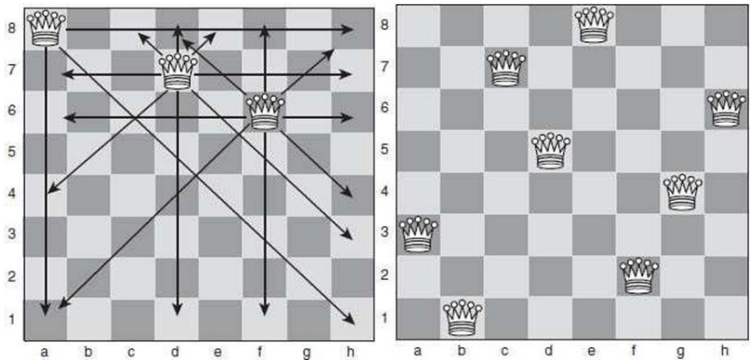
The initial state is how we start out, which is to say we know that the lens is somewhere on the field, but we don't know where. If we use the representation where we examine the field in units of one square foot, then our first action might be to examine the square in the top-left corner of the field. If we do not find the lens there, we could consider the state now to be that we have examined the top-left square and have not found the lens. After a number of actions, the state might be that we have examined 500 squares, and we have now just found the lens in the last square we examined. This is a goal state because it satisfies the goal that we had of finding a contact lens.

Search is a method that can be used by computers to examine a problem space like this in order to find a goal. Often, we want to find the goal as quickly as possible or without using too many resources. A problem space can also be considered to be a search space because in order to solve the problem, we will search the space for a goal state. We will continue to use the term search space to describe this concept. In this chapter, we will look at a number of methods for examining a search space. These methods are called search methods.

□ **The Importance of Search in AI**

- It has already become clear that many of the tasks underlying AI can be phrased in terms of a search for the solution to the problem at hand.
- Many goal based agents are essentially problem solving agents which must decide what to do by searching for a sequence of actions that lead to their solutions.
- For production systems, we have seen the need to search for a sequence of rule applications that lead to the required fact or action.
- For neural network systems, we need to search for the set of connection weights that will result in the required input to output mapping.

□ **Which search algorithm one should use will generally depend on the problem domain?**



There are four important factors to consider:

- Completeness – Is a solution guaranteed to be found if at least one solution exists?
- Optimality – Is the solution found guaranteed to be the best (or lowest cost) solution if there exists more than one solution?
- Time Complexity – The upper bound on the time

required to find a solution, as a function of the complexity of the problem.

- Space Complexity – The upper bound on the storage space (memory) required at any point during the search, as a function of the complexity of the problem.

3.3. Machine Learning

Machine learning is an application of artificial intelligence (AI) that provides systems the ability to automatically learn and improve from experience without being explicitly programmed. **Machine learning focuses on the development of computer programs** that can access data and use it to learn for themselves.

The process of learning begins with observations or data, such as examples, direct experience, or instruction, in order to look for patterns in data and make better decisions in the future based on the examples that we provide. **The primary aim is to allow the computers learn automatically** without human intervention or assistance and adjust actions accordingly.

But, using the classic algorithms of machine learning, text is considered as a sequence of keywords; instead, **an approach based on semantic analysis mimics the human ability to understand the meaning of a text.**

How machine learning works

UC Berkeley (link resides outside IBM) breaks out the learning system of a machine learning algorithm into three main parts.

1. **A Decision Process**: In general, machine learning algorithms are used to make a prediction or classification. Based on some input data, which can be labelled or unlabeled, your algorithm will produce an estimate about a pattern in the data.
2. **An Error Function**: An error function serves to evaluate the prediction of the model. If there are known examples,

an error function can make a comparison to assess the accuracy of the model.

3. **An Model Optimization Process**: If the model can fit better to the data points in the training set, then weights are adjusted to reduce the discrepancy between the known example and the model estimate. The algorithm will repeat this evaluate and optimize process, updating weights autonomously until a threshold of accuracy has been met.

Some Machine Learning Methods

Machine learning algorithms are often categorized as supervised or unsupervised.

- Supervised machine learning algorithms can apply what has been learned in the past to new data using labeled examples to predict future events. Starting from the analysis of a known training dataset, the learning algorithm produces an inferred function to make predictions about the output values. The system is able to provide targets for any new input after sufficient training. The learning algorithm can also compare its output with the correct, intended output and find errors in order to modify the model accordingly.
- In contrast, **unsupervised machine learning algorithms** are used when the information used to train is neither classified nor labeled. Unsupervised learning studies how systems can infer a function to describe a hidden structure from unlabeled data. The system doesn't figure out the right output, but it explores the data and can draw inferences from datasets to describe hidden structures from unlabeled data.
- **Semi-supervised machine learning algorithms** fall somewhere in between supervised and unsupervised learning, since they use both labeled and unlabeled data for training – typically a small amount of labeled data and a large amount of unlabeled data. The systems that use

this method are able to considerably improve learning accuracy. Usually, semi-supervised learning is chosen when the acquired labeled data requires skilled and relevant resources in order to train it / learn from it. Otherwise, acquiring unlabeled data generally doesn't require additional resources.

- **Reinforcement machine learning algorithms** is a learning method that interacts with its environment by producing actions and discovers errors or rewards. Trial and error search and delayed reward are the most relevant characteristics of reinforcement learning. This method allows machines and software agents to automatically determine the ideal behavior within a specific context in order to maximize its performance. Simple reward feedback is required for the agent to learn which action is best; this is known as the reinforcement signal.

Real-world machine learning use cases

Here are just a few examples of machine learning you might encounter every day:

Speech Recognition: It is also known as automatic speech recognition (ASR), computer speech recognition, or speech-to-text, and it is a capability which uses natural language processing (NLP) to process human speech into a written format. Many mobile devices incorporate speech recognition into their systems to conduct voice search—e.g. Siri—or provide more accessibility around texting.

Customer Service: Online chatbots are replacing human agents along the customer journey. They answer frequently asked questions (FAQs) around topics, like shipping, or provide personalized advice, cross-selling products or suggesting sizes for users, changing the way we think about customer engagement across websites and social media platforms. Examples include messaging bots on e-commerce sites with virtual agents,

messaging apps, such as Slack and Facebook Messenger, and tasks usually done by virtual assistants and voice assistants.

Computer Vision: This AI technology enables computers and systems to derive meaningful information from digital images, videos and other visual inputs, and based on those inputs, it can take action. This ability to provide recommendations distinguishes it from image recognition tasks. Powered by convolutional neural networks, computer vision has applications within photo tagging in social media, radiology imaging in healthcare, and self-driving cars within the automotive industry.

Recommendation Engines: Using past consumption behavior data, AI algorithms can help to discover data trends that can be used to develop more effective cross-selling strategies. This is used to make relevant add-on recommendations to customers during the checkout process for online retailers.

Automated stock trading: Designed to optimize stock portfolios, AI-driven high-frequency trading platforms make thousands or even millions of trades per day without human intervention.

Challenges of Machine Learning

As machine learning technology advances, it has certainly made our lives easier. However, implementing machine learning within businesses has also raised a number of ethical concerns surrounding AI technologies. Some of these include:

- **Technological Singularity**

While this topic garners a lot of public attention, many researchers are not concerned with the idea of AI surpassing human intelligence in the near or immediate future. This is also referred to as superintelligence, which Nick Bostrum defines as "any intellect that vastly outperforms the best human brains in practically every field, including scientific creativity, general wisdom, and social skills." Despite the fact that Strong AI and superintelligence is not imminent in society, the idea of it raises

some interesting questions as we consider the use of autonomous systems, like self-driving cars. It's unrealistic to think that a driverless car would never get into a car accident, but who is responsible and liable under those circumstances? Should we still pursue autonomous vehicles, or do we limit the integration of this technology to create only semi-autonomous vehicles which promote safety among drivers? The jury is still out on this, but these are the types of ethical debates that are occurring as new, innovative AI technology develops.

- **AI Impact on Jobs:**

While a lot of public perception around artificial intelligence centers around job loss, this concern should be probably reframed. With every disruptive, new technology, we see that the market demand for specific job roles shift. For example, when we look at the automotive industry, many manufacturers, like GM, are shifting to focus on electric vehicle production to align with green initiatives. The energy industry isn't going away, but the source of energy is shifting from a fuel economy to an electric one. Artificial intelligence should be viewed in a similar manner, where artificial intelligence will shift the demand of jobs to other areas. There will need to be individuals to help manage these systems as data grows and changes every day. There will still need to be resources to address more complex problems within the industries that are most likely to be affected by job demand shifts, like customer service. The important aspect of artificial intelligence and its effect on the job market will be helping individuals transition to these new areas of market demand.

- **Privacy:**

Privacy tends to be discussed in the context of data privacy, data protection and data security, and these concerns have allowed policymakers to make more strides here in recent years. For example, in 2016, GDPR legislation was created to protect the personal data of people in the European Union and European Economic Area, giving individuals more control of their data. In

the United States, individual states are developing policies, such as the California Consumer Privacy Act (CCPA), which require businesses to inform consumers about the collection of their data. This recent legislation has forced companies to rethink how they store and use personally identifiable data (PII). As a result, investments within security have become an increasing priority for businesses as they seek to eliminate any vulnerabilities and opportunities for surveillance, hacking, and cyberattacks.

- **Bias and Discrimination:**

Instances of bias and discrimination across a number of intelligent systems have raised many ethical questions regarding the use of artificial intelligence. How can we safeguard against bias and discrimination when the training data itself can lend itself to bias? While companies typically have well-meaning intentions around their automation efforts, Reuters (link resides outside IBM) highlights some of the unforeseen consequences of incorporating AI into hiring practices. In their effort to automate and simplify a process, Amazon unintentionally biased potential job candidates by gender for open technical roles, and they ultimately had to scrap the project. As events like these surface, Harvard Business Review (link resides outside IBM) has raised other pointed questions around the use of AI within hiring practices, such as what data should you be able to use when evaluating a candidate for a role.

Bias and discrimination aren't limited to the human resources function either; it can be found in a number of applications from facial recognition software to social media algorithms.

As businesses become more aware of the risks with AI, they've also become more active this discussion around AI ethics and values. For example, last year IBM's CEO Arvind Krishna shared that IBM has sunset its general purpose IBM facial recognition and analysis products, emphasizing that "IBM firmly opposes and will not condone uses of any technology, including facial recognition technology offered by other vendors, for mass

surveillance, racial profiling, violations of basic human rights and freedoms, or any purpose which is not consistent with our values and Principles of Trust and Transparency."

To read more about this, check out IBM's policy blog, relaying its point of view on "A Precision Regulation Approach to Controlling Facial Recognition Technology Exports."

- **Accountability**

Since there isn't significant legislation to regulate AI practices, there is no real enforcement mechanism to ensure that ethical AI is practiced. The current incentives for companies to adhere to these guidelines are the negative repercussions of an unethical AI system to the bottom line. To fill the gap, ethical frameworks have emerged as part of a collaboration between ethicists and researchers to govern the construction and distribution of AI models within society. However, at the moment, these only serve to guide, and research (link resides outside IBM) (PDF, 984 KB) shows that the combination of distributed responsibility and lack of foresight into potential consequences isn't necessarily conducive to preventing harm to society.

Machine learning and deep learning are subfields of AI

As a whole, artificial intelligence contains many subfields, including:

- **Machine learning** automates analytical model building. It uses methods from neural networks, statistics, operations research and physics to find hidden insights in data without being explicitly programmed where to look or what to conclude.
- **A neural network** is a kind of machine learning inspired by the workings of the human brain. It's a computing system made up of interconnected units (like neurons) that processes information by responding to external inputs, relaying information between each unit. The

process requires multiple passes at the data to find connections and derive meaning from undefined data.

- **Deep learning** uses huge neural networks with many layers of processing units, taking advantage of advances in computing power and improved training techniques to learn complex patterns in large amounts of data. Common applications include image and speech recognition.
- **Computer vision** relies on pattern recognition and deep learning to recognize what's in a picture or video. When machines can process, analyze and understand images, they can capture images or videos in real time and interpret their surroundings.
- **Natural language processing** is the ability of computers to analyze, understand and generate human language, including speech. The next stage of NLP is natural language interaction, which allows humans to communicate with computers using normal, everyday language to perform tasks.

3.4. Neural Network

What are Artificial Neural Networks (ANNs)?

The inventor of the first neurocomputer, Dr. Robert Hecht-Nielsen, defines a neural network as −

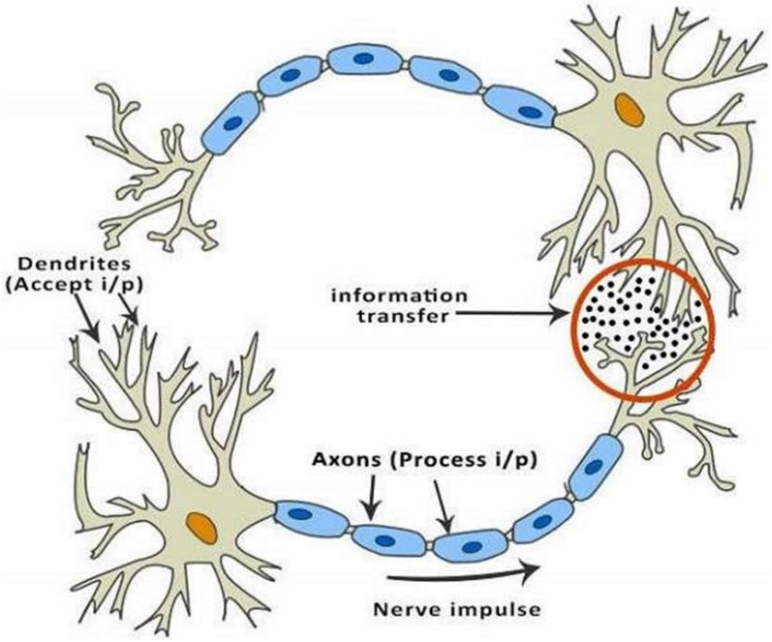
"...a computing system made up of a number of simple, highly interconnected processing elements, which process information by their dynamic state response to external inputs."

Basic Structure of ANNs

The idea of ANNs is based on the belief that working of human brain by making the right connections, can be imitated using silicon and wires as living neurons and dendrites.

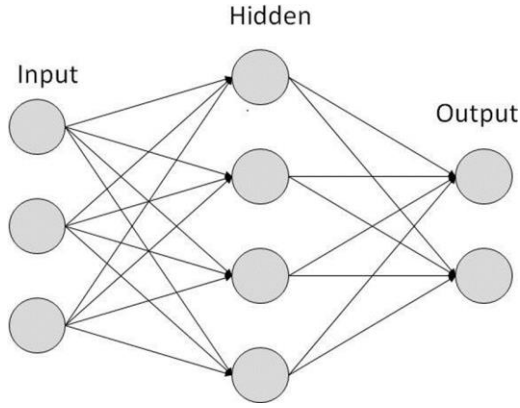
The human brain is composed of 86 billion nerve cells called

neurons. They are connected to other thousand cells by Axons. Stimuli from external environment or inputs from sensory organs are accepted by dendrites. These inputs create electric impulses, which quickly travel through the neural network. A neuron can then send the message to other neuron to handle the issue or does not send it forward.



ANNs are composed of multiple nodes, which imitate biological neurons of human brain. The neurons are connected by links and they interact with each other. The nodes can take input data and perform simple operations on the data. The result of these operations is passed to other neurons. The output at each node is called its activation or node value.

Each link is associated with weight. ANNs are capable of learning, which takes place by altering weight values. The following illustration shows a simple ANN –

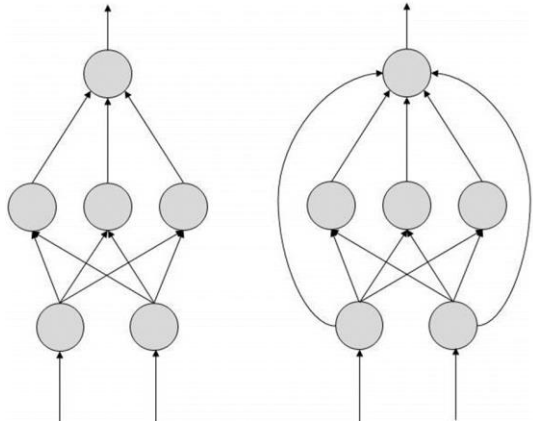


Types of Artificial Neural Networks

There are two Artificial Neural Network topologies − **FeedForward** and **Feedback.**

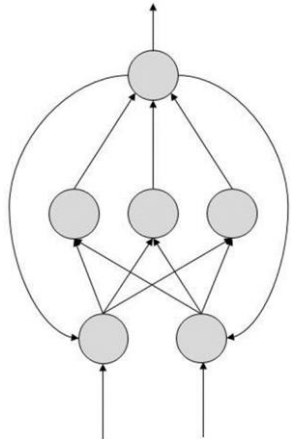
FeedForward ANN

In this ANN, the information flow is unidirectional. A unit sends information to other unit from which it does not receive any information. There are no feedback loops. They are used in pattern generation/recognition/classification. They have fixed inputs and outputs.



FeedBack ANN

Here, feedback loops are allowed. They are used in content addressable memories.



Working of ANNs

In the topology diagrams shown, each arrow represents a connection between two neurons and indicates the pathway for the flow of information. Each connection has a weight, an integer number that controls the signal between the two neurons.

If the network generates a "good or desired" output, there is no need to adjust the weights. However, if the network generates a "poor or undesired" output or an error, then the system alters the weights in order to improve subsequent results.

Machine Learning in ANNs

ANNs are capable of learning and they need to be trained. There are several learning strategies −

- **Supervised Learning** − It involves a teacher that is scholar than the ANN itself. For example, the teacher feeds some example data about which the teacher already knows the answers.

For example, pattern recognizing. The ANN comes up with guesses while recognizing. Then the teacher provides the ANN with the answers. The network then compares it guesses with the teacher's "correct" answers and makes adjustments according to errors.

- **Unsupervised Learning** − It is required when there is no example data set with known answers. For example, searching for a hidden pattern. In this case, clustering i.e. dividing a set of elements into groups according to some unknown pattern is carried out based on the existing data sets present.
- **Reinforcement Learning** − This strategy built on observation. The ANN makes a decision by observing its environment. If the observation is negative, the network adjusts its weights to be able to make a different required decision the next time.

Back Propagation Algorithm

It is the training or learning algorithm. It learns by example. If you submit to the algorithm the example of what you want the network to do, it changes the network's weights so that it can produce desired output for a particular input on finishing the training. Back Propagation networks are ideal for simple Pattern Recognition and Mapping Tasks.

Bayesian Networks (BN)

These are the graphical structures used to represent the probabilistic relationship among a set of random variables. Bayesian networks are also called **Belief Networks** or **Bayes Nets.** BNs reason about uncertain domain.

In these networks, each node represents a random variable with specific propositions. For example, in a medical diagnosis domain, the node Cancer represents the proposition that a patient has cancer.

The edges connecting the nodes represent probabilistic dependencies among those random variables. If out of two nodes, one is affecting the other then they must be directly connected in the directions of the effect. The strength of the relationship between variables is quantified by the probability associated with each node.

There is an only constraint on the arcs in a BN that you cannot return to a node simply by following directed arcs. Hence the BNs are called Directed Acyclic Graphs (DAGs).

BNs are capable of handling multivalued variables simultaneously. The BN variables are composed of two dimensions −

- Range of prepositions
- Probability assigned to each of the prepositions.

Consider a finite set X = {X1, X2, …,Xn} of discrete random variables, where each variable *Xi* may take values from a finite set, denoted by *Val(Xi).* If there is a directed link from variable *Xi* to variable, *Xj,* then variable *Xi* will be a parent of variable *Xj* showing direct dependencies between the variables.

The structure of BN is ideal for combining prior knowledge and observed data. BN can be used to learn the causal relationships and understand various problem domains and to predict future events, even in case of missing data.

Building a Bayesian Network

A knowledge engineer can build a Bayesian network. There are a number of steps the knowledge engineer needs to take while building it.

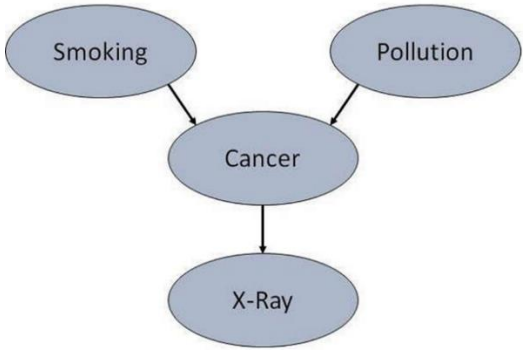
Example problem − *Lung cancer.* A patient has been suffering from breathlessness. He visits the doctor, suspecting he has lung cancer. The doctor knows that barring lung cancer, there are various other possible diseases the patient might have such as tuberculosis and bronchitis.

Create Arcs between Nodes

Topology of the network should capture qualitative relationships between variables.

For example, what causes a patient to have lung cancer? - Pollution and smoking. Then add arcs from node *Pollution* and node *Smoker* to node *Lung-Cancer.*

Similarly if patient has lung cancer, then X-ray result will be positive. Then add arcs from node *Lung-Cancer* to node *X-Ray.*



Applications of Neural Networks

They can perform tasks that are easy for a human but difficult for a machine −

- **Aerospace** − Autopilot aircrafts, aircraft fault detection.
- **Automotive** − Automobile guidance systems.
- **Military** − Weapon orientation and steering, target tracking, object discrimination, facial recognition, signal/image identification.
- **Electronics** − Code sequence prediction, IC chip layout, chip failure analysis, machine vision, voice synthesis.
- **Financial** − Real estate appraisal, loan advisor, mortgage screening, corporate bond rating, portfolio trading program, corporate financial analysis, currency value prediction, document readers, credit application evaluators.

- **Industrial** − Manufacturing process control, product design and analysis, quality inspection systems, welding quality analysis, paper quality prediction, chemical product design analysis, dynamic modeling of chemical process systems, machine maintenance analysis, project bidding, planning, and management.
- **Medical** − Cancer cell analysis, EEG and ECG analysis, prosthetic design, transplant time optimizer.
- **Speech** − Speech recognition, speech classification, text to speech conversion.
- **Telecommunications** − Image and data compression, automated information services, real-time spoken language translation.
- **Transportation** − Truck Brake system diagnosis, vehicle scheduling, routing systems.
- **Software** − Pattern Recognition in facial recognition, optical character recognition, etc.
- **Time Series Prediction** − ANNs are used to make predictions on stocks and natural calamities.
- **Signal Processing** − Neural networks can be trained to process an audio signal and filter it appropriately in the hearing aids.
- **Control** − ANNs are often used to make steering decisions of physical vehicles.
- **Anomaly Detection** − As ANNs are expert at recognizing patterns, they can also be trained to generate an output when something unusual occurs that misfits the pattern.

3.5. Robotics

What is Robotics?

Robotics is a branch of AI, which is composed of Electrical Engineering, Mechanical Engineering, and Computer Science for designing, construction, and application of robots.

"Artificial intelligence will reach human levels by around 2029. Follow that out further to, say, 2045, we will have multiplied the

intelligence, the human biological machine intelligence of our civilization a billion-fold."

-Ray Kurzweil, American inventor

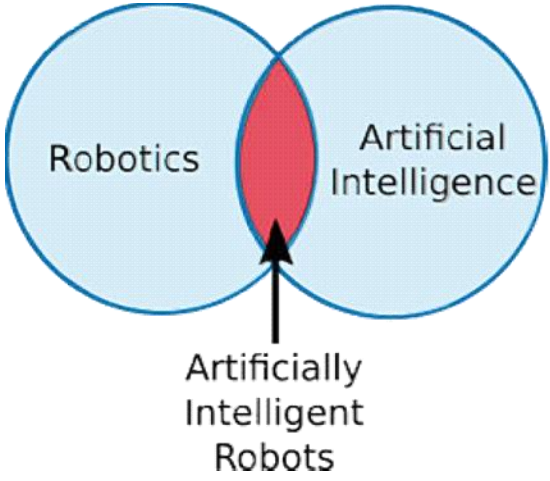
Aspects of Robotics

- The robots have **mechanical construction**, form, or shape designed to accomplish a particular task.
- They have **electrical components** which power and control the machinery.
- They contain some level of **computer program** that determines what, when and how a robot does something.

Are robotics and artificial intelligence the same thing?

The first thing to clarify is that robotics and artificial intelligence are not the same things at all. In fact, the two fields are almost entirely separate.

A Venn diagram of the two fields would look like this:



As you can see, there is one area small where the two fields overlap:

Artificially Intelligent Robots. It is within this overlap that people sometimes confuse the two concepts.

To understand how these three terms relate to each other, let's look at each of them individually.

Robot Locomotion

Locomotion is the mechanism that makes a robot capable of moving in its environment. There are various types of locomotions −

- Legged
- Wheeled
- Combination of Legged and Wheeled Locomotion
- Tracked slip/skid

Legged Locomotion

- This type of locomotion consumes more power while demonstrating walk, jump, trot, hop, climb up or down, etc.
- It requires more number of motors to accomplish a movement. It is suited for rough as well as smooth terrain where irregular or too smooth surface makes it consume more power for a wheeled locomotion. It is little difficult to implement because of stability issues.
- It comes with the variety of one, two, four, and six legs. If a robot has multiple legs then leg coordination is necessary for locomotion.

The total number of possible g its (a periodic sequence of lift and release vents for each of the total legs) a robot can travel depends upon the number of its legs.

If a robot has k legs, then the number of possible events

$N = (2k-1)!$.

In case of a two-legged robot (k=2), the number of possible events is N = (k-1)! = (2*2-1)! = 3!= 6.

Hence there are six possible different events −

- Lifting the Left leg
- Releasing the Left leg
- Lifting the Right leg
- Releasing the Right leg
- Lifting both the legs together
- Releasing both the legs together

In case of k=6 legs, there are 39916800 possible events. Hence the complexity of robots is directly proportional to the number of legs.



Wheeled Locomotion

It requires fewer number of motors to accomplish a movement. It is little easy to implement as there are less stability issues in case of more number of wheels. It is power efficient as compared to legged locomotion.

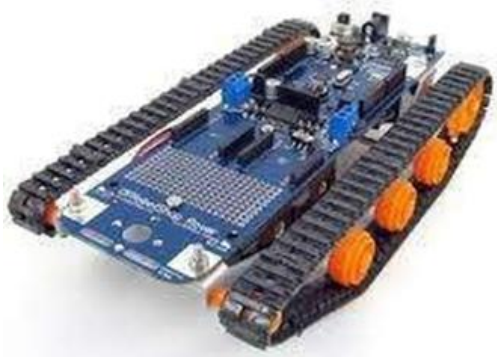
- **Standard wheel** − Rotates around the wheel axle and around the contact

- **Castor wheel** − Rotates around the wheel axle and the offset steering joint.
- **Swedish 45o and Swedish 90o wheels** − Omni-wheel, rotates around the contact point, around the wheel axle, and around the rollers.
- **Ball or spherical wheel** − Omnidirectional wheel, technically difficult to implement.

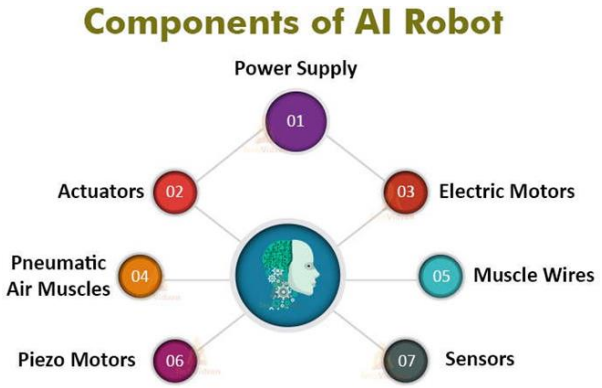


Slip/Skid Locomotion

In this type, the vehicles use tracks as in a tank. The robot is steered by moving the tracks with different speeds in the same or opposite direction. It offers stability because of large contact area of track and ground.



Components of AI Robot



To build an AI robot we need the following parts−

1. **Power Supply**

Every electronic device draws power from an electric source. AI robots are no different from their power sources being batteries, solar power, hydraulic. The robots need power to supply the **voltage** signals that make the motors turn, the sensors operate and therefore the robot brain.

2. **Actuators**

Actuators are the components responsible for converting the electric energy of the source to They became an important a part of the automated system, as they assist with controlling equipment using **hydraulic**, **pneumatic** or in a number of the cases **electronic** signals.

3. **Electric motors (AC/DC)**

Motors have the application for the rotation movement required by the robot.

The smaller motors and drives are **integrated** into robotics systems, helping to bring new levels of efficiency and productivity to more compact robotic solutions

4. **Pneumatic Air Muscles**

These are arm devices that extend or contract and are operated with the help of **pressurized** air. These are very lightweight because their main element may be a thin membrane.

This enables them to be directly connected to the structure they power, which is a plus when considering the replacement of a defective muscle.

5. **Muscle Wires**

Muscle wires are a temperature-dependent conductor of electricity that turns harder when an electric current is passed through it. It has a unique ability to **contract** on demand.

6. **Piezo Motors**

These types of motors undergo rotational movement on the application of an electric field.

Piezoelectric motors use the converse piezoelectricity of piezoelectric sensors, during which deformation or vibration of the piezoelectric material produces an electrical charge.

A circuit makes acoustic or ultrasonic vibrations within the piezoelectric material, which produce **linear** or **rotation**.

7. **Sensors**

Sensors are devices to detect various signals in the surroundings. A sensor makes "sense" of property — it turns something about the physical world into data upon which a system can **act**.

Difference in Robot Systems and AI Programs

The difference between Robotics and AI is as follows

- **AI Programs**
    - AI programs are in the form of software.

- The method of input is in the form of code.
- They can be accessed upon desktops and laptops.

- **Robots**
  - Robots are a combination of both hardware and software.
  - The method of inputs is in the form of analog signals.
  - Specific to the work, specialized robots have to be assigned.

Types of Robots

Mechanical bots come in all shapes and sizes to efficiently carry out the task for which they are designed. All robots vary in design, functionality and degree of autonomy. From the 0.2 millimeter-long "RoboBee" to the 200 meter-long robotic shipping vessel "Vindskip," robots are emerging to carry out tasks that humans simply can't.

Generally, there are five types of robots:

1) Pre-Programmed Robots

Pre-programmed robots operate in a controlled environment where they do simple, monotonous tasks. An example of a pre-programmed robot would be a mechanical arm on an automotive assembly line. The arm serves one function — to weld a door on, to insert a certain part into the engine, etc. — and its job is to perform that task longer, faster and more efficiently than a human.

2) Humanoid Robots

Humanoid robots are robots that look like and/or mimic human behavior. These robots usually perform human-like activities (like running, jumping and carrying objects), and are sometimes designed to look like us, even having human faces and expressions. Two of the most prominent examples of humanoid

robots are Hanson Robotics' Sophia (in the video above) and Boston Dynamics' Atlas.

3) Autonomous Robots

Autonomous robots operate independently of human operators. These robots are usually designed to carry out tasks in open environments that do not require human supervision. They are quite unique because they use sensors to perceive the world around them, and then employ decision- making structures (usually a computer) to take the optimal next step based on their data and mission. An example of an autonomous robot would be the Roomba vacuum cleaner, which uses sensors to roam freely throughout a home.

Examples of Autonomous Robots

- Cleaning Bots (for example, Roomba)
- Lawn Trimming Bots
- Hospitality Bots
- Autonomous Drones
- Medical Assistant Bots

4) Teleoperated Robots

Teleoperated robots are semi-autonomous bots that use a wireless network to enable human control from a safe distance. These robots usually work in extreme geographical conditions, weather, circumstances, etc. Examples of teleoperated robots are the human-controlled submarines used to fix underwater pipe leaks during the BP oil spill or drones used to detect landmines on a battlefield.

5) Augmenting Robots

Augmenting robots either enhance current human capabilities or replace the capabilities a human may have lost. The field of robotics for human augmentation is a field where science fiction could become reality very soon, with bots that have the ability to

redefine the definition of humanity by making humans faster and stronger. Some examples of current augmenting robots are robotic prosthetic limbs or exoskeletons used to lift hefty weights.

Applications of Robotics

- Helping fight forest fires
- Working alongside humans in manufacturing plants (known as co-bots)
- Robots that offer companionship to elderly individuals
- Surgical assistants
- Last-mile package and food order delivery
- Autonomous household robots that carry out tasks like vacuuming and mowing the grass
- Assisting with finding items and carrying them throughout warehouses Used during search-and-rescue missions after natural disasters
- Landmine detectors in war zones

Expert System

What are Expert Systems?

The expert systems are the computer applications developed to solve complex problems in a particular domain, at the level of extra-ordinary human intelligence and expertise.

Characteristics of Expert Systems

- High performance
- Understandable
- Reliable
- Highly responsive

Capabilities of Expert Systems

The expert systems are capable of −

- Advising

- Instructing and assisting human in decision making
- Demonstrating
- Deriving a solution
- Diagnosing
- Explaining
- Interpreting input
- Predicting results
- Justifying the conclusion
- Suggesting alternative options to a problem

They are incapable of −

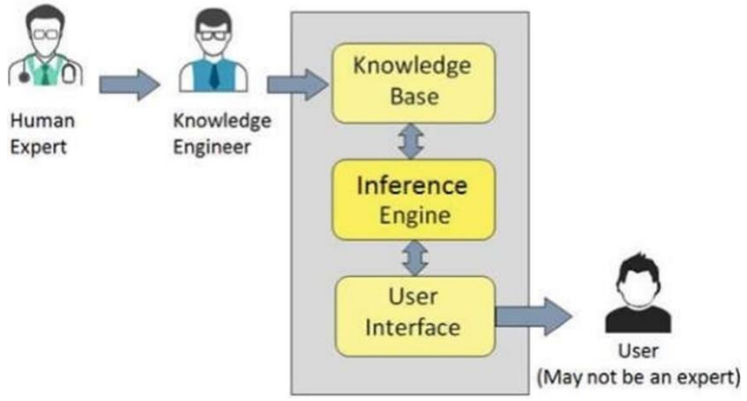
- Substituting human decision makers
- Possessing human capabilities
- Producing accurate output for inadequate knowledge base
- Refining their own knowledge

Components of Expert Systems

The components of ES include −

- Knowledge Base
- Inference Engine
- User Interface

Let us see them one by one briefly −



Knowledge Base

It contains domain-specific and high-quality knowledge.

Knowledge is required to exhibit intelligence. The success of any ES majorly depends upon the collection of highly accurate and precise knowledge.

What is Knowledge?

The data is collection of facts. The information is organized as data and facts about the task domain. **Data, information,** and **past experience** combined together are termed as knowledge.

Components of Knowledge Base

The knowledge base of an ES is a store of both, factual and heuristic knowledge.

- **Factual Knowledge** − It is the information widely accepted by the Knowledge Engineers and scholars in the task domain.
- **Heuristic Knowledge** − It is about practice, accurate judgement, one's ability of evaluation, and guessing.

Knowledge representation

It is the method used to organize and formalize the knowledge in the knowledge base. It is in the form of IF-THEN-ELSE rules.

Knowledge Acquisition

The success of any expert system majorly depends on the quality, completeness, and accuracy of the information stored in the knowledge base.

The knowledge base is formed by readings from various experts, scholars, and the **Knowledge Engineers**. The knowledge engineer is a person with the qualities of empathy, quick learning, and case analyzing skills.

He acquires information from subject expert by recording, interviewing, and observing him at work, etc. He then categorizes and organizes the information in a meaningful way, in the form of IF-THEN-ELSE rules, to be used by interference machine. The knowledge engineer also monitors the development of the ES.

Inference Engine

Use of efficient procedures and rules by the Inference Engine is essential in deducting a correct, flawless solution.

In case of knowledge-based ES, the Inference Engine acquires and manipulates the knowledge from the knowledge base to arrive at a particular solution.

In case of rule based ES, it −

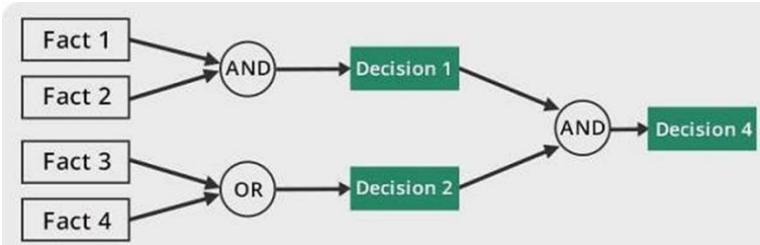
- Applies rules repeatedly to the facts, which are obtained from earlier rule application.
- Adds new knowledge into the knowledge base if required.
- Resolves rules conflict when multiple rules are applicable to a particular case. To recommend a solution, the Inference Engine uses the following strategies −
- Forward Chaining
- Backward Chaining

Forward Chaining

It is a strategy of an expert system to answer the question, **"What can happen next?"**

Here, the Inference Engine follows the chain of conditions and derivations and finally deduces the outcome. It considers all the facts and rules, and sorts them before concluding to a solution.

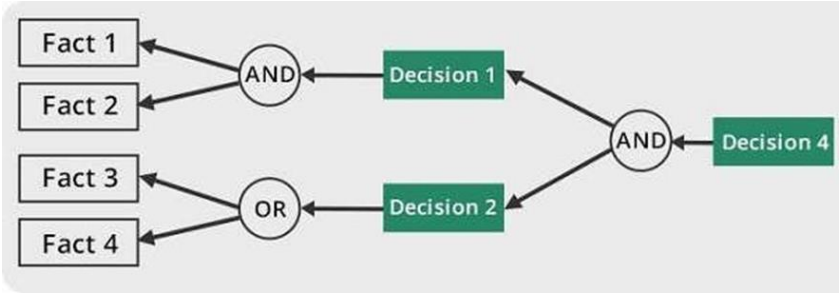
This strategy is followed for working on conclusion, result, or effect. For example, prediction of share market status as an effect of changes in interest rates.



Backward Chaining

With this strategy, an expert system finds out the answer to the question, **"Why this happened?"**

On the basis of what has already happened, the Inference Engine tries to find out which conditions could have happened in the past for this result. This strategy is followed for finding out cause or reason. For example, diagnosis of blood cancer in humans.



User Interface

User interface provides interaction between user of the ES and the ES itself. It is generally Natural Language Processing so as to be used by the user who is well-versed in the task domain. The user of the ES need not be necessarily an expert in Artificial Intelligence.

It explains how the ES has arrived at a particular recommendation. The explanation may appear in the following forms −

- Natural language displayed on screen.
- Verbal narrations in natural language.
- Listing of rule numbers displayed on the screen.

The user interface makes it easy to trace the credibility of the deductions.

Requirements of Efficient ES User Interface

- It should help users to accomplish their goals in shortest possible way.
- It should be designed to work for user's existing or desired work practices.
- Its technology should be adaptable to user's requirements; not the other way round.
- It should make efficient use of user input.

Expert Systems Limitations

No technology can offer easy and complete solution. Large systems are costly, require significant development time, and computer resources. ESs have their limitations which include −

- Limitations of the technology
- Difficult knowledge acquisition
- ES are difficult to maintain
- High development costs

Applications of Expert System

The following table shows where ES can be applied.

Application	Description
Design Domain	Camera lens design, automobile design.
Medical Domain	Diagnosis Systems to deduce cause of disease from observed data, conduction medical operations on humans.
Monitoring Systems	Comparing data continuously with observed system or with prescribed

	behavior such as leakage monitoring in long petroleum pipeline.
Process Control Systems	Controlling a physical process based on monitoring.
Knowledge Domain	Finding out faults in vehicles, computers.
Finance/Commerce	Detection of possible fraud, suspicious transactions, stock market trading, Airline scheduling, cargo scheduling.

Expert System Technology

There are several levels of ES technologies available. Expert systems technologies include −

- **Expert System Development Environment** − The ES development environment includes hardware and tools. They are −
    - Workstations, minicomputers, mainframes.
    - High level Symbolic Programming Languages such as **LIS**t **P**rogramming (LISP) and **PRO**grammation en **LOG**ique (PROLOG).
    - Large databases.
- **Tools** − They reduce the effort and cost involved in developing an expert system to large extent.
    - Powerful editors and debugging tools with multi-windows.
    - They provide rapid prototyping
    - Have Inbuilt definitions of model, knowledge representation, and inference design.

- **Shells** − A shell is nothing but an expert system without knowledge base. A shell provides the developers with knowledge acquisition, inference engine, user interface, and explanation facility. For example, few shells are given below −

    - Java Expert System Shell (JESS) that provides fully developed Java API for creating an expert system.

    - *Vidwan*, a shell developed at the National Centre for Software Technology, Mumbai in 1993. It enables knowledge encoding in the form of IF-THEN rules.

Development of Expert Systems: General Steps

The process of ES development is iterative. Steps in developing the ES include −

Identify Problem Domain

- The problem must be suitable for an expert system to solve it.
- Find the experts in task domain for the ES project.
- Establish cost-effectiveness of the system.

Design the System

- Identify the ES Technology
- Know and establish the degree of integration with the other systems and databases.
- Realize how the concepts can represent the domain knowledge best.

Develop the Prototype

From Knowledge Base: The knowledge engineer works to −

- Acquire domain knowledge from the expert.
- Represent it in the form of If-THEN-ELSE rules.

Test and Refine the Prototype

- The knowledge engineer uses sample cases to test the prototype for any deficiencies in performance.
- End users test the prototypes of the ES.

Develop and Complete the ES

- Test and ensure the interaction of the ES with all elements of its environment, including end users, databases, and other information systems.
- Document the ES project well.
- Train the user to use ES.

Maintain the System

- Keep the knowledge base up-to-date by regular review and update.
- Cater for new interfaces with other information systems, as those systems evolve.

Benefits of Expert Systems

- **Availability** − They are easily available due to mass production of software.
- **Less Production Cost** − Production cost is reasonable. This makes them affordable.
- **Speed** − They offer great speed. They reduce the amount of work an individual puts in.
- **Less Error Rate** − Error rate is low as compared to human errors.
- **Reducing Risk** − They can work in the environment dangerous to humans.
- **Steady response** − They work steadily without getting motional, tensed or fatigued.

Advantages :

- Low accessibility cost.

- Fast response.
- Not affected by emotions, unlike humans.
- Low error rate.
- Capable of explaining how they reached a solution.

Disadvantages :

- The expert system has no emotions.
- Common sense is the main issue of the expert system.
- It is developed for a specific domain.
- It needs to be updated manually. It does not learn itself.
- Not capable to explain the logic behind the decision.

Applications :

The application of an expert system can be found in almost all areas of business or government. They include areas such as –

- Different types of medical diagnosis like internal medicine, blood diseases and show on.
- Diagnosis of the complex electronic and electromechanical system.
- Diagnosis of a software development project.
- Planning experiment in biology, chemistry and molecular genetics.
- Forecasting crop damage.
- Diagnosis of the diesel-electric locomotive system.
- Identification of chemical compound structure.
- Scheduling of customer order, computer resources and various manufacturing task.
- Assessment of geologic structure from dip meter logs.
- Assessment of space structure through satellite and robot.
- The design of VLSI system.
- Teaching students specialize task.
- Assessment of log including civil case evaluation, product liability etc.

3.6. Fuzzy Logic

What is Fuzzy Logic?

Fuzzy Logic (FL) is a method of reasoning that resembles human reasoning. The approach of FL imitates the way of decision making in humans that involves all intermediate possibilities between digital values YES and NO.

The conventional logic block that a computer can understand takes precise input and produces a definite output as TRUE or FALSE, which is equivalent to human's YES or NO.

The inventor of fuzzy logic, Lotfi Zadeh, observed that unlike computers, the human decision making includes a range of possibilities between YES and NO, such as –

- CERTAINLY YES
- POSSIBLY YES
- CANNOT SAY
- POSSIBLY NO
- CERTAINLY NO

The fuzzy logic works on the levels of possibilities of input to achieve the definite output.

Implementation

- It can be implemented in systems with various sizes and capabilities ranging from small micro-controllers to large, networked, workstation-based control systems.
- It can be implemented in hardware, software, or a combination of both.

Why Fuzzy Logic?

Fuzzy logic is useful for commercial and practical purposes.

- It can control machines and consumer products.

- It may not give accurate reasoning, but acceptable reasoning.
- Fuzzy logic helps to deal with the uncertainty in engineering.

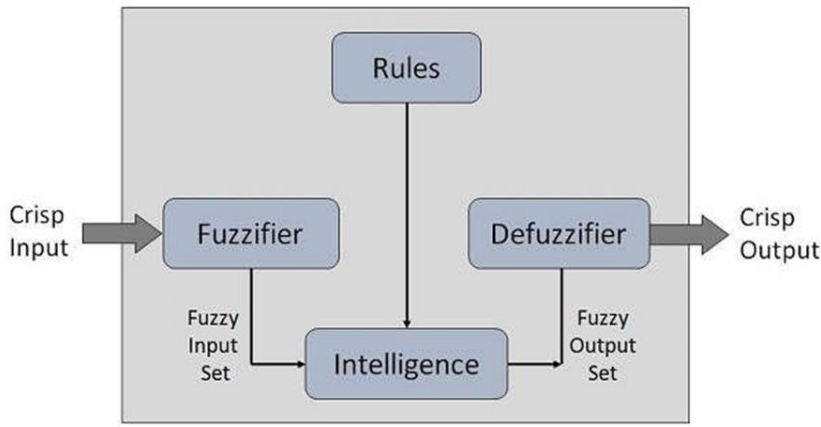
Fuzzy Logic Systems Architecture

It has four main parts as shown −

Fuzzification Module − It transforms the system inputs, which are crisp numbers, into fuzzy sets. It splits the input signal into five steps such as –

LP	x is Large Positive
MP	x is Medium Positive
S	x is Small
MN	x is Medium Negative
LN	x is Large Negative

- **Knowledge Base** − It stores IF-THEN rules provided by experts.
- **Inference Engine** − It simulates the human reasoning process by making fuzzy inference on the inputs and IF-THEN rules.
- **Defuzzification Module** − It transforms the fuzzy set obtained by the inference engine into a crisp value.



The **membership functions work on** fuzzy sets of variables.

Membership Function

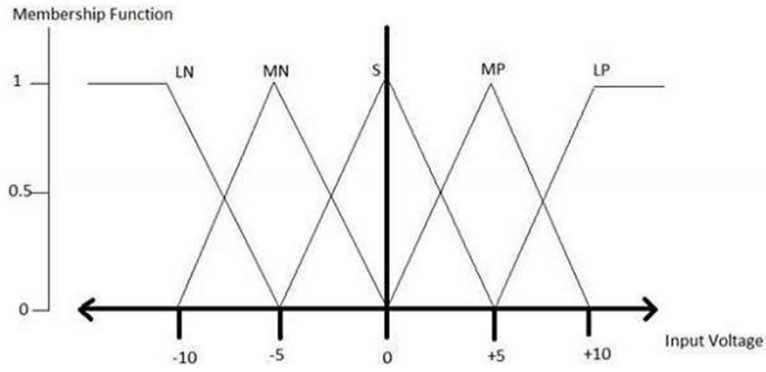
Membership functions allow you to quantify linguistic term and represent a fuzzy set graphically. A **membership function** for a fuzzy *set A* on the universe of discourse X is defined as μA:X → [0,1].

Here, each element of *X* is mapped to a value between 0 and 1. It is called **membership value** or **degree of membership**. It quantifies the degree of membership of the element in *X* to the fuzzy set *A*.

- x axis represents the universe of discourse.
- y axis represents the degrees of membership in the [0, 1] interval.

There can be multiple membership functions applicable to fuzzify a numerical value. Simple membership functions are used as use of complex functions does not add more precision in the output.

All membership functions for **LP, MP, S, MN,** and **LN** are shown as below –

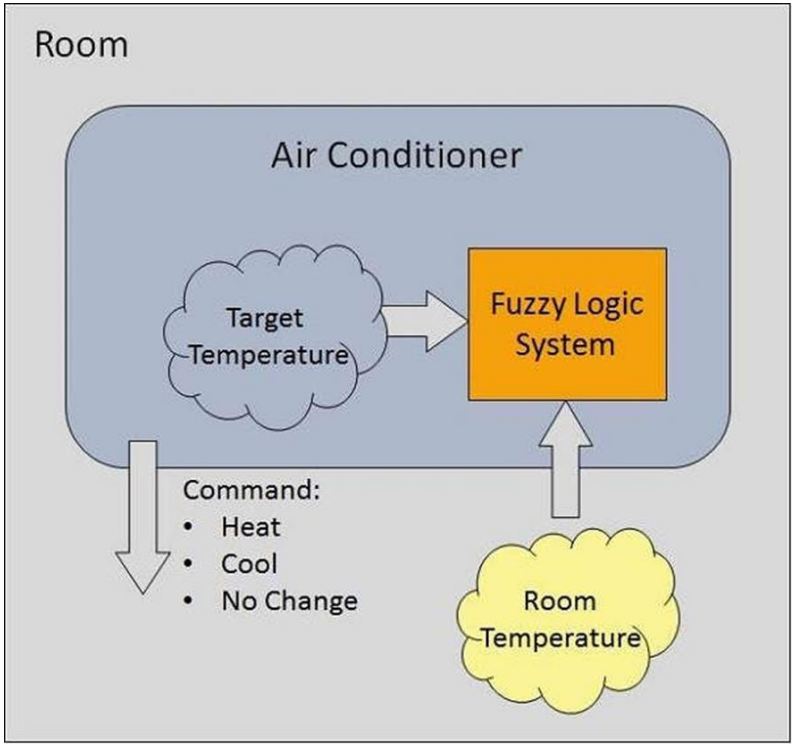


The triangular membership function shapes are most common among various other membership function shapes such as trapezoidal, singleton, and Gaussian.

Here, the input to 5-level fuzzifier varies from -10 volts to +10 volts. Hence the corresponding output also changes.

Example of a Fuzzy Logic System

Let us consider an air conditioning system with 5-level fuzzy logic system. This system adjusts the temperature of air conditioner by comparing the room temperature and the target temperature value.



Algorithm

- Define linguistic Variables and terms (start)
- Construct membership functions for them. (start)
- Construct knowledge base of rules (start)
- Convert crisp data into fuzzy data sets using membership functions. (fuzzification)

- Evaluate rules in the rule base. (Inference Engine)
- Combine results from each rule. (Inference Engine)
- Convert output data into non-fuzzy values. (defuzzification)

Development

Step 1 – Define linguistic variables and terms

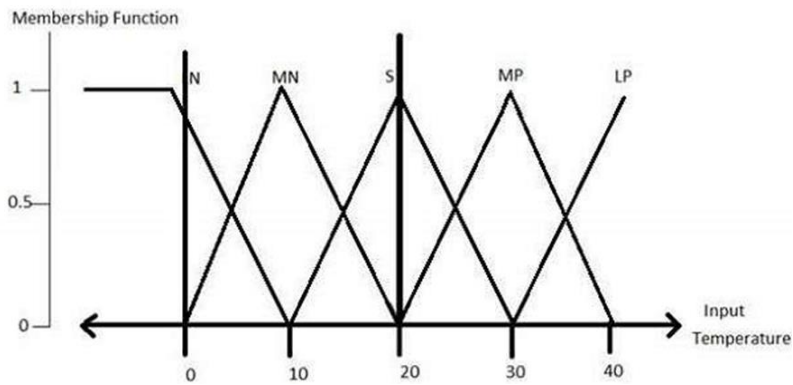
Linguistic variables are input and output variables in the form of simple words or sentences. For room temperature, cold, warm, hot, etc., are linguistic terms.

Temperature (t) = {very-cold, cold, warm, very-warm, hot}

Every member of this set is a linguistic term and it can cover some portion of overall temperature values.

Step 2 – Construct membership functions for them

The membership functions of temperature variable are as shown –



Step3 – Construct knowledge base rules

Create a matrix of room temperature values versus target temperature values that an air conditioning system is expected to provide.

RoomTemp./Target	Very_Cold	Cold	Warm	Hot	Very_Hot
Very_Cold	No_Change	Heat	Heat	Heat	Heat
Cold	Cool	No_Change	Heat	Heat	Heat
Warm	Cool	Cool	No_Change	Heat	Heat
Hot	Cool	Cool	Cool	No_Change	Heat
Very_Hot	Cool	Cool	Cool	Cool	No_Change

Build a set of rules into the knowledge base in the form of IF-THEN-ELSE structures.

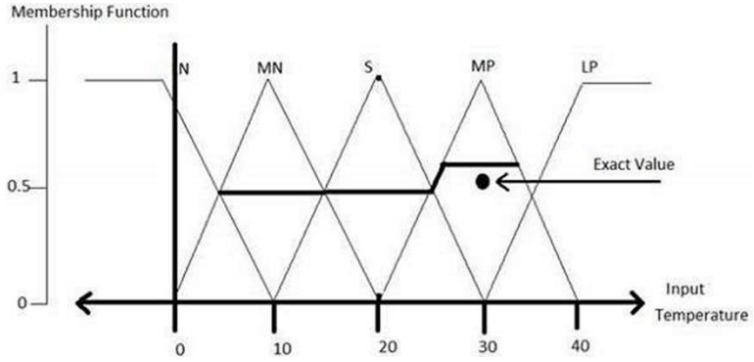
Sr. No.	Condition	Action
1	IF temperature=(Cold OR Very_Cold) AND target=Warm THEN	Heat
2	IF temperature=(Hot OR Very_Hot) AND target=Warm THEN	Cool
3	IF (temperature=Warm) AND (target=Warm) THEN	No_Change

Step 4 – Obtain fuzzy value

Fuzzy set operations perform evaluation of rules. The operations used for OR and AND are Max and Min respectively. Combine all results of evaluation to form a final result. This result is a fuzzy value.

Step 5 – Perform defuzzification

Defuzzification is then performed according to membership function for output variable.



Application Areas of Fuzzy Logic

The key application areas of fuzzy logic are as given −

Automotive Systems

- Automatic Gearboxes
- Four-Wheel Steering
- Vehicle environment control

Consumer Electronic Goods

- Hi-Fi Systems
- Photocopiers
- Still and Video Cameras
- Television

Domestic Goods

- Microwave Ovens
- Refrigerators
- Toasters
- Vacuum Cleaners
- Washing Machines

Environment Control

- Air Conditioners/Dryers/Heaters
- Humidifiers

Advantages of FLSs

- Mathematical concepts within fuzzy reasoning are very simple.
- You can modify a FLS by just adding or deleting rules due to flexibility of fuzzy logic.
- Fuzzy logic Systems can take imprecise, distorted, noisy input information.
- FLSs are easy to construct and understand.
- Fuzzy logic is a solution to complex problems in all fields of life, including medicine, as it resembles human reasoning and decision making.

Disadvantages of FLSs

- There is no systematic approach to fuzzy system designing.
- They are understandable only when simple.
- They are suitable for the problems which do not need high accuracy.

3.7. Natural Language Processing

Natural Language Processing (NLP) refers to AI method of communicating with an intelligent system using a natural language such as English.

Processing of Natural Language is required when you want an intelligent system like robot to perform as per your instructions, when you want to hear decision from a dialogue based clinical expert system, etc.

The field of NLP involves making computers to perform useful tasks with the natural languages humans use. The input and output of an NLP system can be −

- Speech
- Written Text

Components of NLP

There are two components of NLP as given −

Natural Language Understanding (NLU)

Understanding involves the following tasks −

- Mapping the given input in natural language into useful representations.
- Analyzing different aspects of the language.

Natural Language Generation (NLG)

It is the process of producing meaningful phrases and sentences in the form of natural language from some internal representation.

It involves −

- **Text planning** − It includes retrieving the relevant content from knowledge base.
- **Sentence planning** − It includes choosing required words, forming meaningful phrases, setting tone of the sentence.
- **Text Realization** − It is mapping sentence plan into sentence structure. The NLU is harder than NLG.

Difficulties in NLU

NL has an extremely rich form and structure.

It is very ambiguous. There can be different levels of ambiguity −

- **Lexical ambiguity** − It is at very primitive level such as word-level.
- For example, treating the word “board” as noun or verb?
- **Syntax Level ambiguity** − A sentence can be parsed in different ways.

For example, “He lifted the beetle with red cap.” − Did he use cap to lift the beetle or he lifted a beetle that had red cap?

- **Referential ambiguity** − Referring to something using pronouns. For example, Rima went to Gauri. She said, “I am tired.” − Exactly who is tired?
- One input can mean different meanings.
- Many inputs can mean the same thing.

NLP Terminology

- **Phonology** − It is study of organizing sound systematically.
- **Morphology** − It is a study of construction of words from primitive meaningful units.
- **Morpheme** − It is primitive unit of meaning in a language.
- **Syntax** − It refers to arranging words to make a sentence. It also involves determining the structural role of words in the sentence and in phrases.
- **Semantics** − It is concerned with the meaning of words and how to combine words into meaningful phrases and sentences.
- **Pragmatics** − It deals with using and understanding sentences in different situations and how the interpretation of the sentence is affected.
- **Discourse** − It deals with how the immediately preceding sentence can affect the interpretation of the next sentence.
- **World Knowledge** − It includes the general knowledge about the world.

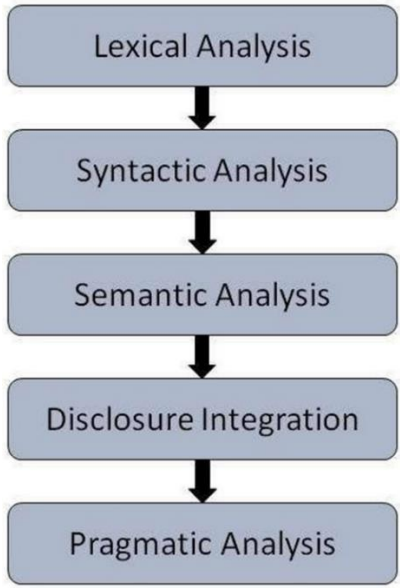
Steps in NLP

There are general five steps −

- **Lexical Analysis** − It involves identifying and analyzing the structure of words. Lexicon of a language means the collection of words and phrases in a language. Lexical analysis is dividing the whole chunk of txt into

paragraphs, sentences, and words.

- **Syntactic Analysis (Parsing)** − It involves analysis of words in the sentence for grammar and arranging words in a manner that shows the relationship among the words. The sentence such as "The school goes to boy" is rejected by English syntactic analyzer.



- **Semantic Analysis** − It draws the exact meaning or the dictionary meaning from the text. The text is checked for meaningfulness. It is done by mapping syntactic structures and objects in the task domain. The semantic analyzer disregards sentence such as "hot ice- cream".
- **Discourse Integration** − The meaning of any sentence depends upon the meaning of the sentence just before it. In addition, it also brings about the meaning of immediately succeeding sentence.
- **Pragmatic Analysis** − During this, what was said is re-interpreted on what it actually meant. It involves deriving those aspects of language which require real world knowledge.

Implementation Aspects of Syntactic Analysis

There are a number of algorithms researchers have developed for syntactic analysis, but we consider only the following simple methods −

- Context-Free Grammar
- Top-Down Parser

Let us see them in detail –

Context-Free Grammar

It is the grammar that consists rules with a single symbol on the left-hand side of the rewrite rules. Let us create grammar to parse a sentence −

"The bird pecks the grains" **Articles (DET)** − a | an | the **Nouns** − bird | birds | grain | grains

Noun Phrase (NP) − Article + Noun | Article + Adjective + Noun

= DET N | DET ADJ N

Verbs − pecks | pecking | pecked

Verb Phrase (VP) − NP V | V NP

Adjectives (ADJ) − beautiful | small | chirping

The parse tree breaks down the sentence into structured parts so that the computer can easily understand and process it. In order for the parsing algorithm to construct this parse tree, a set of rewrite rules, which describe what tree structures are legal, need to be constructed.

These rules say that a certain symbol may be expanded in the tree by a sequence of other symbols. According to first order logic rule, if there are two strings Noun Phrase (NP) and Verb Phrase (VP), then the string combined by NP followed by VP is a sentence. The

rewrite rules for the sentence are as follows −

S → NP VP

NP → DET N | DET ADJ N VP → V NP

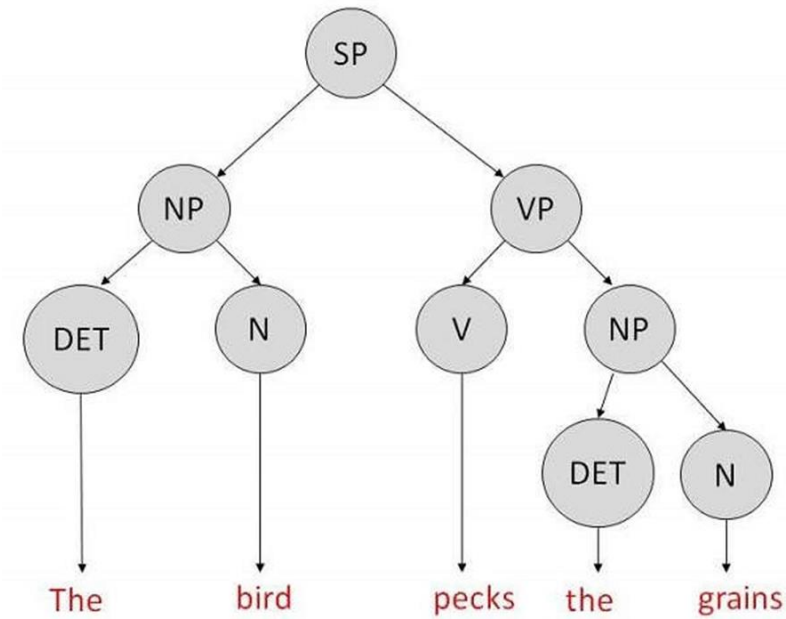
Lexocon −

DET → a | the

ADJ → beautiful | perching

N → bird | birds | grain | grains V → peck | pecks | pecking

The parse tree can be created as shown –



Now consider the above rewrite rules. Since V can be replaced by both, "peck" or "pecks", sentences such as "The bird peck the grains" can be wrongly permitted. i. e. the subject-verb agreement error is approved as correct.

Merit − The simplest style of grammar, therefore widely used one.

Demerits −

- They are not highly precise. For example, "The grains peck the bird", is a syntactically correct according to parser, but even if it makes no sense, parser takes it as a correct sentence.
- To bring out high precision, multiple sets of grammar need to be prepared. It may require a completely different sets of rules for parsing singular and plural variations, passive sentences, etc., which can lead to creation of huge set of rules that are unmanageable.

Top-Down Parser

Here, the parser starts with the S symbol and attempts to rewrite it into a sequence of *terminal symbols* that matches the classes of the words in the input sentence until it consists entirely of terminal symbols.

These are then checked with the input sentence to see if it matched. If not, the process is started over again with a different set of rules. This is repeated until a specific rule is found which describes the structure of the sentence.

Merit − It is simple to implement.

Demerits −

- It is inefficient, as the search process has to be repeated if an error occurs.
- Slow speed of working.

3.8. GENERAL APPLICATION OF AI

AI Facts and Figures

- According to Statista, revenue from the artificial

intelligence (AI) software market worldwide is expected to reach 126 billion dollars by 2025.

- As per Gartner, 37% of organizations have implemented AI in some form. The percentage of enterprises employing AI grew 270% over the past four years.
- According to Servion Global Solutions, by 2025, 95% of customer interactions will be powered by AI.
- A recent 2020 report from Statista reveals that the global AI software market is expected to grow approximately 54% year-on-year and is expected to reach a forecast size of USD $22.6 billion.

Let's now take a look at how AI is used in different domains.

What Are the Applications of Artificial Intelligence?

Here is the list of the top 12 Artificial Intelligence (AI) Applications:

AI Application in E-Commerce

- ✓ **Personalized Shopping**

Artificial Intelligence technology is used to create recommendation engines through which you can engage better with your customers. These recommendations are made in accordance with their browsing history, preference, and interests. It helps in improving your relationship with your customers and their loyalty towards your brand.

- ✓ **AI-powered Assistants**

Virtual shopping assistants and chatbots help improve the user experience while shopping online. Natural Language Processing is used to make the conversation sound as human and personal as possible. Moreover, these assistants can have real-time engagement with your customers. Did you know that on amazon.com, soon, customer service could be handled by chatbots?

- ✓ **Fraud Prevention**

Credit card frauds and fake reviews are two of the most significant issues that E-Commerce companies deal with. By considering the usage patterns, AI can help reduce the possibility of credit card frauds taking place. Many customers prefer to buy a product or service based on customer reviews. AI can help identify and handle fake reviews.

1. AI Applications in Navigation

Based on research from MIT, GPS technology can provide users with accurate, timely, and detailed information to improve safety. The technology uses a combination of Convolutional Neural Network and Graph Neural Network, which makes lives easier for users by automatically detecting the number of lanes and road types behind obstructions on the roads. AI is heavily used by Uber and many logistics companies to improve operational efficiency, analyze road traffic, and optimize routes.

2. AI Applications in Robotics

Robotics is another field where artificial intelligence applications are commonly used. Robots powered by AI use real-time updates to sense obstacles in its path and pre-plan its journey instantly.

It can be used for -

- Carrying goods in hospitals, factories, and warehouses
- Cleaning offices and large equipment
- Inventory management

3. AI Applications in Human Resource

Did you know that companies use intelligent software to ease the hiring process?

Artificial Intelligence helps with blind hiring. Using machine learning software, you can examine applications based on specific

parameters. AI drive systems can scan job candidates' profiles, and resumes to provide recruiters an understanding of the talent pool they must choose from.

4. AI Applications in Healthcare

Artificial Intelligence finds diverse applications in the healthcare sector. AI applications are used in healthcare to build sophisticated machines that can detect diseases and identify cancer cells. Artificial Intelligence can help analyze chronic conditions with lab and other medical data to ensure early diagnosis. AI uses the combination of historical data and medical intelligence for the discovery of new drugs.

5. AI Applications in Agriculture

Artificial Intelligence is used to identify defects and nutrient deficiencies in the soil. This is done using computer vision, robotics, and machine learning applications, AI can analyze where weeds are growing. AI bots can help to harvest crops at a higher volume and faster pace than human laborers.

6. AI Applications in Gaming

Another sector where Artificial Intelligence applications have found prominence is the gaming sector. AI can be used to create smart, human-like NPCs to interact with the players.

It can also be used to predict human behavior using which game design and testing can be improved. The Alien Isolation games released in 2014 uses AI to stalk the player throughout the game. The game uses two Artificial Intelligence systems - 'Director AI' that frequently knows your location and the 'Alien AI,' driven by sensors and behaviors that continuously hunt the player.

7. AI Applications in Automobiles

Artificial Intelligence is used to build self-driving vehicles. AI can be used along with the vehicle's camera, radar, cloud services,

GPS, and control signals to operate the vehicle. AI can improve the in-vehicle experience and provide additional systems like emergency braking, blind-spot monitoring, and driver-assist steering.

8. AI Applications in Social Media

- **Instagram**

On Instagram, AI considers your likes and the accounts you follow to determine what posts you are shown on your explore tab.

- **Facebook**

Artificial Intelligence is also used along with a tool called Deep Text. With this tool, Facebook can understand conversations better. It can be used to translate posts from different languages automatically.

- **Twitter**

AI is used by Twitter for fraud detection, removing propaganda, and hateful content. Twitter also uses AI to recommend tweets that users might enjoy, based on what type of tweets they engage with.

9. AI Applications in Marketing

Artificial intelligence (AI) applications are popular in the marketing domain as well.

- Using AI, marketers can deliver highly targeted and personalized ads with the help of behavioral analysis, pattern recognition, etc. It also helps with retargeting audiences at the right time to ensure better results and reduced feelings of distrust and annoyance.
- AI can help with content marketing in a way that matches the brand's style and voice. It can be used to handle routine tasks like performance, campaign reports, and much more.
- Chatbots powered by AI, Natural Language Processing,

Natural Language Generation, and Natural Language Understanding can analyze the user's language and respond in the ways humans do.

- AI can provide users with real-time personalizations based on their behavior and can be used to edit and optimize marketing campaigns to fit a local market's needs.

10. AI Applications in Chatbots

AI chatbots can comprehend natural language and respond to people online who use the "live chat" feature that many organizations provide for customer service. AI chatbots are effective with the use of machine learning, and can be integrated in an array of websites and applications. AI chatbots can eventually build a database of answers, in addition to pulling information from an established selection of integrated answers. As AI continues to improve, these chatbots can effectively resolve customer issues, respond to simple inquiries, improve customer service, and provide 24/7 support. All in all, these AI chatbots can help to improve customer satisfaction.

AI Applications in Finance

It has been reported that 80 percent of banks recognize the benefits that AI can provide. Whether it’s personal finance, corporate finance, or consumer finance, the highly evolved technology that is offered through AI can help to significantly improve a wide range of financial services. For example, customers looking for help regarding wealth management solutions can easily get the information they need through SMS text messaging or online chat, all AI-powered. Artificial intelligence can also detect changes in transaction patterns and other potential red flags that can signify fraud, which humans can easily miss, and thus saving businesses and individuals from significant loss. Aside from fraud detection and task automation, AI can also better predict and assess loan risks.

4

Introduction of IOT

IoT comprises things that have unique identities and are connected to internet. By 2020 there will be a total of 50 billion devices /things connected to internet. IoT is not limited to just connecting things to the internet but also allow things to communicate and exchange data.

Definition:

A dynamic global n/w infrastructure with self-configuring capabilities based on standard and interoperable communication protocols where physical and virtual —things‖ have identities, physical attributes and virtual personalities and use intelligent interfaces, and are seamlessly integrated into information n/w, often communicate data associated with users and their environments.

Characteristics:

1) **Dynamic & Self Adapting**: IoT devices and systems may have the capability to dynamically adapt with the changing contexts and take actions based on their operating conditions, user's context or sensed environment.

   Eg: the surveillance system is adapting itself based on context and changing conditions.

2) **Self-Configuring:** allowing a large number of devices to work together to provide certain functionality.

3) **Inter Operable Communication Protocols:** support a number of interoperable communication protocols Ans can communicate with other devices and also with infrastructure.
4) **Unique Identity:** Each IoT device has a unique identity and a unique identifier (IP address).
5) **Integrated into Information Network:** that allow them to communicate and exchange data with other devices and systems.

There are two ways of building IoT:

1) Form a separate internetwork including only physical objects.
2) Make the Internet ever more expansive, but this requires hard-core technologies such as rigorous cloud computing and rapid big data storage (expensive).

In the near future, IoT will become broader and more complex in terms of scope. It will change the world in terms of

"anytime, anyplace, anything in connectivity."

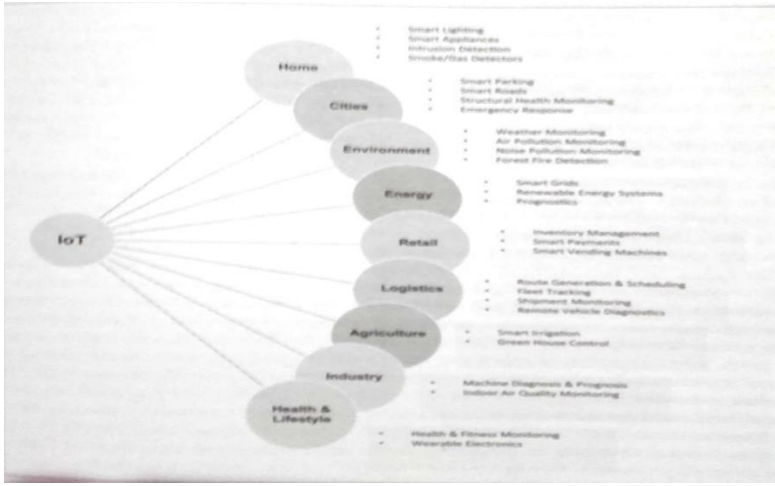
Application Domains: IoT is currently found in four different popular domains:

1) Manufacturing/Industrial business - 40.2%
2) Healthcare - 30.3%
3) Security - 7.7%
4) Retail - 8.3%

Applications of IoT:

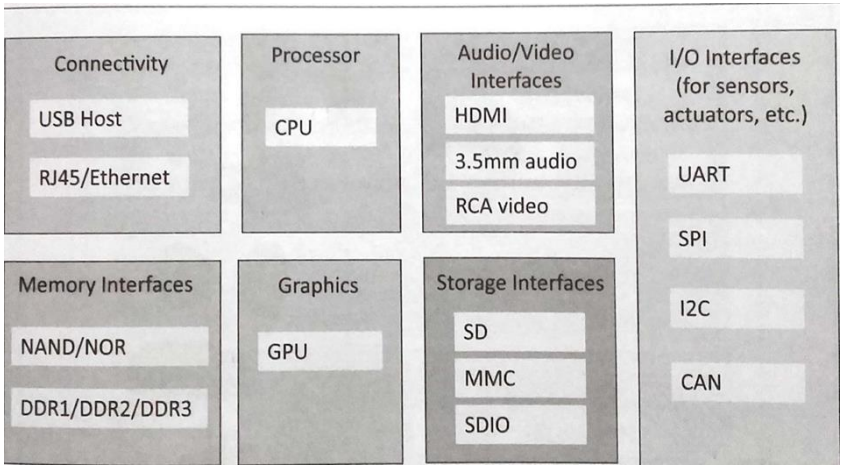
1) Home
2) Cities
3) Environment
4) Energy
5) Retail
6) Logistics

7) Agriculture
8) Industry
9) Health & Lifestyle



4.1. Physical Design of IOT

Things in IoT:



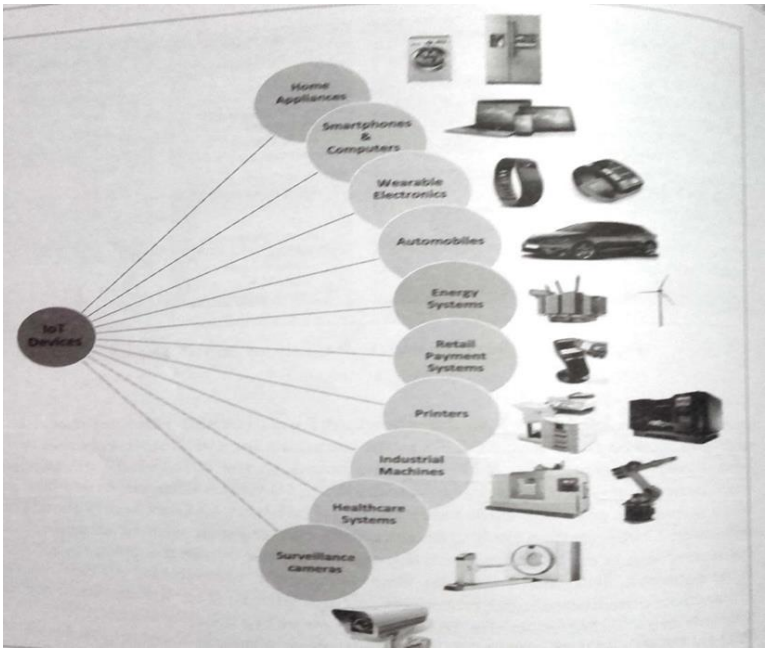
The things in IoT refers to IoT devices which have unique identities and perform remote sensing, actuating and monitoring

capabilities. IoT devices can exchange dat with other connected devices applications. It collects data from other devices and process data either locally or remotely.

An IoT device may consist of several interfaces for communication to other devices both wired and wireless.

These includes

1) I/O interfaces for sensors,
2) Interfaces for internet connectivity
3) Memory and storage interfaces
4) audio/video interfaces.



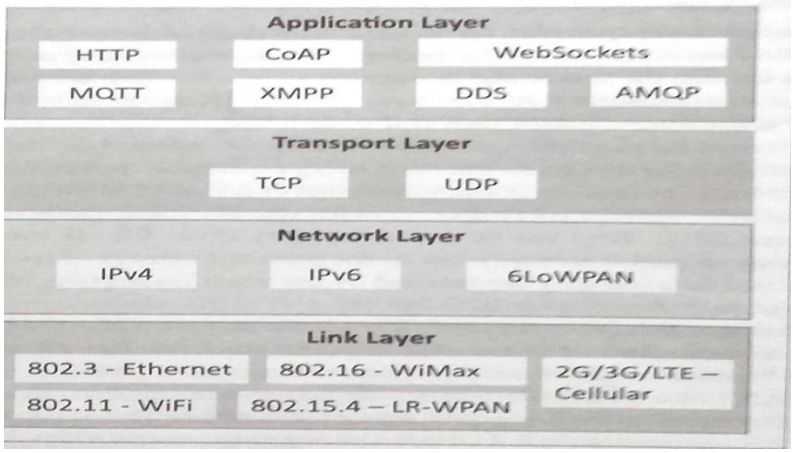
1) IoT Protocols:

a) Link Layer: Protocols determine how data is physically sent over the network ‘s physical layer or medium. Local network connect to which host is attached. Hosts on the same link

exchange data packets over the link layer using link layer protocols. Link layer determines how packets are coded and signaled by the h/w device over the medium to which the host is attached.

Protocols:

- 802.3-Ethernet: IEEE802.3 is collection of wired Ethernet standards for the link layer. Eg: 802.3 uses co-axial cable; 802.3i uses copper twisted pair connection; 802.3j uses fiber optic connection; 802.3ae uses Ethernet over fiber.
- 802.11-WiFi: IEEE802.11 is a collection of wireless LAN (WLAN) communication standards including extensive description of link layer. Eg: 802.11a operates in 5GHz band, 802.11b and 802.11g operates in 2.4GHz band, 802.11n operates in 2.4/5GHz band, 802.11ac operates in 5GHz band, 802.11ad operates in 60Ghzband.



- 802.16 - WiMax: IEEE802.16 is a collection of wireless broadband standards including exclusive description of link layer. WiMax provide data rates from 1.5 Mb/s to 1Gb/s.
- 802.15.4-LR-WPAN: IEEE802.15.4 is a collection of

standards for low rate wireless personal area network(LR-WPAN). Basis for high level communication protocols such as ZigBee. Provides data rate from 40kb/s to250kb/s.

- 2G/3G/4G-Mobile Communication: Data rates from 9.6kb/s(2G) to up to100Mb/s(4G).

B) **Network/Internet Layer:** Responsible for sending IP datagrams from source n/w to destination n/w. Performs the host addressing and packet routing. Datagrams contains source and destination address.

Protocols:

- **IPv4:** Internet Protocol version4 is used to identify the devices on a n/w using a hierarchical addressing scheme. 32 bit address. Allows total of 2**32addresses.
- **IPv6:** Internet Protocol version6 uses 128 bit address scheme and allows 2**128 addresses.
- **6LOWPAN:**(IPv6overLowpower Wireless Personal Area Network) operates in 2.4 GHz frequency range and data transfer 250 kb/s.

C) **Transport Layer:** Provides end-to-end message transfer capability independent of the underlying n/w. Set up on connection with ACK as in TCP and without ACK as in UDP. Provides functions such as error control, segmentation, flow control and congestion control.

Protocols:

- **TCP:** Transmission Control Protocol used by web browsers (along with HTTP and HTTPS), email (along with SMTP, FTP). Connection oriented and stateless protocol. IP Protocol deals with sending packets, TCP ensures reliable transmission of protocols in order.

Avoids n/w congestion and congestion collapse.

- **UDP:** User Datagram Protocol is connectionless protocol. Useful in time sensitive applications, very small data units to exchange. Transaction oriented and stateless protocol. Does not provide guaranteed delivery.

D) **Application Layer:** Defines how the applications interface with lower layer protocols to send data over the n/w. Enables process-to-process communication using ports.

Protocols:

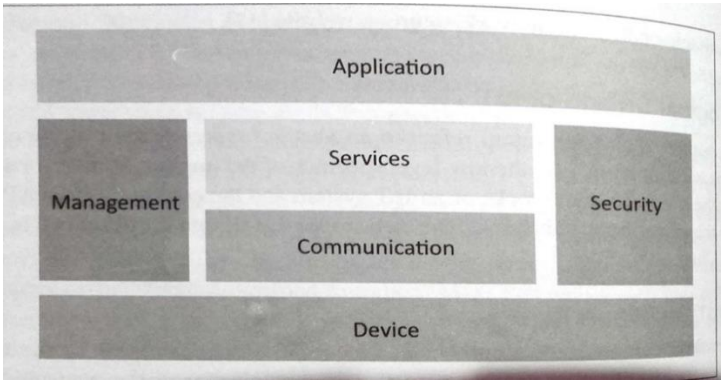
- **HTTP:** Hyper Text Transfer Protocol that forms foundation of WWW. Follow request- response model Stateless protocol.
- **CoAP:** Constrained Application Protocol for machine-to-machine(M2M) applications with constrained devices, constrained environment and constrained n/w. Uses client- server architecture.
- **WebSocket:** allows full duplex communication over a single socket connection.
- **MQTT:** Message Queue Telemetry Transport is light weight messaging protocol based on publish-subscribe model. Uses client server architecture. Well suited for constrained environment.
- **XMPP:** Extensible Message and Presence Protocol for real time communication and streaming XML data between network entities. Support client-server and server-server communication.
- **DDS:** Data Distribution Service is data centric middleware standards for device-to-device or machine-to-machine communication. Uses publish-subscribe model.
- **AMQP:** Advanced Message Queuing Protocol is open application layer protocol for business messaging. Supports both point-to-point and publish-subscribe model.

4.2. Logical Design of IoT

Refers to an abstract represent of entities and processes without going into the low level specifies of implementation.

1) IoT Functional Blocks
2) IoT Communication Models
3) IoT Comm. APIs

IoT Functional Blocks: Provide the system the capabilities for identification, sensing, actuation, communication and management.



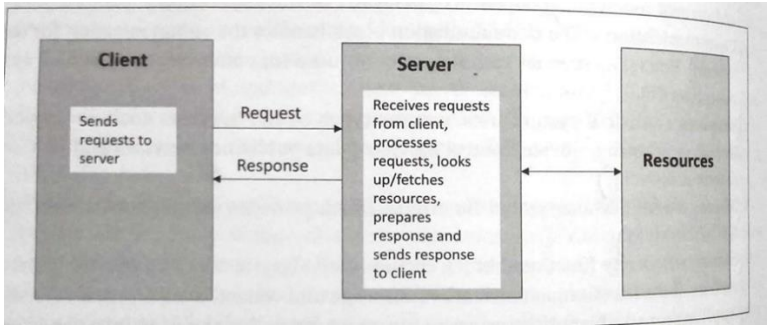
- **Device:** An IoT system comprises of devices that provide sensing, actuation, and monitoring and control functions.
- **Communication:** handles the communication for IoT system.
- **Services:** for device monitoring, device control services, data publishing services and services for device discovery.
- **Management:** Provides various functions to govern the IoT system.
- **Security:** Secures IoT system and priority functions such as authentication, authorization, message and context integrity and data security.
- **Application:** IoT application provide an interface that the

users can use to control and monitor various aspects of IoT system.

1) IoT Communication Models:

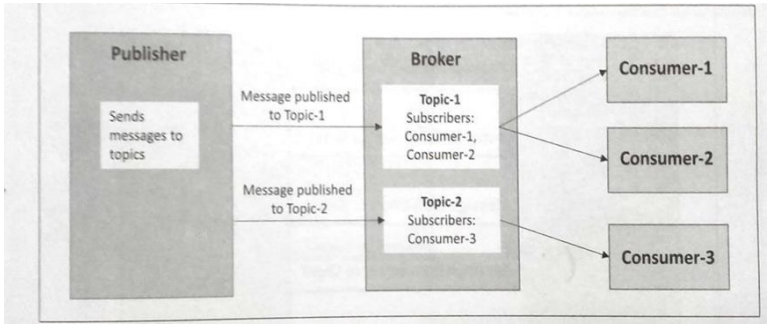
- Request-Response
- Publish-Subscribe
- Push-Pull
- Exclusive Pair

Request-Response Model:



In which the client sends request to the server and the server replies to requests. Is a stateless communication model and each request-response pair is independent of others.

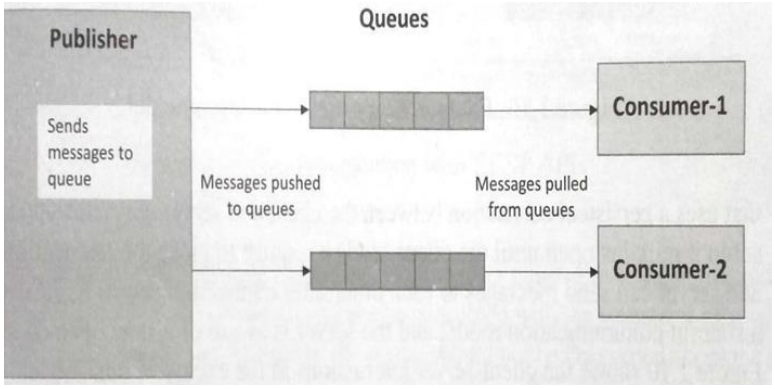
Publish-Subscribe Model:



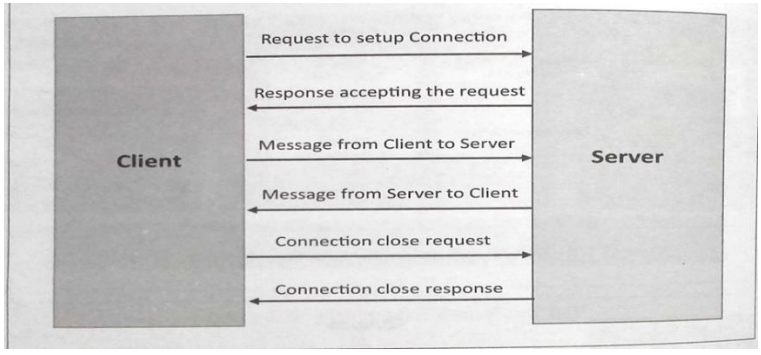
Involves publishers, brokers and consumers. Publishers are

source of data. Publishers send data to the topics which are managed by the broker. Publishers are not aware of the consumers. Consumers subscribe to the topics which are managed by the broker. When the broker receives data for a topic from the publisher, it sends the data to all the subscribed consumers.

Push-Pull Model: in which data producers push data to queues and consumers pull data from the queues. Producers do not need to aware of the consumers. Queues help in decoupling the message between the producers and consumers.



1) **Exclusive Pair:** is bi-directional, fully duplex communication model that uses a persistent connection between the client and server. Once connection is set up it remains



open until the client send a request to close the connection. Is a stateful communication model and server is aware of all the open connections.

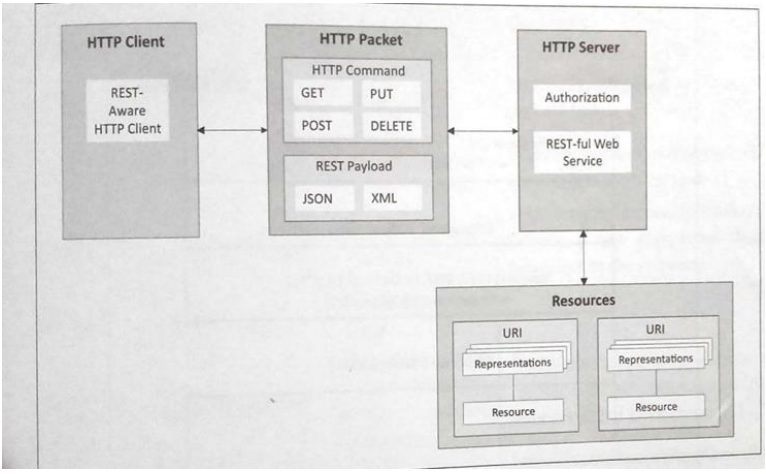
2) IoT Communication APIs:

a) REST based communication APIs (Request-Response Based Model)

b) WebSocket based Communication APIs (Exclusive PairBasedModel)

a) REST based communication APIs: Representational State Transfer (REST) is a set of architectural principles by which we can design web services and web APIs that focus on a system's resources and have resource states are addressed and transferred.

The REST architectural constraints: Fig. shows communication between client server with REST APIs.



Client-Server: The principle behind client-server constraint is the separation of concerns. Separation allows client and server to be independently developed and updated.

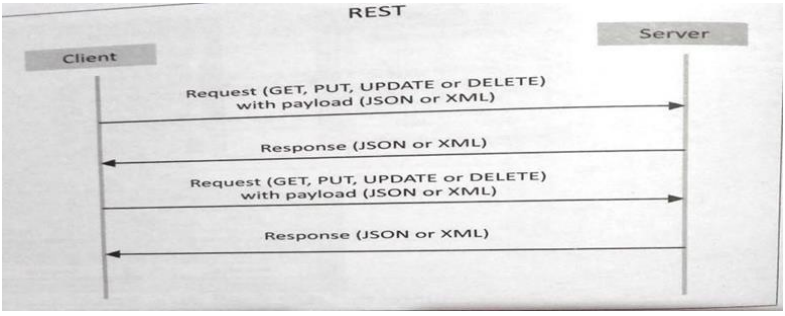
Stateless: Each request from client to server must contain all the info. Necessary to understand the request, and cannot take advantage of any stored context on the server.

Cache-able: Cache constraint requires that the data within a response to a request be implicitly or explicitly labeled as cache-able or non-cacheable. If a response is cache-able, then a client cache is given the right to reuse that response data for later, equivalent requests.

Layered System: constraints the behavior of components such that each component cannot see beyond the immediate layer with which they are interacting.

User Interface: constraint requires that the method of communication between a client and a server must be uniform.

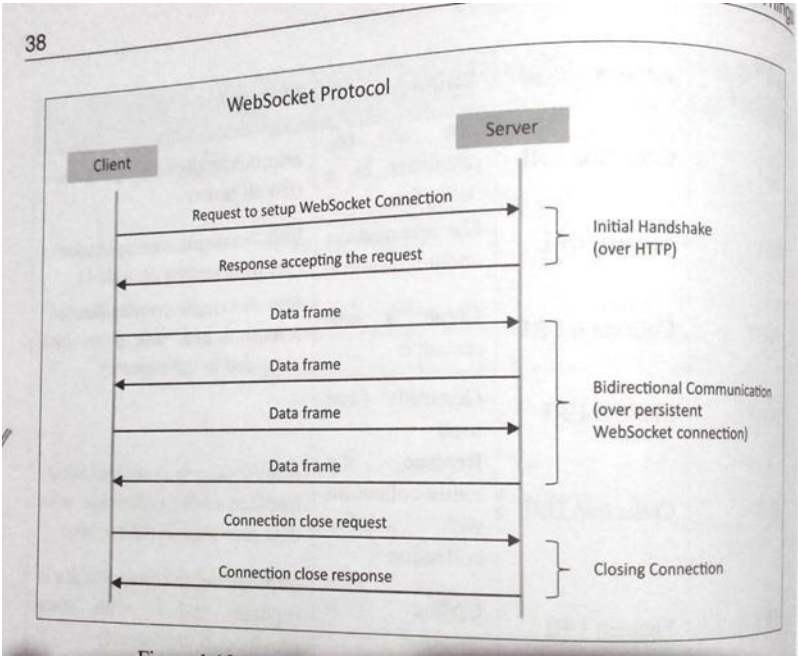
Code on Demand: Servers can provide executable code or scripts for clients to execute in their context. This constraint is the only one that is optional.



Request-Response model used by REST:

RESTful webservice is a collection of resources which are represented by URIs. RESTful web API has a base URI (http://example.com/api/tasks/). The clients and requests to these URIs using the methods defined by the HTTP protocol (GET, PUT, POST or DELETE). A RESTful web service can support various internet media types.

b) WebSocket Based Communication APIs: WebSocket APIs allow bi-directional, full duplex communication between clients and servers. WebSocket APIs follow the exclusive pair communication model.



IoT Enabling Technologies

IoT is enabled by several technologies including Wireless Sensor Networks, Cloud Computing, Big Data Analytics, Embedded Systems, Security Protocols and architectures, Communication Protocols, Web Services, Mobile internet and semantic search engines.

1) **Wireless Sensor Network (WSN):** Comprises of distributed devices with sensors which are used to monitor the environmental and physical conditions. Zig Bee is one of the most popular wireless technologies used byWSNs.

WSNs used in IoT systems are described as follows:

- Weather Monitoring System: in which nodes collect temp, humidity and other data, which is aggregated and analyzed.
- Indoor air quality monitoring systems: to collect data on the indoor air quality and concentration of various gases.
- Soil Moisture Monitoring Systems: to monitor soil moisture at various locations.
- Surveillance Systems: use WSNs for collecting surveillance data (motion data detection).
- Smart Grids: use WSNs for monitoring grids at various points.
- Structural Health Monitoring Systems: Use WSNs to monitor the health of structures (building, bridges) by collecting vibrations from sensor nodes deployed at various points in the structure.

2) **Cloud Computing:** Services are offered to users in different forms.

- Infrastructure-as-a-service (IaaS): provides users the ability to provision computing and storage resources. These resources are provided to the users as a virtual machine instance and virtual storage.
- Platform-as-a-Service (PaaS): provides users the ability to develop and deploy application in cloud using the development tools, APIs, software libraries and services provided by the cloud service provider.
- Software-as-a-Service (SaaS): provides the user a complete software application or the user interface to the application itself.

3) **Big Data Analytics:** Some examples of big data generated by IoT are

- Sensor data generated by IoT systems.
- Machine sensor data collected from sensors established in

industrial and energy systems.
- Health and fitness data generated IoT devices.
- Data generated by IoT systems for location and tracking vehicles.
- Data generated by retail inventory monitoring systems.

4) **Communication Protocols:** form the back-bone of IoT systems and enable network connectivity and coupling to applications.

   - Allow devices to exchange data over network.
   - Define the exchange formats, data encoding addressing schemes for device and routing of packets from source to destination.
   - It includes sequence control, flow control and retransmission of lost packets.

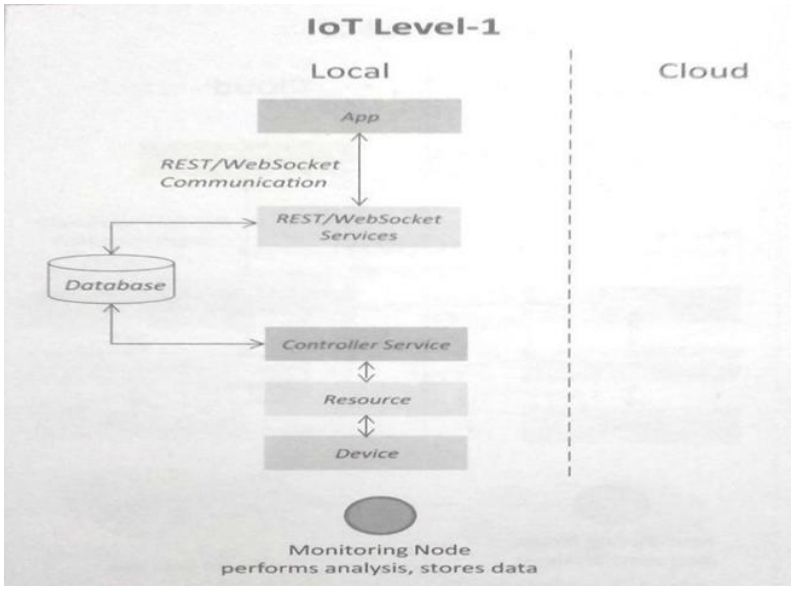
5) **Embedded Systems:** is a computer system that has computer hardware and software embedded to perform specific tasks. Embedded System range from low cost miniaturized devices such as digital watches to devices such as digital cameras, POS terminals, vending machines, appliances.,

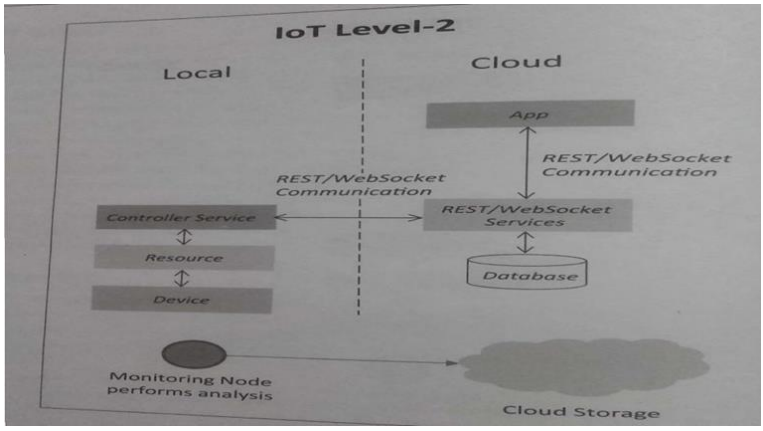
IoT Levels and Deployment Templates

IoT Level1: System has a single node that performs sensing and/or actuation, stores data, performs analysis and host the application as shown in fig. Suitable for modeling low cost and low complexity solutions where the data involved is not big and analysis requirement are not computationally intensive. An e.g., of IoT Level1 is Home automation.

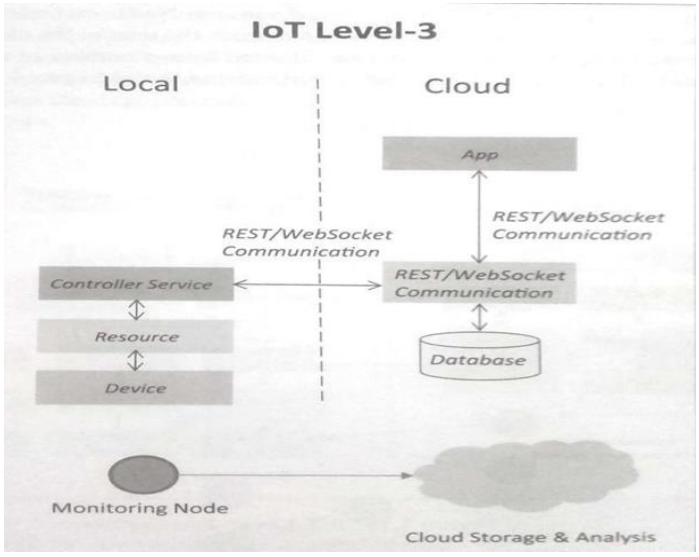
IoT Level2: has a single node that performs sensing and/or actuating and local analysis as shown in fig. Data is stored in cloud and application is usually cloud based. Level2 IoT systems are suitable for solutions where data are involved is big, however, the primary analysis requirement is not computationally intensive and can be done locally itself. An e,g., of Level2 IoT

system for Smart Irrigation.

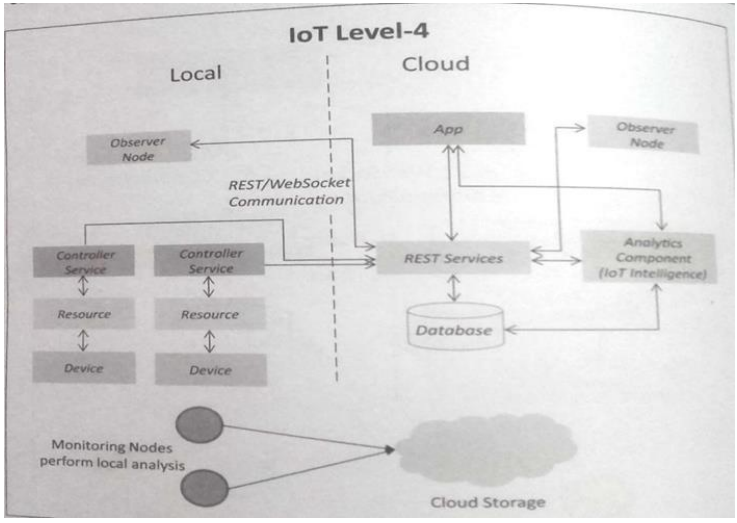




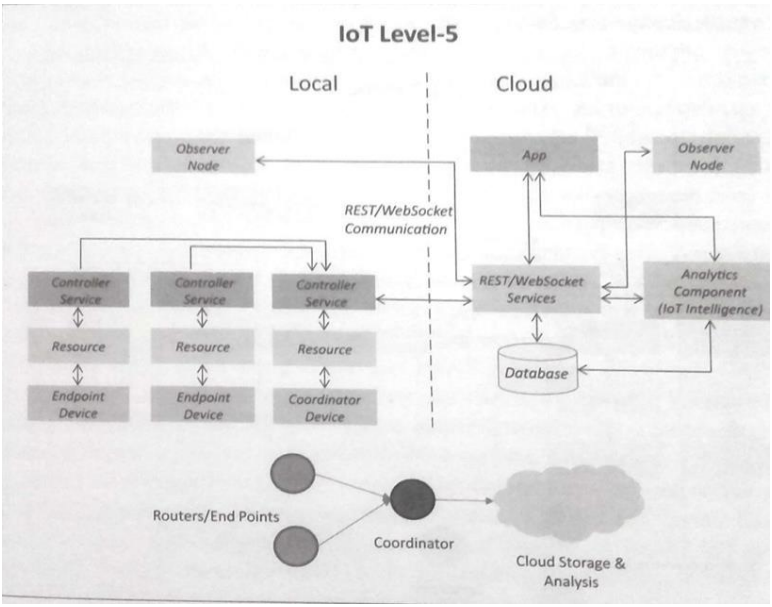
IoT Level3: system has a single node. Data is stored and analyzed in the cloud application is cloud based as shown in fig. Level3 IoT systems are suitable for solutions where the data involved is big and analysis requirements are computationally intensive. An example of IoT level3 system for tracking package handling.



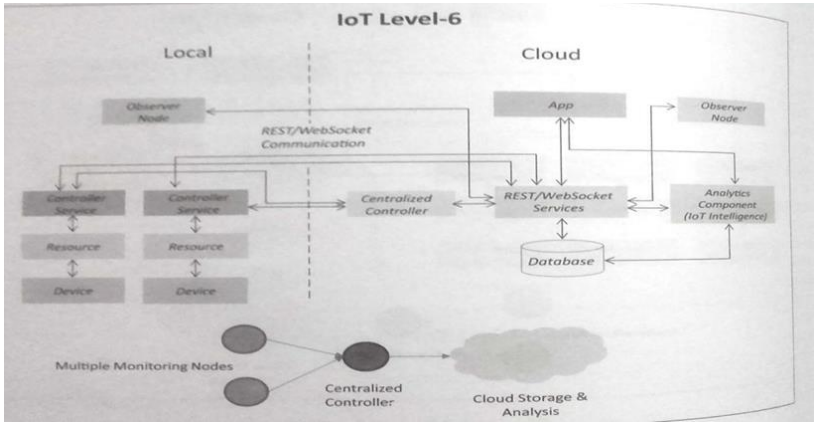
IoT Level4: System has multiple nodes that perform local analysis. Data is stored in the cloud and application is cloud based as shown in fig. Level4 contains local and cloud based observer nodes which can subscribe to and receive information collected in the cloud from IoT devices. An example of a Level4 IoT system for Noise Monitoring.



IoT Level5: System has multiple end nodes and one coordinator node as shown in fig. The end nodes that perform sensing and/or actuation. Coordinator node collects data from the end nodes and send ‘s to the cloud. Data is stored and analyzed in the cloud and application is cloud based. Level5 IoT systems are suitable for solution based on wireless sensor network, in which data involved is big and analysis requirements are computationally intensive. An example of Level5 system for Forest Fire Detection.



IoT Level6: System has multiple independent end nodes that perform sensing and/or actuation and sensed data to the cloud. Data is stored in the cloud and application is cloud based as shown in fig. The analytics component analyses the data and stores the result in the cloud data base. The results are visualized with cloud based application. The centralized controller is aware of the status of all the end nodes and sends control commands to nodes. An example of a Level6 IoT system for Weather Monitoring System.



DOMAIN SPECIFIC IoTs

1) Home Automation:

a) **Smart Lighting:** helps in saving energy by adapting the lighting to the ambient conditions and switching on/off or diming the light when needed.

b) **Smart Appliances:** make the management easier and also provide status information to the user remotely.

c) **Intrusion Detection:** use security cameras and sensors (PIR sensors and door sensors) to detect intrusion and raise alerts. Alerts can be in the form of SMS or email sent to the user.

d) **Smoke/Gas Detectors:** Smoke detectors are installed in homes and buildings to detect smoke that is typically an early sign of fire. Alerts raised by smoke detectors can be in the form of signals to a fire alarm system. Gas detectors can detect the presence of harmful gases such as CO, LPG etc.,

2) Cities:

a) **Smart Parking:** make the search for parking space easier and convenient for drivers. Smart parking are powered by IoT systems that detect the no. of empty parking slots

and send information over internet to smart application backends.

b) **Smart Lighting:** for roads, parks and buildings can help in saving energy.

c) **Smart Roads:** Equipped with sensors can provide information on driving condition, travel time estimating and alert in case of poor driving conditions, traffic condition and accidents.

d) **Structural Health Monitoring:** uses a network of sensors to monitor the vibration levels in the structures such as bridges and buildings.

e) **Surveillance:** The video feeds from surveillance cameras can be aggregated in cloud based scalable storage solution.

f) **Emergency Response:** IoT systems for fire detection, gas and water leakage detection can help in generating alerts and minimizing their effects on the critical infrastructures.

3) Environment:

a) **Weather Monitoring:** Systems collect data from a no. of sensors attached and send the data to cloud based applications and storage back ends. The data collected in cloud can then be analyzed and visualized by cloud-based applications.

b) **Air Pollution Monitoring:** System can monitor emission of harmful gases (CO2, CO, NO, NO2 etc.,) by factories and automobiles using gaseous and meteorological sensors. The collected data can be analyzed to make informed decisions on pollutions control approaches.

c) **Noise Pollution Monitoring:** Due to growing urban development, noise levels in cities have increased and even become alarmingly high in some cities. IoT based noise pollution monitoring systems use a no. of noise monitoring systems that are deployed at different places in a city. The data on noise levels from the station is

collected on servers or in the cloud. The collected data is then aggregated to generate noise maps.

d) **Forest Fire Detection:** Forest fire can cause damage to natural resources, property and human life. Early detection of forest fire can help in minimizing damage.

e) **River Flood Detection:** River floods can cause damage to natural and human resources and human life. Early warnings of floods can be given by monitoring the water level and flow rate. IoT based river flood monitoring system uses a no. of sensor nodes that monitor the water level and flow rate sensors.

4) Energy:

a) **Smart Grids:** is a data communication network integrated with the electrical grids that collects and analyze data captured in near-real-time about power transmission, distribution and consumption. Smart grid technology provides predictive information and recommendations to utilities, their suppliers, and their customers on how best to manage power. By using IoT based sensing and measurement technologies, the health of equipment and integrity of the grid can be evaluated.

b) **Renewable Energy Systems:** IoT based systems integrated with the transformers at the point of interconnection measure the electrical variables and how much power is fed into the grid. For wind energy systems, closed-loop controls can be used to regulate the voltage at point of interconnection which coordinate wind turbine outputs and provides power support.

c) **Prognostics:** In systems such as power grids, real-time information is collected using specialized electrical sensors called Phasor Measurement Units (PMUs) at the substations. The information received from PMUs must be monitored in real-time for estimating the state of the system and for predicting failures.

5) **Retail:**

a) **Inventory Management:** IoT systems enable remote monitoring of inventory using data collected by RFID readers.
b) **Smart Payments:** Solutions such as contact-less payments powered by technologies such as Near Field Communication (NFC) and Bluetooth.
c) **Smart Vending Machines:** Sensors in a smart vending machines monitors its operations and send the data to cloud which can be used for predictive maintenance.

6) **Logistics:**

a) **Route generation & scheduling:** IoT based system backed by cloud can provide first response to the route generation queries and can be scaled upto serve a large transportation network.
b) **Fleet Tracking:** Use GPS to track locations of vehicles in real-time.
c) **Shipment Monitoring:** IoT based shipment monitoring systems use sensors such as temp, humidity, to monitor the conditions and send data to cloud, where it can be analyzed to detect food spoilage.
d) **Remote Vehicle Diagnostics:** Systems use on-board IoT devices for collecting data on Vehicle operations (speed, RPM etc.,) and status of various vehicle subsystems.

7) **Agriculture:**

a) **Smart Irrigation:** to determine moisture amount in soil.

b) **Green House Control:** to improve productivity.

8) **Industry:**

a) Machine diagnosis and prognosis

b) Indoor Air Quality Monitoring

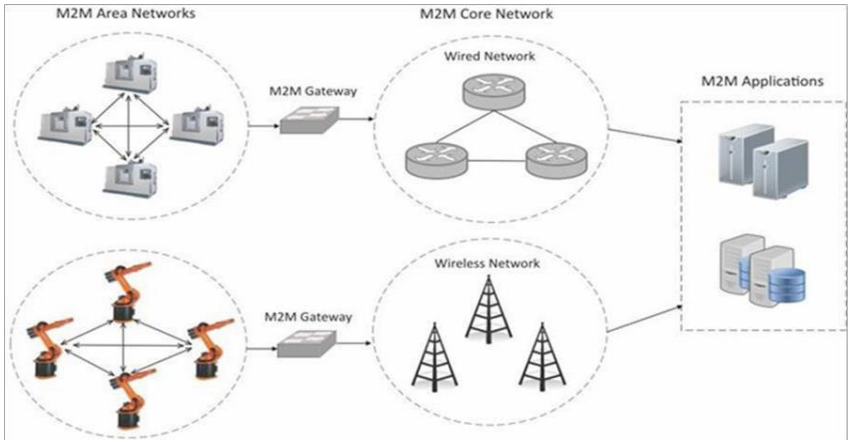
9) Health and Life Style:

a) Health & Fitness Monitoring
b) Wearable Electronic,

M2M: Machine-to-Machine (M2M) refers to networking of machines (or devices) for the purpose of remote monitoring and control and data exchange.

- Term which is often synonymous with IoT is Machine-to-Machine (M2M).
- IoT and M2M are often used interchangeably.

Fig. Shows the end-to-end architecture of M2M systems comprises of M2M area networks, communication networks and application foeman.



- An M2M area network comprises of machines (or M2M nodes) which have embedded network modules for sensing, actuation and communicating various communication protocols can be used for M2M LAN such as ZigBee, Bluetooth, M-bus, Wireless M-Bus etc., These protocols provide connectivity between M2M nodes within an M2M area network.
- The communication network provides connectivity to

remote M2M area networks. The communication network provides connectivity to remote M2M area network. The communication network can use either wired or wireless network (IP based). While the M2M are networks use either proprietarily or non-IP bead communication protocols, the communication network uses IP-based network. Since non-IP based protocols are used within M2M area network, the M2M nodes within one network cannot communicate with nodes in an external network.

- To enable the communication between remote M2M are network, M2M gateways are used.

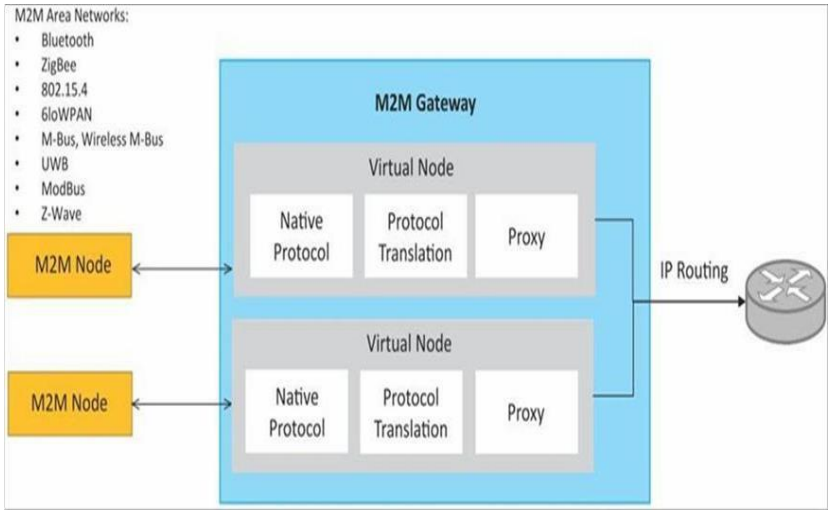


Fig. Shows a block diagram of an M2M gateway. The communication between M2M nodes and the M2M gateway is based on the communication protocols which are naive to the M2M are network. M2M gateway performs protocol translations to enable Ip-connectivity for M2M are networks. M2M gateway acts as a proxy performing translations from/to native protocols to/from Internet Protocol (IP). With an M2M gateway, each mode in an M2M area network appears as a virtualized node for external M2M area networks.

- **Differences between IoT and M2M**

1) Communication Protocols:

- □ Commonly uses M2M protocols include ZigBee, Bluetooth, ModBus , M- Bus, Wireless M-Bustec.,
- □ In IoT uses HTTP, CoAP, WebSocket, MQTT, XMPP, DDS, AMQP etc.,

2) Machines in M2M Vs Things inIoT:

- □ Machines in M2M will be homogenous whereas Things in IoT will be heterogeneous.

3) Hardware Vs Software Emphasis:

- □ the emphasis of M2M is more on hardware with embedded modules, the emphasis of IoT is more on software.

4) Data Collection &Analysis

- □ M2M data is collected in point solutions and often in on-premises storage infrastructure.
- □ The data in IoT is collected in the cloud (can be public, private or hybrid cloud).

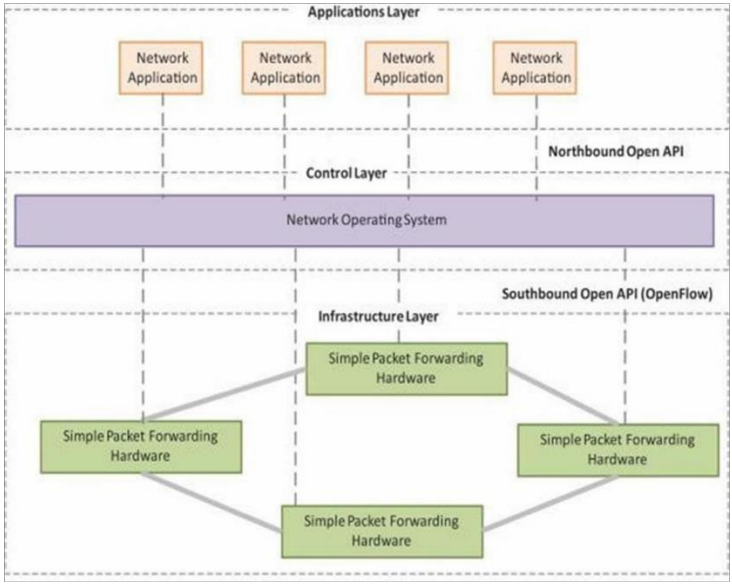
5) Applications

- □ M2M data is collected in point solutions and can be accessed by on-premises applications such as diagnosis applications, service management applications, and on- premises enterprise applications.
- □ IoT data is collected in the cloud and can be accessed by cloud applications such as analytics applications, enterprise applications, remote diagnosis and management applications.

SDN and NVF for IoT

Software Defined Networking (SDN):

- Software-Defined Networking (SDN) is networking architecture that separates the control plane from the data plane and centralizes the network controller.
- Software-based SDN controllers maintain a unified view of the network
- The underlying infrastructure in SDN uses simple packet forwarding hardware as opposed to specialized hardware in conventional networks.



SDN Architecture

Key elements of SDN:

1) Centralized Network Controller

With decoupled control and data planes and centralized network controller, the network administrators can rapidly configure the network.

2) Programmable Open APIs

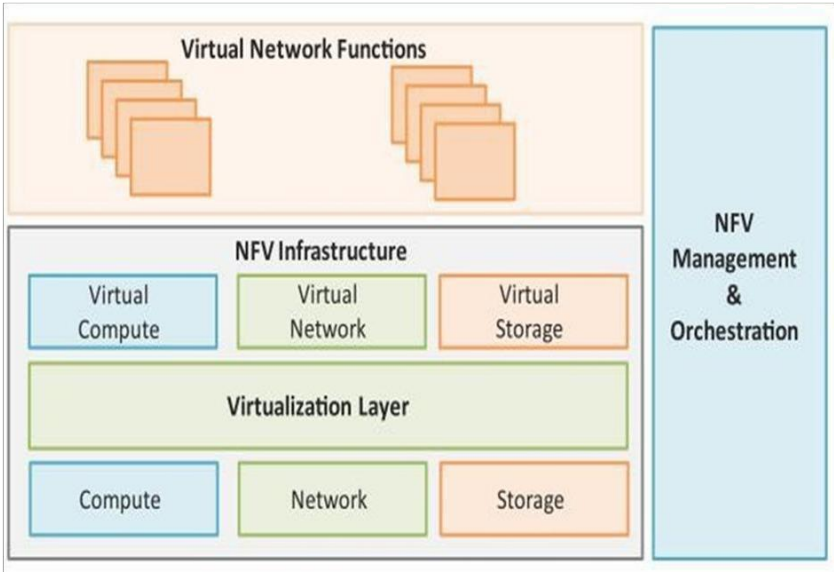
SDN architecture supports programmable open APIs for interface between the SDN application and control layers (Northbound interface).

3) Standard Communication Interface (OpenFlow)

SDN architecture uses a standard communication interface between the control and infrastructure layers (Southbound interface). OpenFlow, which is defined by the Open Networking Foundation (ONF) is the broadly accepted SDN protocol for the South bound interface.

4) Network Function Virtualization (NFV)

- Network Function Virtualization (NFV) is a technology that leverages virtualization to consolidate the heterogeneous network devices onto industry standard high-volume servers, switches and storage.
- NFV is complementary to SDN as NFV can provide the infrastructure on which SDN can run.



Key elements of NFV:

NFV Architecture

1) **Virtualized Network Function(VNF):**

VNF is a software implementation of a network function which is capable of running over the NFV Infrastructure (NFVI).

2) **NFV Infrastructure (NFVI):**

NFVI includes compute, network and storage resources that are virtualized.

3) **NFV Management and Orchestration:**

NFV Management and Orchestration focuses on all virtualization-specific management tasks and covers the orchestration and life-cycle management of physical and/or software resources that support the infrastructure virtualization, and the life-cycle management of VNFs.

Need for IoT Systems Management

Managing multiple devices within a single system requires advanced management capabilities.

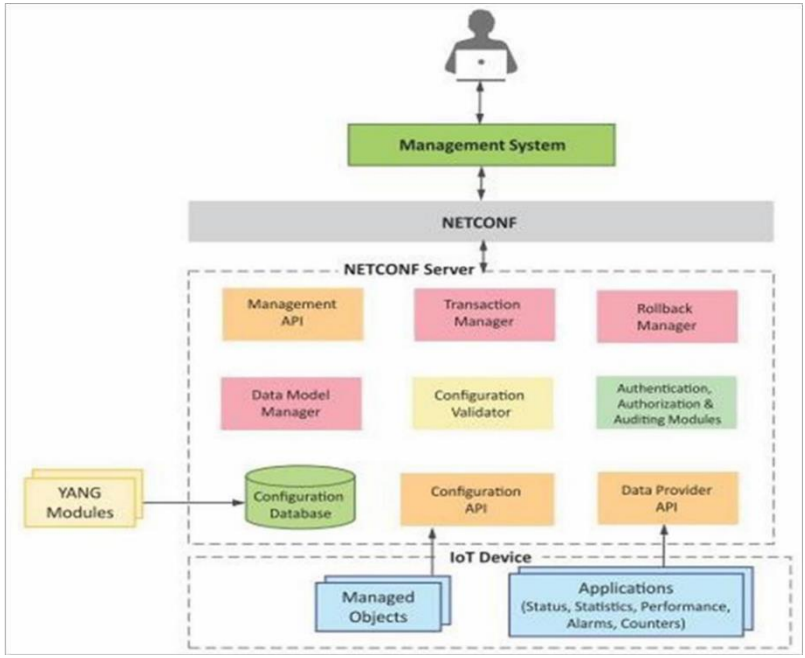
1) **Automating Configuration:** IoT system management capabilities can help in automating the system configuration.
2) **Monitoring Operational & Statistical Data:** Management systems can help in monitoring operational and statistical data of a system. This data can be used for fault diagnosis or prognosis.
3) **Improved Reliability:** A management system that allows validating the system configurations before they are put into effect can help in improving the system reliability.
4) **System Wide Configurations:** For IoT systems that consists of multiple devices or nodes, ensuring system

wide configuration can be critical for the correct functioning of the system.

5) **Multiple System Configurations:** For some systems it may be desirable to have multiple valid configurations which are applied at different times or in certain conditions.
6) **Retrieving & Reusing Configurations:** Management systems which have the capability of retrieving configurations from devices can help in reusing the configurations for other devices of the same type.

IoT Systems Management with NETCONF-YANG

YANG is a data modeling language used to model configuration and state data manipulated by the NETCONF protocol.



The generic approach of IoT device management with NETCONF-YANG. Roles of various components are:

1) Management System
2) Management API
3) Transaction Manager
4) Rollback Manager
5) Data Model Manager
6) Configuration Validator
7) Configuration Database
8) Configuration API
9) Data Provider API

1) **Management System:** The operator uses a management system to send NETCONF messages to configure the IoT device and receives state information and notifications from the device as NETCONF messages.
2) **Management API:** allows management application to start NETCONF sessions.
3) **Transaction Manager:** executes all the NETCONF transactions and ensures that ACID properties hold true for the transactions.
4) **Rollback Manager:** is responsible for generating all the transactions necessary to rollback a current configuration to its original state.
5) **Data Model Manager:** Keeps track of all the YANG data models and the corresponding managed objects. Also keeps track of the applications which provide data for each part of a antimodel.
6) **Configuration Validator:** checks if the resulting configuration after applying a transaction would be a valid configuration.
7) **Configuration Database:** contains both configuration and operational data.
8) **Configuration API :** Using the configuration API the application on the IoT device can be read configuration data from the configuration datastore and write operational data to the operational data store.
9) **Data Provider API:** Applications on the IoT device can register for callbacks for various events using the Data

Provider API. Through the Data Provider API, the applications can report statistics and operational data.

Steps for IoT device Management with NETCONF-YANG

1) Create a YANG model of the system that defines the configuration and state data of the system.
2) Complete the YANG model with the ‗Inctool' which comes with Libnetconf.
3) Fill in the IoT device management code in the Trans API module.
4) Build the callbacks C file to generate the library file.
5) Load the YANG module and the Trans API module into the Netopeer server using Netopeer manager tool.
6) The operator can now connect from the management system to the Netopeer server using the NetopeerCLI.
7) Operator can issue NETCONF commands from the Netopeer CLI. Command can be issued to change the configuration data, get operational data or execute an RPC on the IoT device

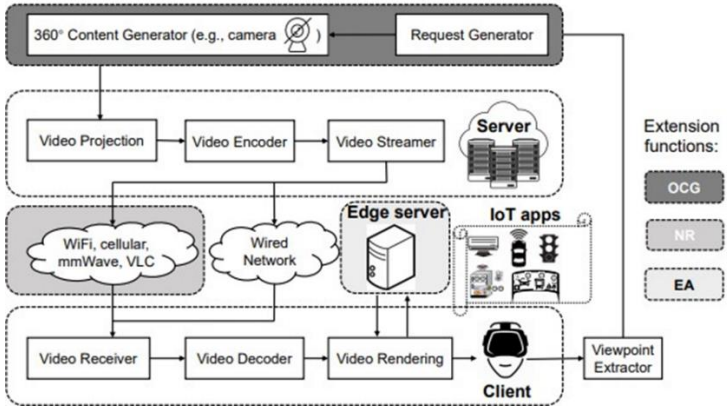
4.3. VR: VIRTUAL REALITY

Introduction to Virtual Reality is the first course in the Virtual Reality Specialization. A learner with no previous experience in Virtual Reality and/or game programming will be able to evaluate existing VR applications, and design, test, and implement their own VR experiences/games using Unity by the end of the specialization.

Imagination is to Technology as Fuel is to Fire. Imagination and purpose together drive technology. It is due to these that technology today is evolving at an exponential rate. Virtual Reality on one hand places the viewer inside a moment or a place, made possible by visual and sound technology that man oeuvres the brain into believing it is somewhere else. It is an experience of a world that does not exist. Sounds cool, right.

Virtual Reality tricks one's mind using computers that allow one to experience and more interestingly, interact with a 3D world. This is made possible by putting on a head-mounted display which sends a form of input tracking. The display is split between the eyes and thus creates a stereoscopic 3D effect with stereo sound to give you a graphic experience. The technology feeds in the images of the objects taken at slightly different angles which creates an impression of depth and solidity. The LCD or OLED panels inside are refracted by lenses completely fills the field of vision with what is to be displayed and experienced. Together with the technology and the input tracking, it creates an immersive and exciting believable world that the computer generates.

VR displays are available in various forms. Ranging from the ones that already contain the display, splitting the feed for each eye using a cable to transfer the feed to the console, to the more affordable ones which depend upon the VR mode and applications on Smartphones. The HTC Vive, the Oculus Rift and Sony PlayStation VR are a few of the head-mounts that use this setup. One can create one's own Virtual Reality Box at home, along with a smartphone-compatible with the VR mode. Irrespective of the use, Virtual Reality produces a set of data that can be used to develop models, communication, training methods and interaction. In simple words, the possibilities are endless.



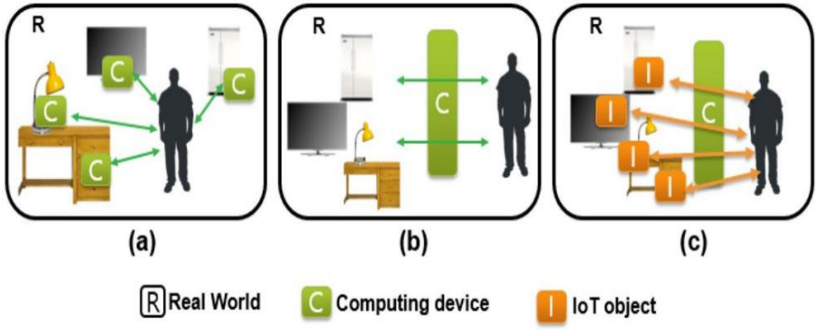
4.4. VR Applications

Automotive industry- VR allows engineers and designers to experiment easily with the look and build of a vehicle before commissioning expensive prototypes. Brands such as BMW and Jaguar Land Rover already use VR to hold early design and engineering reviews to check the visual design and object obscuration of the vehicle - all before any money has been spent on physically manufacturing the parts.

Healthcare- Healthcare in an important application where VR can have a significant impact. Healthcare professionals now use virtual models to prepare themselves for working on real bodies and VR has even been used as pain relief for burn injuries.

Retail-The problem with online shopping is that we can't try on the clothes we want before we buy them, which results in us buying two sizes and sending one back, or ordering one size and praying it fits your shape and size. This could soon change with body-scanning technology in VR, which would allow us to try on clothes in the virtual world to see what they would look like in person.

Tourism-Imagine being able to try your holiday before you buy it. That's exactly what the future could hold. The industry is taking the first steps to enabling you to go on guided virtual tours of hotels, restaurants and tourist landmarks.



Real estate-You can look round properties from the comfort of your [existing] home - no estate agent or sacrificing your weekend needed. This allows people to explore houses online and then only view the ones you're most likely to love in person. Matterport are leading the way and have already scanned thousands of homes using their scanning hardware. Etc.

4.5. Smart Systems

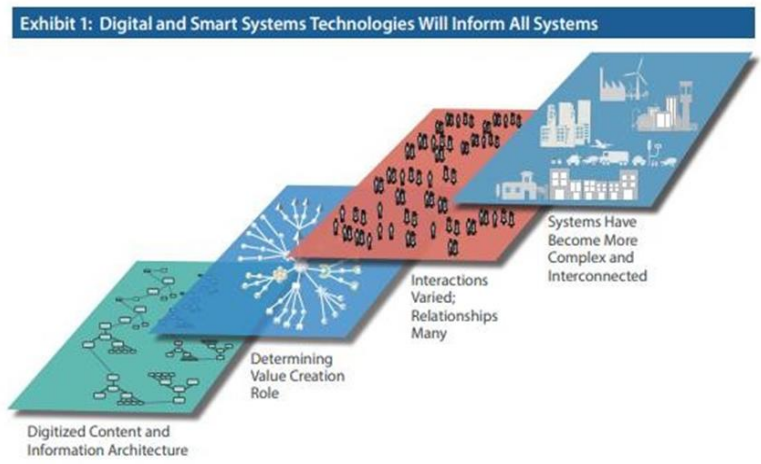
Introduction

For quite a few years now, Harbor Research has focused most of its research and consulting on what we call "Smart Systems"—the convergence of pervasive or embedded computing with the packet-switching "network of networks" called the Internet. These days, many people refer to this phenomenon as "the Internet of Things (IoT)." We prefer "Smart Systems" over other terms in common use because it captures the profound enormity of the phenomenon - something much greater in scope than just machine connectivity.

Whatever we chose to call it -- "Smart Systems" or "Pervasive Computing" or "The Internet of Things" — we are referring to digital microprocessors and sensors embedded in everyday objects. But even this makes too many assumptions about what the smart systems phenomenon will be. Encoded information in physical objects is also smart— even without intrinsic computing ability. Seen in this way, a printed bar code, a house key, or even the pages of a technical manual can have the status of an "information device" on a network.

For that matter, all of these characterizations do not even begin to address the human-machine dimension of collaboration. But very few people are thinking about smart connected systems on that level. Current technologists are operating with outdated models of data, networking and information management that were conceived in the mainframe and client-server eras and cannot serve the needs of a truly connected world. "Smart Systems"

should automatically be understood as "real-time networked information" but it isn't. We have now entered the age when everyday objects will communicate with, and control, other objects over a global data network—24/7/365. It's not "the future," it's now—this year, next year—and thus it is vitally important that business leaders understand this phenomenon, its effects on their business, and what they should do right now to position themselves for opportunities that are literally just around the corner.

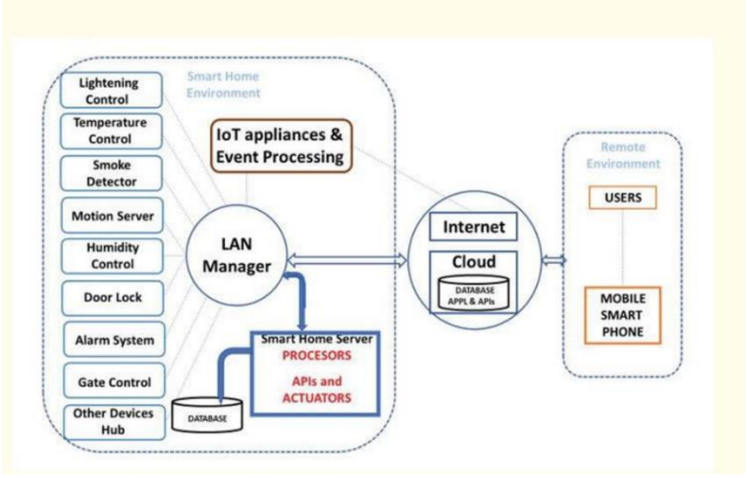


When telephones first came into existence, all calls were routed through switchboards and had to be connected by a live operator. It was long ago forecast that if telephone traffic continued to grow in this way, soon everybody in the world would have to be a switchboard operator. Of course, that did not happen because automation was built into the systems to handle common tasks like connecting calls. We are quickly approaching analogous circumstances in the IoT arena with the proliferation of connected devices. The tools we are working with today to make products "smart" were not designed to handle the diversity of devices, the scope of interactions and the massive volume of data-points generated from devices. Each new device requires too much

customization and development resources just to perform the same basic tasks efficiently and effectively

Peer-to-peer information, social networking and pervasive computing are combining to create new modes of collaboration and decision making. People, information, and technology are becoming more connected, distributed and pervasive enabling the convergence of physical and virtual worlds. Social networking technologies are moving to the enterprise and will be embraced and experienced differently than in the consumer space. Network awareness will include knowledge, people and things. These forces are informing a new trend we call "Smart Systems." In its simplest form, Smart Systems is a new generation of computing systems and information architecture that when combined with artificial intelligence, machine learning and Internet of Things technologies are breaking away from today's information, computing and telecom (ICT) paradigms to enable intelligent real-world physical systems to be integrated onto networks and the data from machines, sensors, video streams, maps, people, news-feeds and more to become an integral part of all information systems. This new paradigm is driving all information systems and, more importantly, their interactions towards real-time, state-based, context-sensitive capabilities that integrate people, processes, physical equipment and knowledge to enable collective awareness and better decision making -- thus, Smart Systems. The foundation of Smart Systems is based on leveraging embedded computing, software and networking technology to deliver smart, remotely monitored goods that will support entirely new modes of customer- device interaction and service delivery. The core platforms that inform Smart Systems combine new innovations in software and information architectures with data collection, aggregation, integration and management tools. These "data" technologies will work together in unprecedented ways to solve more complex business problems than previous generations of computing. Since the beginning of computing there have essentially been three waves of technology and architecture: mainframe computing, personal computing, and network

computing. This next generation of "Smart Systems" technology will add significant new capabilities to computing and network systems.



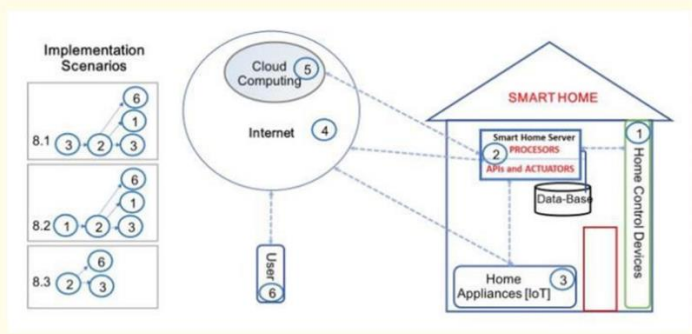


Figure 5.
Advanced smart home implementations chart.

These new capabilities will revolve around realtime situational awareness and automated analysis. As a result, technology moves beyond just proposing task solutions — such as executing a work order or a sales order — to sensing what is happening in the world around it, analyzing that new information for exceptions, risks and alternatives, and taking actions. The three previous waves of technology each have had significant impacts on productivity and

efficiencies; mainframes standardized transactions; personal computing placed processing power into the hands of professionals; and, networked systems enabled business process automation. What is important about this next wave of Smart Systems is the combined impact of these innovation cycles.

EMBEDDED SYSTEM

An embedded system is a combination of computer hardware and software designed for a specific function. Embedded systems may also function within a larger system. The systems can be programmable or have a fixed functionality. Industrial machines, consumer electronics, agricultural and processing industry devices, automobiles, medical equipment, cameras, digital watches, household appliances, airplanes, vending machines and toys, as well as mobile devices, are possible locations for an embedded system.

While embedded systems are computing systems, they can range from having no user interface (UI) -- for example, on devices designed to perform a single task -- to complex graphical user interfaces (GUIs), such as in mobile devices. User interfaces can include buttons, LEDs (light- emitting diodes) and touchscreen sensing. Some systems use remote user interfaces as well.

Markets and Markets, a business-to-business (B2B) research firm, predicted that the embedded market will be worth $116.2 billion by 2025. Chip manufacturers for embedded systems include many well-known technology companies, such as Apple, IBM, Intel and Texas Instruments. The expected growth is partially due to the continued investment in artificial intelligence (AI), mobile computing and the need for chips designed for high-level processing.

Examples of embedded systems

Embedded systems are used in a wide range of technologies across an array of industries. Some examples include:

- **Automobiles.** Modern cars commonly consist of many computers (sometimes as many as 100), or embedded systems, designed to perform different tasks within the vehicle. Some of these systems perform basic utility functions and others provide entertainment or user-facing functions. Some embedded systems in consumer vehicles include cruise control, backup sensors, suspension control, navigation systems and airbag systems.
- **Mobile phones.** These consist of many embedded systems, including GUI software and hardware, operating systems (OSes), cameras, microphones, and USB (Universal Serial Bus) I/O (input/output) modules.
- **Industrial machines.** They can contain embedded systems, like sensors, and can be embedded systems themselves. Industrial machines often have embedded automation systems that perform specific monitoring and control functions.
- **Medical equipment.** These may contain embedded systems like sensors and control mechanisms. Medical equipment, such as industrial machines, also must be very user-friendly so that human health isn't jeopardized by preventable machine mistakes. This means they'll often include a more complex OS and GUI designed for an appropriate UI.

How does an embedded system work?

Embedded systems always function as part of a complete device -- that's what's meant by the term embedded. They are low-cost, low-power-consuming, small computers that are embedded in other mechanical or electrical systems. Generally, they comprise a processor, power supply, and memory and communication ports. Embedded systems use the communication ports to transmit data between the processor and peripheral devices -- often, other embedded systems -- using a communication protocol. The processor interprets this data with the help of minimal software

stored on the memory. The software is usually highly specific to the function that the embedded system serves.

The processor may be a microprocessor or microcontroller. Microcontrollers are simply microprocessors with peripheral interfaces and integrated memory included. Microprocessors use separate integrated circuits for memory and peripherals instead of including them on the chip. Both can be used, but microprocessors typically require more support circuitry than microcontrollers because there is less integrated into the microprocessor. The term system on a chip (SoC) is often used. SoCs include multiple processors and interfaces on a single chip. They are often used for high-volume embedded systems. Some example SoC types are the application-specific integrated circuit (ASIC) and the field-programmable gate array (FPGA).

Often, embedded systems are used in real-time operating environments and use a real-time operating system (RTOS) to communicate with the hardware. Near-real-time approaches are suitable at higher levels of chip capability, defined by designers who have increasingly decided the systems are generally fast enough and the tasks tolerant of slight variations in reaction. In these instances, stripped-down versions of the Linux operating system are commonly deployed, although other OSes have been pared down to run on embedded systems, including Embedded Java and Windows IoT (formerly Windows Embedded).

Characteristics of embedded systems

The main characteristic of embedded systems is that they are task-specific. Additionally, embedded systems can include the following characteristics:

- Typically, consist of hardware, software and firmware;
- Can be embedded in a larger system to perform a specific function, as they are built for specialized tasks within the system, not various tasks;
- Can be either microprocessor-based or microcontroller-

based -- both are integrated circuits that give the system compute power;
- Are often used for sensing and real-time computing in internet of things (iot) devices, which are devices that are internet-connected and do not require a user to operate;
- Can vary in complexity and in function, which affects the type of software, firmware and hardware they use; and
- Are often required to perform their function under a time constraint to keep the larger system functioning properly.

Structure of embedded systems

Embedded systems vary in complexity but, generally, consist of three main elements:

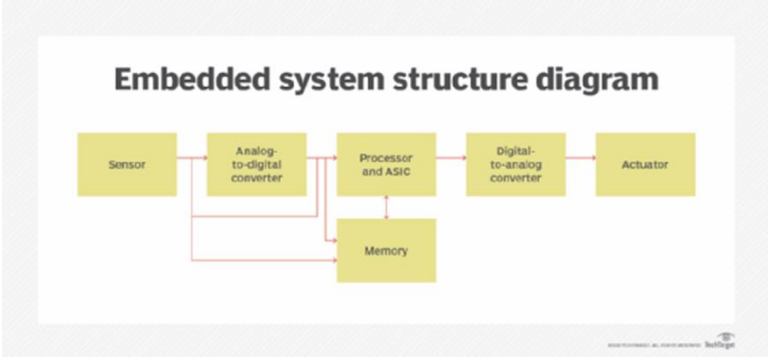
- **Hardware.** The hardware of embedded systems is based around microprocessors and microcontrollers. Microprocessors are very similar to microcontrollers and, typically, refer to a CPU (central processing unit) that is integrated with other basic computing components such as memory chips and digital signal processors (DSPs). Microcontrollers have those components built into one chip.
- **Software and firmware.** Software for embedded systems can vary in complexity. However, industrial-grade microcontrollers and embedded IoT systems usually run very simple software that requires little memory.
- **Real-time operating system.** These are not always included in embedded systems, especially smaller-scale systems. RTOSes define how the system works by supervising the software and setting rules during program execution.

In terms of hardware, a basic embedded system would consist of the following elements:

- **Sensors** convert physical sense data into an electrical signal.

- **Analog-to-digital (A-D) converters** change an analog electrical signal into a digital one.
- **Processors** process digital signals and store them in memory.
- **Digital-to-analog (D-A) converters** change the digital data from the processor into analog data.
- **Actuators** compare actual output to memory-stored output and choose the correct one.

The sensor reads external inputs, the converters make that input readable to the processor, and the processor turns that information into useful output for the embedded system.



A diagram of the basic structure and flow of information in embedded systems.

Types of embedded systems

There are a few basic embedded system types, which differ in their functional requirements. They are:

- **Mobile embedded systems** are small-sized systems that are designed to be portable. Digital cameras are an example of this.
- **Networked embedded systems** are connected to a network to provide output to other systems. Examples include home security systems and point of sale (POS) systems.

- **Standalone embedded systems** are not reliant on a host system. Like any embedded system, they perform a specialized task. However, they do not necessarily belong to a host system, unlike other embedded systems. A calculator or MP3 player is an example of this.
- **Real-time embedded systems** give the required output in a defined time interval. They are often used in medical, industrial and military sectors because they are responsible for time- critical tasks. A traffic control system is an example of this.

Embedded systems can also be categorized by their performance requirements:

- **Small-scale embedded systems** often use no more than an 8-bit microcontroller.
- **Medium-scale embedded systems** use a larger microcontroller (16-32 bit) and often link microcontrollers together.
- **Sophisticated-scale embedded systems** often use several algorithms that result in software and hardware complexities and may require more complex software, a configurable processor and/or a programmable logic array.

There are several common embedded system software architectures, which become necessary as embedded systems grow and become more complex in scale. These include:

- **Simple control loops** call subroutines, which manage a specific part of the hardware or embedded programming.
- **Interrupt controlled systems** have two loops: a main one and a secondary one. Interruptions in the loops trigger tasks.
- **Cooperative multitasking** is essentially a simple control loop located in an application programming interface (API).
- **Preemptive multitasking or multithreading** is often

used with an RTOS and features synchronization and task switching strategies.

Very large-scale integration, or *VLSI*, is a term that describes the complexity of an integrated circuit (IC). VLSI is the process of embedding hundreds of thousands of transistors into a chip, whereas LSI (large-scale integration) microchips contain thousands of transistors, MSI (medium-scale integration) contains hundreds of transistors, and SSI (small-scale integration) contains tens of transistors. ULSI, or ultra-large-scale integration, refers to placing millions of transistors on a chip.

VLSI circuits are common features of embedded systems. Many ICs in embedded systems are VLSIs, and the use of the VLSI acronym has largely fallen out of favor.

Debugging embedded systems

One area where embedded systems part ways with the operating systems and development environments of other larger-scale computers is in the area of debugging. Usually, developers working with desktop computer environments have systems that can run both the code being developed and separate debugger applications that can monitor the embedded system programmers generally cannot, however.

Some programming languages run on microcontrollers with enough efficiency that rudimentary interactive debugging is available directly on the chip. Additionally, processors often have CPU debuggers that can be controlled -- and, thus, control program execution -- via a JTAG or similar debugging port.

In many instances, however, programmers need tools that attach a separate debugging system to the target system via a serial or other port. In this scenario, the programmer can see the source code on the screen of a general-purpose computer, just as would be the case in the debugging of software on a desktop computer. A separate, frequently used approach is to run software on a PC

that emulates the physical chip in software. This is essentially making it possible to debug the performance of the software as if it were running on an actual physical chip.

Broadly speaking, embedded systems have received more attention to testing and debugging because a great number of devices using embedded controls are designed for use, especially in situations where safety and reliability are top priorities.

History of embedded systems

Embedded systems date back to the 1960s. Charles Stark Draper developed an integrated circuit in 1961 to reduce the size and weight of the Apollo Guidance Computer, the digital system installed on the Apollo Command Module and Lunar Module. The first computer to use ICs, it helped astronauts collect real-time flight data.

In 1965, Autonetics, now a part of Boeing, developed the D-17B, the computer used in the Minuteman I missile guidance system. It is widely recognized as the first mass-produced embedded system. When the Minuteman II went into production in 1966, the D-17B was replaced with the NS- 17 missile guidance system, known for its high-volume use of integrated circuits. In 1968, the first embedded system for a vehicle was released; the Volkswagen 1600 used a microprocessor to control its electronic fuel injection system.

By the late 1960s and early 1970s, the price of integrated circuits dropped and usage surged. The first microcontroller was developed by Texas Instruments in 1971. The TMS1000 series, which became commercially available in 1974, contained a 4-bit processor, read-only memory (ROM) and random-access memory (RAM), and it cost around $2 apiece in bulk orders.

Also, in 1971, Intel released what is widely recognized as the first commercially available processor, the 4004. The 4-bit microprocessor was designed for use in calculators and small

electronics, though it required eternal memory and support chips. The 8-bit Intel 8008, released in 1972, had 16 KB of memory; the Intel 8080 followed in 1974 with 64 KB of memory. The 8080's successor, the x86 series, was released in 1978 and is still largely in use today.

In 1987, the first embedded operating system, the real-time VxWorks, was released by Wind River, followed by Microsoft's Windows Embedded CE in 1996. By the late 1990s, the first embedded Linux products began to appear. Today, Linux is used in almost all embedded devices.

Embedded system trends

While some embedded systems can be relatively simple, they are becoming more complex, and more and more of them are now able to either supplant human decision-making or offer capabilities beyond what a human could provide. For instance, some aviation systems, including those used in drones, are able to integrate sensor data and act upon that information faster than a human could, permitting new kinds of operating features.

The embedded system is expected to continue growing rapidly, driven in large part by the internet of things. Expanding IoT applications, such as wearables, drones, smart homes, smart buildings, video surveillance, 3D printers and smart transportation, are expected to fuel embedded system growth.

5

History of Centralized, De-Centralized and Distributed Computing

5.1. Centralized Systems

Centralized systems are systems that use client/server architecture where one or more client nodes are directly connected to a central server. This is the most commonly used type of system in many organizations where a client sends a request to a company server and receives the response.

In a centralized system, all users are connected to a central network owner or "server". The central owner stores data, which other users can access, and also user information. This user information may include user profiles, user-generated content, and more. A centralized system is easy to set up and can be developed quickly.

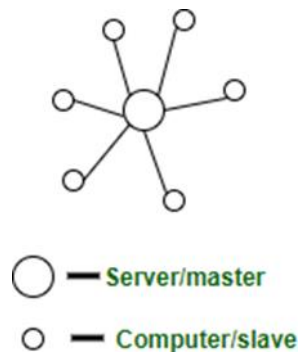


Figure – Centralized system visualization

But this system has an important limitation. If the server crashes, the system no longer works properly and users cannot access the data. Because a centralized system needs a central owner to connect all the other users and devices, the availability of the network depends on this owner. Add to that the obvious security concerns that arise when one owner stores (and can access) user data, and it's easy to understand why centralized systems are no longer the first choice for many organizations.

Example –

Wikipedia. Consider a massive server to which we send our requests and the server responds with the article that we requested. Suppose we enter the search term 'junk food' in the Wikipedia search bar. This search term is sent as a request to the Wikipedia servers (mostly located in Virginia, U.S.A) which then responds back with the articles based on relevance. In this situation, we are the client node, Wikipedia servers are the central server.

Characteristics of Centralized System –

- **Presence of a global clock:** As the entire system consists of a central node (a server/ a master) and many client nodes(a computer/ a slave), all client nodes sync up with the global clock(the clock of the central node).
- **One single central unit:** One single central unit which serves/coordinates all the other nodes in the system.
- **Dependent failure of components:** Central node failure causes the entire system to fail. This makes sense because when the server is down, no other entity is there to send/receive responses/requests.

Scaling –

Only vertical scaling on the central server is possible. Horizontal scaling will contradict the single central unit characteristic of this system of a single central entity.

Components of Centralized System –

Components of Centralized System are,

- Node (Computer, Mobile, etc.).
- Server.
- Communication link (Cables, Wi-Fi, etc.).

Architecture of Centralized System –

Client-Server architecture. The central node that serves the other nodes in the system is the server node and all the other nodes are the client nodes.

Limitations of Centralized System –

- Can't scale up vertically after a certain limit – After a limit, even if you increase the hardware and software capabilities of the server node, the performance will not increase appreciably leading to a cost/benefit ratio < 1.
- Bottlenecks can appear when the traffic spikes – as the server can only have a finite number of open ports to which can listen to connections from client nodes. So, when high traffic occurs like a shopping sale, the server can essentially suffer a Denial-of-Service attack or Distributed Denial-of-Service attack.

Advantages of Centralized System –

- Easy to physically secure. It is easy to secure and service the server and client nodes by virtue of their location
- Smooth and elegant personal experience – A client has a dedicated system which he uses(for example, a personal computer) and the company has a similar system which can be modified to suit custom needs
- Dedicated resources (memory, CPU cores, etc)
- More cost-efficient for small systems up to a certain limit – As the central systems take fewer funds to set up, they have an edge when small systems have to be built

- Quick updates are possible – Only one machine to update.
- Easy detachment of a node from the system. Just remove the connection of the client node from the server and voila! Node detached.

Disadvantages of Centralized System –

- Highly dependent on the network connectivity – The system can fail if the nodes lose connectivity as there is only one central node.
- No graceful degradation of the system – abrupt failure of the entire system
- Less possibility of data backup. If the server node fails and there is no backup, you lose the data straight away
- Difficult server maintenance – There is only one server node and due to availability reasons, it is inefficient and unprofessional to take the server down for maintenance. So, updates have to be done on- the-fly(hot updates) which is difficult and the system could break.

Applications of Centralized System –

- Application development – Very easy to set up a central server and send client requests. Modern technology these days do come with default test servers which can be launched with a couple of commands. For example, Express server, Django server.
- Data analysis – Easy to do data analysis when all the data is in one place and available for analysis
- Personal computing.

Use Cases –

- Centralized databases – all the data in one server for use.
- Single-player games like Need For Speed, GTA Vice City – an entire game in one system(commonly, a Personal Computer)
- Application development by deploying test servers

leading to easy debugging, easy deployment, easy simulation

- Personal Computers

Organizations Using – National Informatics Center (India), IBM

5.2. Decentralized Systems

These are other types of systems that have been gaining a lot of popularity, primarily because of the massive hype of Bitcoin. Now many organizations are trying to find the application of such systems.

In decentralized systems, every node makes its own decision. The final behavior of the system is the aggregate of the decisions of the individual nodes. Note that there is no single entity that receives and responds to the request.

As its name implies, decentralized systems don't have one central owner. Instead, they use multiple central owners, each of which usually stores a copy of the resources users can access.

A decentralized system can be just as vulnerable to crashes as a centralized one. However, it is by design more tolerant to faults. That's because when one or more central owners or servers fail, the others can continue to provide data access to users.

Resources remain active if at least one of the central servers continue to operate. Usually, this means that system owners can repair faulty servers and address any other problems while the system itself continues to run as usual.

Server crashes in a decentralized system may affect the performance and limit access to some data. But in terms of overall system uptime, this system offers a big improvement over a centralized system.

Another advantage of this design is that the access time to the data is often faster. That's because owners can create nodes in different

regions or areas where user activity is high.

However, decentralized systems are still prone to the same security and privacy risks to users as centralized systems. And while their fault tolerance is higher, this comes at a price. Maintaining a decentralized system is usually more expensive.

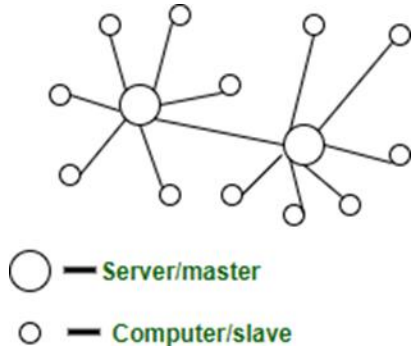


Figure – Decentralized system visualization

Example –

Bitcoin. Let's take Bitcoin for example because it is the most popular use case of decentralized systems. No single entity/organization owns the bitcoin network. The network is a sum of all the nodes who talk to each other for maintaining the amount of bitcoin every account holder has.

Characteristics of Decentralized System –

- **Lack of a global clock:** Every node is independent of each other and hence, has different clocks that they run and follow.
- **Multiple central units (Computers/Nodes/Servers):** More than one central unit which can listen for connections from other nodes
- **Dependent failure of components:** one central node failure causes a part of the system to fail; not the whole system

Scaling –

Vertical scaling is possible. Each node can add resources (hardware, software) to itself to increase the performance leading to an increase in the performance of the entire system.

Components –

Components of Decentralized System are,

- Node (Computer, Mobile, etc.)
- Communication link (Cables, Wi-Fi, etc.)

Architecture of Decentralized System –

- Peer-to-peer architecture – all nodes are peers of each other. No one node has supremacy over other nodes
- Master-slave architecture – one node can become a master by voting and help in coordinating of a part of the system but this does not mean the node has supremacy over the other node which it is coordinating

Limitations of Decentralized System –

- May lead to the problem of coordination at the enterprise level – When every node is the owner of its own behavior, its difficult to achieve collective tasks
- Not suitable for small systems – Not beneficial to build and operate small decentralized systems because of the low cost/benefit ratio
- No way to regulate a node on the system – no superior node overseeing the behavior of subordinate nodes

Advantages of Decentralized System –

- Minimal problem of performance bottlenecks occurring – The entire load gets balanced on all the nodes; leading to minimal to no bottleneck situations
- High availability – Some nodes(computers, mobiles,

servers) are always available/online for work, leading to high availability
- More autonomy and control over resources – As each node controls its own behavior, it has better autonomy leading to more control over resources

Disadvantages of Decentralized System –

- Difficult to achieve global big tasks – No chain of command to command others to perform certain tasks
- No regulatory oversight
- Difficult to know which node failed – Each node must be pinged for availability checking and partitioning of work has to be done to actually find out which node failed by checking the expected output with what the node generated
- Difficult to know which node responded – When a request is served by a decentralized system, the request is actually served by one of the nodes in the system but it is actually difficult to find out which node indeed served the request.

Applications of Decentralized System –

- Private networks – peer nodes joined with each other to make a private network.
- Cryptocurrency – Nodes joined to become a part of a system in which digital currency is exchanged without any trace and location of who sent what to whom. However, in bitcoin, we can see the public address and amount of bitcoin transferred, but those public addresses are mutable and hence difficult to trace.

Use Cases –

- Blockchain
- Decentralized databases – Entire databases split into parts and distributed to different nodes for storage and

use. For example, records with names starting from 'A' to 'K' in one node, 'L' to 'N' in the second node, and 'O' to 'Z' in the third node

- Cryptocurrency

Organizations Using – Bitcoin, Tor network

5.3. Distributed Systems

In decentralized systems, every node makes its own decision. The final behaviour of the system is the aggregate of the decisions of the individual nodes. Note that there is no single entity that receives and responds to the request.

A distributed system is similar to a decentralized one in that it doesn't have a single central owner. But going a step further, it eliminates centralization. In a distributed system, users have equal access to data, though user privileges can be enabled when needed. The best example of a vast, distributed system is the internet itself.

The distributed system enables users to share ownership of the data. Hardware and software resources are also allocated between users, which in some cases may improve the performance of the system. A distributed system is safe from the independent failure of components, which can improve its uptime considerably.

Distributed systems have evolved as a result of the limitations of the other systems. With increasing security, data storage, and privacy concerns, and the constant need for improving performance, distributed systems are the natural choice for many organizations.

It's no surprise then those technologies using the distributed system – most notably the blockchain – are changing many industries.

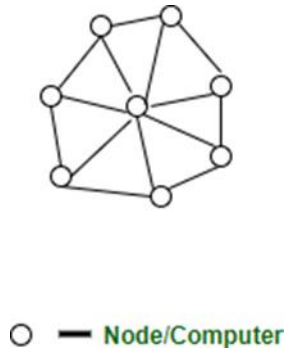


Figure – Distributed system visualization

Example –

Google search system. Each request is worked upon by hundreds of computers which crawl the web and return the relevant results. To the user, Google appears to be one system, but it actually is multiple computers working together to accomplish one single task (return the results to the search query).

Characteristics of Distributed System – :

- **Concurrency of components:** Nodes apply consensus protocols to agree on the same values/transactions/commands/logs.
- **Lack of a global clock:** All nodes maintain their own clock.
- **Independent failure of components:** In a distributed system, nodes fail independently without having a significant effect on the entire system. If one node fails, the entire system sans the failed node continues to work.

Scaling –

Horizontal and vertical scaling is possible.

Components of Distributed System – Components of Distributed System are,

- Node (Computer, Mobile, etc.)
- Communication link (Cables, Wi-Fi, etc.)

Architecture of Distributed System –

- Peer-to-peer – all nodes are peers of each other and work towards a common goal
- Client-server – some nodes become server nodes for the role of coordinator, arbiter, etc.
- N-tier architecture – different parts of an application are distributed in different nodes of the systems and these nodes work together to function as an application for the user/client

Limitations of Distributed System –

- Difficult to design and debug algorithms for the system. These algorithms are difficult because of the absence of a common clock; so no temporal ordering of commands/logs can take place. Nodes can have different latencies which have to be kept in mind while designing such algorithms. The complexity increases with the increase in the number of nodes. Visit this link for more information
- No common clock causes difficulty in the temporal ordering of events/transactions
- Difficult for a node to get the global view of the system and hence take informed decisions based on the state of other nodes in the system

Advantages of Distributed System –

- Low latency than a centralized system – Distributed systems have low latency because of high geographical spread, hence leading to less time to get a response

Disadvantages of Distributed System –

- Difficult to achieve consensus

- The conventional way of logging events by absolute time they occur is not possible here

Applications of Distributed System –

- Cluster computing – a technique in which many computers are coupled together to work so that they achieve global goals. The computer cluster acts as if they were a single computer
- Grid computing – All the resources are pooled together for sharing in this kind of computing turning the systems into a powerful supercomputer; essentially.

Use Cases –

- SOA-based systems
- Multiplayer online games

Organizations Using –Apple, Google, Facebook.

Centralized vs Decentralized vs Distributed Systems Comparison

The following head-to-head comparison focuses on key points like fault tolerance, maintenance, scalability, development, and evolution. For each of these, we are using simple ratings like low, moderate, and high.

💪 **Fault tolerance:**

- Low: Centralized systems
- Moderate: Decentralized systems
- High: Distributed systems

🔧 **Maintenance:**

- Low: Centralized systems
- Moderate: Decentralized systems
- High: Distributed systems

🚀 **Scalability:**

- Low: Centralized systems
- Moderate: Decentralized systems
- High: Distributed systems

💰 **Development:**

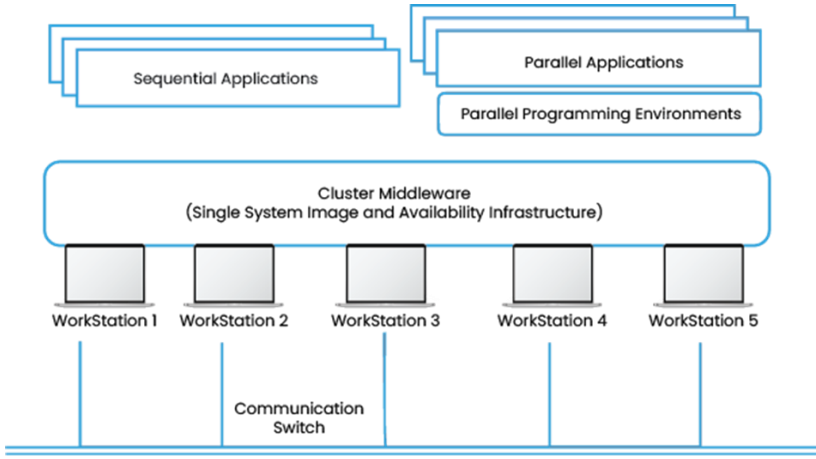
- Moderate: Decentralized systems, Distributed systems
- High: Centralized systems

☑ **Evolution:**

- Low: Centralized systems
- High: Decentralized systems, Distributed systems

5.4. Cluster Computing

Defining Cluster Computing

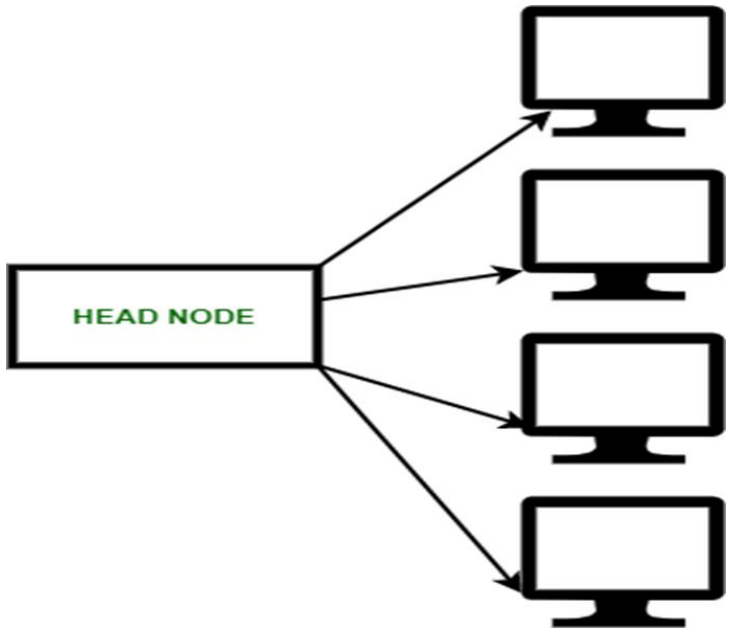


In the most basic form, Cluster computing depicts a system that consists of two or more computers or systems, often known as nodes. These nodes work together for executing applications and performing other tasks. The users using nodes have an apprehension that only a single system responds to them, creating

an illusion of a single resource called virtual machines. This concept is defined as the transparency of the system. Other essential features that are required for constructing such platforms include- **reliability, load balancing, and performance.**

5.5. Cluster Computing Architecture

Introduction : Cluster computing is a collection of tightly or loosely connected computers that work together so that they act as a single entity. The connected computers execute operations all together thus creating the idea of a single system. The clusters are generally connected through fast local area networks (LANs)

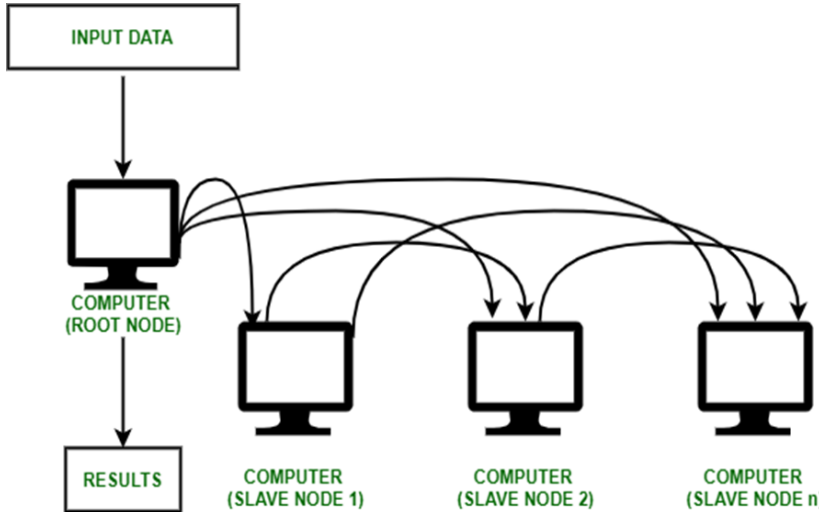


Cluster Computing

Why is Cluster Computing important?

1) Cluster computing gives a relatively inexpensive, unconventional to the large server or mainframe computer solutions.

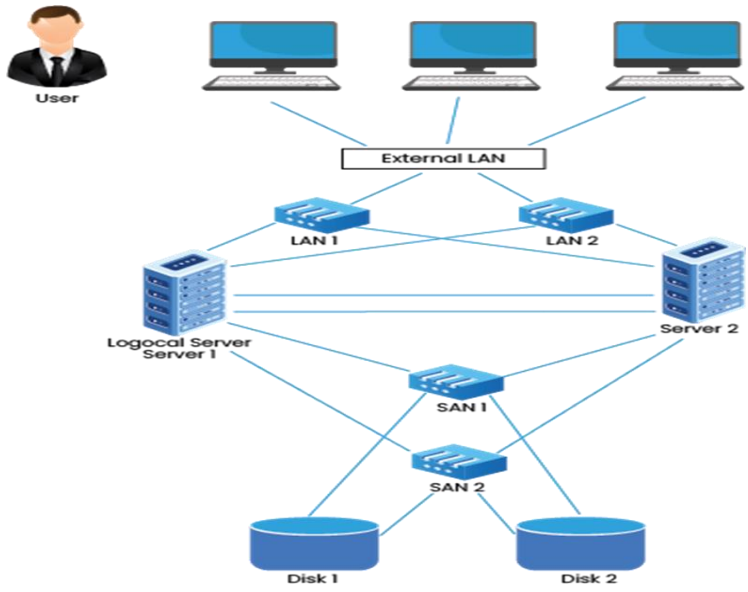
2) It resolves the demand for content criticality and process services in a faster way.
3) Many organizations and IT companies are implementing cluster computing to augment their scalability, availability, processing speed and resource management at economic prices.
4) It ensures that computational power is always available.
5) It provides a single general strategy for the implementation and application of parallel high-performance systems independent of certain hardware vendors and their product decisions.



A Simple Cluster Computing Layout

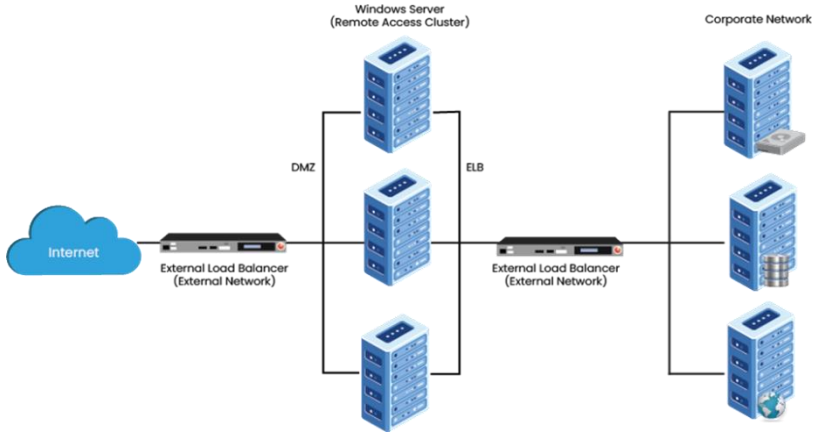
Types of Cluster

High Availability (HA) and Failover Clusters – These cluster models create availability of services and resources in an uninterrupted method using the system's implicit redundancy. The basic idea in this form of Cluster is that if a node fails, then applications and services can be made available to other nodes. These types of Clusters serve as the base for critical missions, mails, files, and application servers.



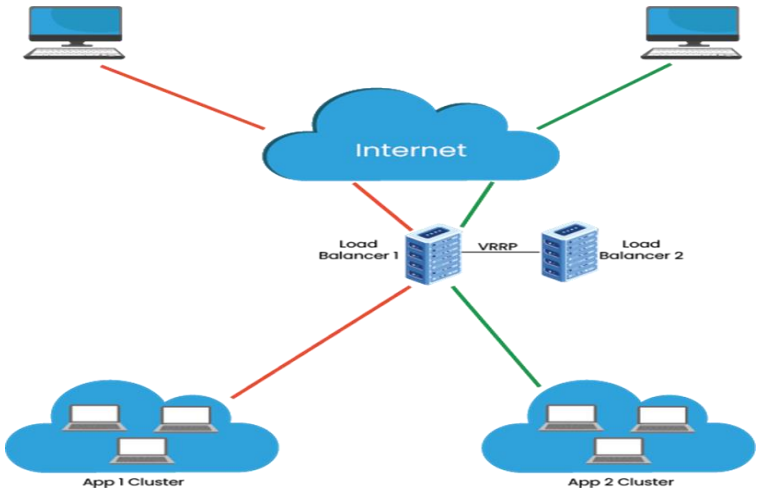
HA Cluster

Load Balancing Clusters – This Cluster distributes all the incoming traffic/requests for resources from nodes that run the same programs and machines. In this Cluster model, all the nodes are responsible for tracking orders, and if a node fails, then the requests are distributed amongst all the nodes available. Such a solution is usually used on web server farms.

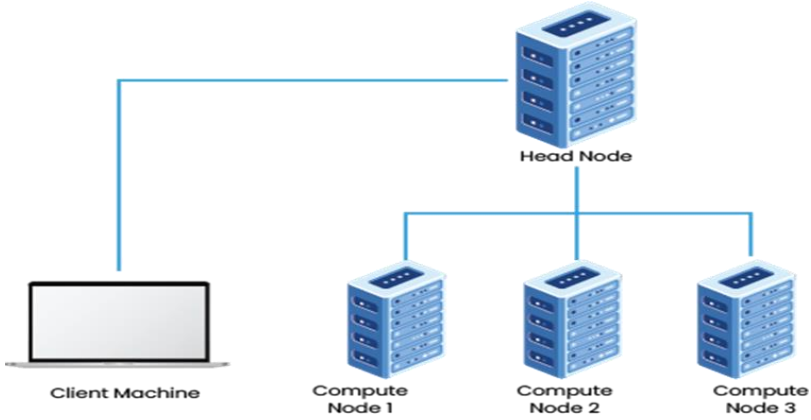


Load Balancing Cluster

HA & Load Balancing Clusters – This cluster model combines both clusters' features, resulting in increased availability and scalability of services and resources. This type of Clusters is commonly used for email, web, news, and FTP servers.



HA & Load Balancing Cluster



Distributed & Parallel Processing Clusters – This Cluster model enhances availability and performance for applications that have

large computational tasks. A large computational task gets divided into smaller tasks and distributed across the stations. Such Clusters are usually used for scientific computing or financial analysis that require high processing power.

Distributed & Parallel Processing Cluster

Need for Cluster Computing

Clusters or combinations of clusters are used when the content is critical, and services need to be available. Internet Service Providers or ISPs and e-commerce sites demand high availability and load balancing in a scalable manner. The parallel clusters are being extensively used in the film industry as they need high-quality graphics and animations. Talking about the Beowulf clusters, they are dominantly used in science, engineering, and finance to perform various critical projects. Researchers, organizations, and businesses use clusters to demand enhanced scalability, resource management, availability, and processing at affordable prices.

Benefits of Cluster Computing

Cluster computing offers a wide array of benefits. Some of these include the following.

Cost-Effectiveness – Compared with the mainframe systems, cluster computing is considered to be much more cost-effective. These computing systems offer enhanced performance with respect to the mainframe computer devices.

Processing Speed – The processing speed of cluster computing is justified with that of the mainframe systems and other supercomputers present in the world.

Expandability – Scalability and expandability are another set of advantages that cluster computing offers. Cluster computing represents an opportunity for adding any number of additional resources and systems to the existing computing network.

Increased Resource Availability – Availability plays a vital role in cluster computing systems. Failure of any connected active node can be easily passed on to other active nodes on the server, ensuring high availability.

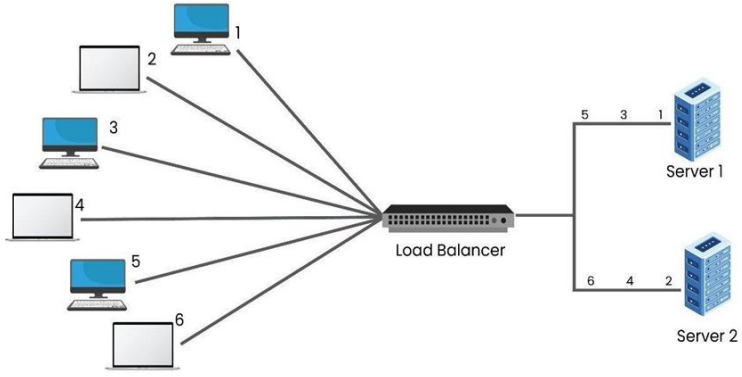
Improved Flexibility – In cluster computing, superior specifications can be upgraded and extended by adding newer nodes to the existing server.

Cluster Computing Load Balancing Algorithms

Load balancing is way more than a simple redirection of client traffic to other servers. To have the proper implementation, one needs equipment with characteristics like- permanent communication check, server verification, and redundancy. These characteristics are essential for supporting traffic scalability on networking and subsequently eliminating the bottleneck or single-point failure.

Some of the common load balancing algorithms in Cluster computing are,

Round Robin- In Cluster computing, the Round Robin algorithm is the most commonly used one. It can be easily implemented and understood. This algorithm is best known for clusters having similar specifications. Users can opt for different algorithms in case servers have other specifications.

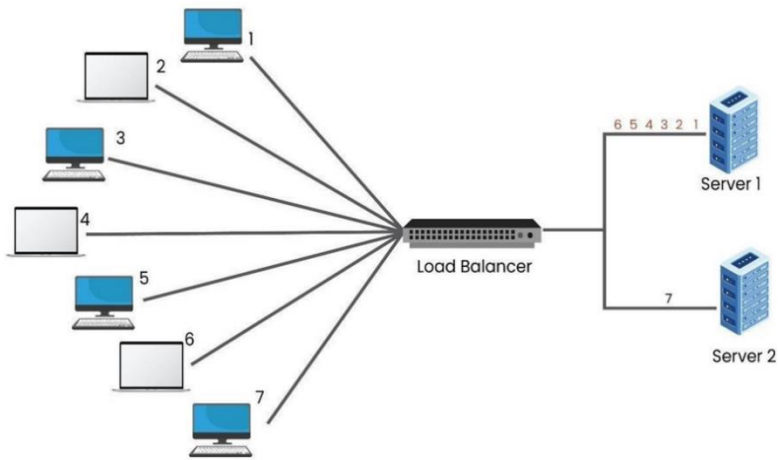


Round Robin Algorithm

In the above diagram, there are two servers awaiting requests behind the load balancer. Upon the arrival of the first request, the load balancer will forward the request to **Server 1** and after getting the second request, it is allotted to **Server 2.** As there are only two servers in the network, the next request will again be forwarded to **Server 1**—this way, the requests are sent to both the servers in a cyclic format.

It is important to note that the servers aren't differentiated by the load balancer in the Round Robin algorithm, resulting in equal distribution of requests. If one of the servers has different specifications and workloads, then the higher workload will be directly affected by receiving cyclic requests.

Weighted Round Robin – The Weighted Round Robin algorithm is much like the Round Robin, however, with a slight change. In this algorithm, the server having higher specifications will be allotted more number of requests. So, how does the load balancer know which server has a higher specification? While setting up the load balancer, users might assign 'weight' to each node. The node having higher specifications will be allotted a higher weight.

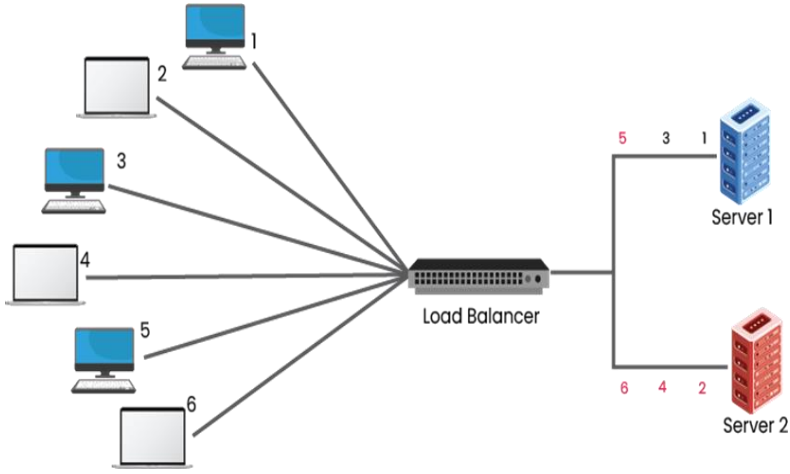


Weighted Round Robin Algorithm

In the above diagram, if **Server 1** has a capacity 6 times more than that of Server 2, the user must assign a weight of 6 to Server 1 and 1 to **Server 2.** After receiving the load balancer's requests, the first-6 will be assigned to Server 1, whereas the 7th request will be allotted to Server 2. In the case of more clients coming in, the same sequence is going to be followed.

Capacity may not be the sole criteria for selecting the Weighted Round Robin algorithm. It can be used if one of the servers in the network has a lower number of connections than using an equally capable server handling critical applications, whose overloading is not a feasible option.

Least Connections – The Least Connections Algorithm considers the number of connections that each server currently has. Whenever there is a connection request, the load balancer determines which of the servers has the least number of connections and then the new connection will be assigned to that server.

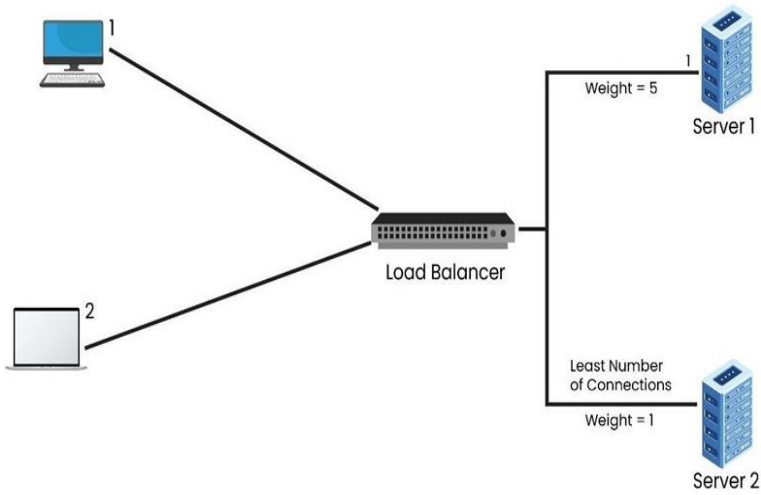


Least Connections Algorithm

There can be situations when two clusters with the same set of

specifications and servers get overloaded much faster. A possible reason could be clients opting for ***Server 2*** *to remain connected for longer than* ***Server 1.*** *This results in total connections in Server 1 to increase, while Server 1 remains the same, causing the resources of Server 2 to run out faster.*

Weighted Least Connections- In this algorithm, a 'weight' is introduced based on each server's capacities. Similar to the Weighted Round Robin, the user needs to define each of the servers. The load balancer implementing the Least Connections algorithm considers two things– the **weights of each server** and the **number of clients** that are attached to each server.



Weighted Least Connections

Random – In this algorithm, clients and servers are mapped by random, i.e., using a random number generator. In case the load balancer has a huge request load, then the Random algorithm distributes requests evenly to all the servers. Like Round Robin, the Random algorithm works well with servers having similar configurations.

5.6. GRID COMPUTING

Grid Computing can be defined as a network of computers working together to perform a task that would rather be difficult for a single machine. All machines on that network work under the same protocol to act as a virtual supercomputer. The task that they work on may include analyzing huge datasets or simulating situations that require high computing power. Computers on the network contribute resources like processing power and storage capacity to the network.

Grid Computing is a subset of distributed computing, where a virtual supercomputer comprises machines on a network connected by some bus, mostly Ethernet or sometimes the Internet. It can also be seen as a form of Parallel Computing where instead of many CPU cores on a single machine, it contains multiple cores spread across various locations. The concept of grid computing isn't new, but it is not yet perfected as there are no standard rules and protocols established and accepted by people.

Working:

A Grid computing network mainly consists of these three types of machines

1. **Control Node:**

A computer, usually a server or a group of servers which administrates the whole network and keeps the account of the resources in the network pool.

2. **Provider:**

The computer contributes its resources to the network resource pool.

3. **User:**

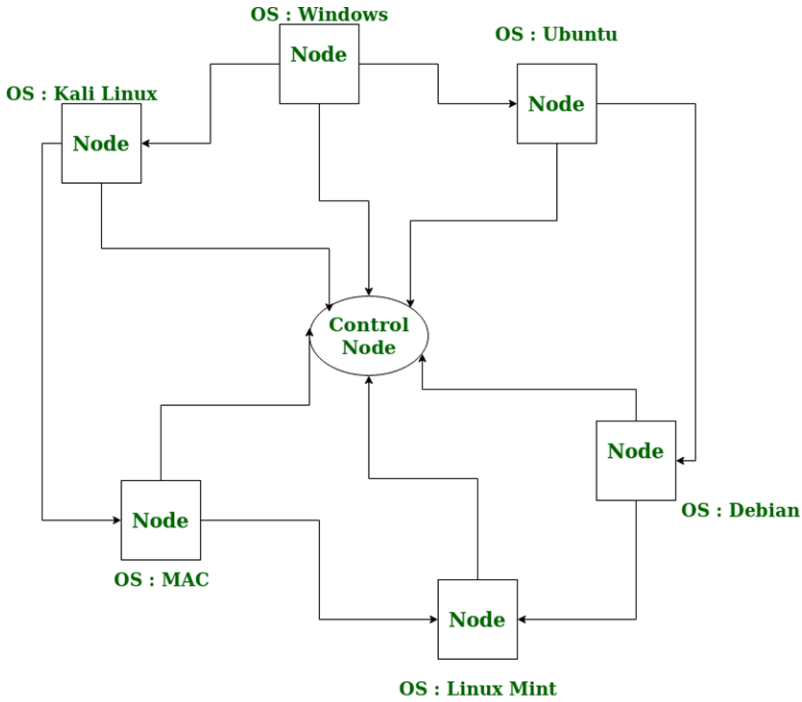
The computer that uses the resources on the network.

When a computer makes a request for resources to the control node, the control node gives the user access to the resources available on the network. When it is not in use it should ideally contribute its resources to the network. Hence a normal computer on the node can swing in between being a user or a provider based on its needs. The nodes may consist of machines with similar platforms using the same OS called homogeneous networks, else machines with different platforms running on various different OSs called heterogeneous networks. This is the distinguishing part of grid computing from other distributed computing architectures.

For controlling the network and its resources a software/networking protocol is used generally known as Middleware. This is responsible for administrating the network and the control nodes are merely its executors. As a grid computing system should use only unused resources of a computer, it is the job of the control node that any provider is not overloaded with tasks.

Another job of the middleware is to authorize any process that is being executed on the network. In a grid computing system, a provider gives permission to the user to run anything on its computer, hence it is a huge security threat for the network. Hence a middleware should ensure that there is no unwanted task being executed on the network.

The meaning of the term Grid Computing has changed over the years, according to "The Grid: Blueprint for a new computing infrastructure" by Ian Foster and Carl Kesselman published in 1999, the idea was to consume computing power like electricity is consumed from a power grid. This idea is similar to the current concept of cloud computing, whereas now grid computing is viewed as a distributed collaborative network. Currently, grid computing is being used in various institutions to solve a lot of mathematical, analytical, and physics problems.



Topology in Grid Computing

Types of Grids

There are various types of grid computing. Let us take a look at them in brief:

- **Computational grid:** This is a type of grid that acts as a mediator of many computers in a given network to solve one single problem at a time.
- **Data grid:** The grid that deals with the sharing and managing the distributed data in a controlled manner is term as a data grid.
- **Collaborative grid:** Such types of grids helps in solving collective problems.
- **Manuscript grid**: These are the type of grids that work when the task is either in the form of images or continuous blocks.

- **Modular grid:** Modular grids are useful when volume does not provide a sole solution to the problems.

Advantages of Grid Computing:

1) It is not centralized, as there are no servers required, except the control node which is just used for controlling and not for processing.
2) Multiple heterogeneous machines i.e. machines with different Operating Systems can use a single grid computing network.
3) Tasks can be performed parallelly across various physical locations and the users don't have to pay for them (with money).

Disadvantages of Grid Computing:

1) The software of the grid is still in the involution stage.
2) A super-fast interconnect between computers resources is the need of hour.
3) Licensing across many servers may make it prohibitive for some applications.
4) Many groups are reluctant with sharing resources.

Uses of Grid Computing

Grid computing plays a very important role when a task requires more than one expert to complete the job. After working individually on their area of expertise, the experts can collect the data, note down their findings, and gather them to take out the final results. Here are a few grid computing examples where grid computing is used:

1) **Movie**: The film industry has been using the process of grid computing for a long time now. Grid computing not only helps in special effects in the films but also ensures faster production.
2) **Gaming**: The gaming industry also uses grid computing techniques in applications like in-game cut scenes,

multiplayer game hosting, and much more.

3) **Life sciences:** Life sciences make use of grid computing for data calculation and analysis.
4) **Engineering and design:** Grid computing has been very useful in engineering and designing in analyzing real-time data, experimenting with new models and methods, and verifying the existing models.
5) **Government**: Government has been using the procedures involved in grid computing for various defence and safety reasons of the country.
6) **Supercomputing**: It is another such area where grid computing has performed exceptionally well and flourishes even more. Various fields where grid computing plays an important role are:
7) Distributed Supercomputing
8) High-throughput Supercomputing
9) On-demand Supercomputing
10) Data-intensive Supercomputing
11) Collaborative Supercomputing

5.7. Introduction To Cloud Computing- Cloud Issues and Challenges

The impact of cloud computing on industry and end users would be difficult to overstate: many aspects of everyday life have been transformed by the omnipresence of software that runs on cloud networks. By leveraging cloud computing, startups and businesses are able to optimize costs and increase their offerings without purchasing and managing all the hardware and software. Independent developers are empowered to launch globally-available apps and online services. Researchers can share and analyze data at scales once reserved only for highly-funded projects. And internet users can quickly access software and storage to create, share, and store digital media in quantities that extend far beyond the computing capacity of their personal devices.

Despite the growing presence of cloud computing, its details remain obscure to many. What exactly is the cloud, how does one use it, and what are its benefits for businesses, developers, researchers, government, healthcare practitioners, and students? In this conceptual article, we'll provide a general overview of cloud computing, its history, delivery models, offerings, and risks.

By the end of this article, you should have an understanding of how the cloud can help support business, research, education, and community infrastructure and how to get started using the cloud for your own projects.

What is Cloud Computing?

Cloud computing is the delivery of computing resources *as a service*, meaning that the resources are owned and managed by the cloud provider rather than the end user. Those resources may include anything from browser-based software applications (such as Tik Tok or Netflix), third party data storage for photos and other digital media (such as iCloud or Dropbox), or third-party servers used to support the computing infrastructure of a business, research, or personal project.

Before the broad proliferation of cloud computing, businesses and general computer users typically had to buy and maintain the software and hardware that they wished to use. With the growing availability of cloud- based applications, storage, services, and machines, businesses and consumers now have access to a wealth of on-demand computing resources as internet-accessed services. Shifting from on-premise software and hardware to networked remote and distributed resources means cloud users no longer have to invest the labor, capital, or expertise required for buying and maintaining these computing resources themselves. This unprecedented access to computing resources has given rise to a new wave of cloud-based businesses, changed IT practices across industries, and transformed many everyday computer-assisted practices. With the cloud, individuals can now work with

colleagues over video meetings and other collaborative platforms, access entertainment and educational content on demand, communicate with household appliances, hail a cab with a mobile device, and rent a vacation room in someone's house.

Defining Cloud Computing

The National Institute of Standards and Technology (NIST), a non-regulatory agency of the United States Department of Commerce with a mission to advance innovation, defines cloud computing as:

"a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction."

NIST lists the following as the five essential characteristics of cloud computing:

- **On-demand self-service**: Cloud resources can be accessed or provisioned without human interaction. With this model, consumers can gain immediate access to cloud services upon signup. Organizations can also create mechanisms for allowing employees, customers, or partners to access internal cloud services on demand according to predetermined logics without needing to go through IT services.
- **Broad network access:** Users can access cloud services and resources through any device and in any networked location provided that they have permission.
- **Resource pooling**: Cloud provider resources are shared by multiple tenants while keeping the data of individual clients hidden from other clients.
- **Rapid elasticity:** Unlike on-premise hardware and software, cloud computing resources can be rapidly increased, decreased, or otherwise modified based on the

cloud user's changing needs.

- **Measured service:** Usage of cloud resources is metered so that businesses and other cloud users need only pay for the resources they use in any given billing cycle.

These characteristics offer a wide variety of transformative opportunities for businesses and individuals alike, which we'll discuss later in the section Benefits of Cloud Computing. To gain some additional context, let's briefly review the emergence of cloud computing.

History of Cloud Computing

Many aspects of cloud computing can be traced as far back as the 1950s, when universities and companies rented out computation time on mainframe computers. At the time, renting was one of the only ways to access computing resources as computing technology was too large and expensive to be owned or managed by individuals. By the 1960s, computer scientists like John McCarthy of Stanford University and J.C.R Licklider of The U.S. Department of Defense Advanced Research Projects Agency (ARPA) began proposing ideas that anticipated some of the major features of cloud computing today, such as the conceptualization of computing as a public utility and the possibility of a network of computers that would allow people to access data and programs from anywhere in the world.

Cloud computing, however, didn't become a mainstream reality and a popular term until the first decade of the 21st century. This decade saw the launch of cloud services like Amazon's Elastic Compute (EC2) and Simple Storage Service (S3) in 2006, Heroku in 2007, Google Cloud Platform in 2008, Alibaba Cloud in 2009, Windows Azure (now Microsoft Azure) in 2010, IBM's Smart Cloud in 2011, and Digital Ocean in 2011. These services allowed existing businesses to optimize costs by migrating their in-house IT infrastructure to cloud-based resources and provided independent developers and small developer teams resources for creating and deploying apps. Cloud-based applications, known

as Software as a Service (SasS) — which we'll discuss in greater detail in the Cloud Delivery Models section — also became popular during this time period. Unlike on-premise software, or software that users need to physically install and maintain on their machines, SaaS increased the availability of applications by allowing users to access them from a variety of devices on demand.

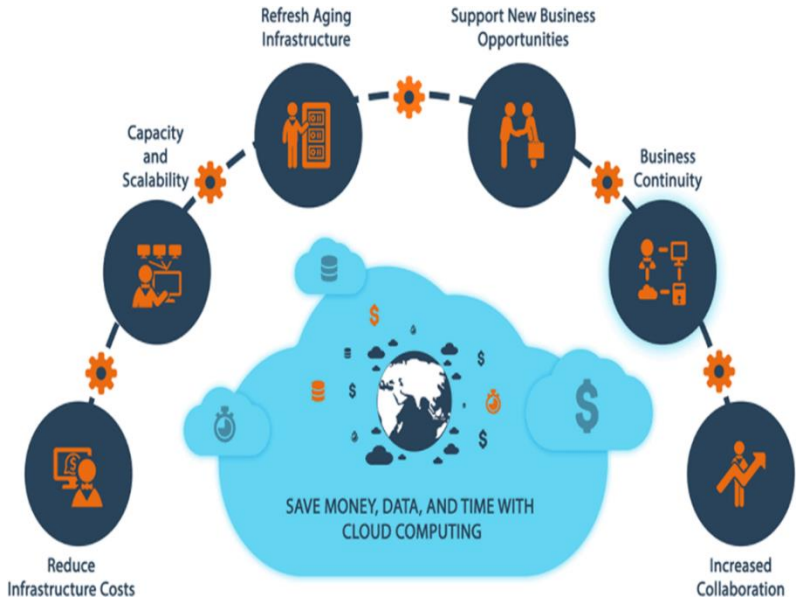
Some of these cloud-based applications — such as Google's productivity apps (Gmail, Drive, and Docs) and Microsoft 365 (a cloud-based version of the Microsoft Office Suite) — were offered by the same companies that launched cloud infrastructure services, while other pre-existing software products, such as Adobe Creative Cloud, were launched as cloud-based applications using the services of cloud providers. New SaaS products and businesses also emerged based on the novel opportunities of these cloud providers, such as Netflix's streaming services in 2007, the music platform Spotify in 2008, the file-hosting service Dropbox in 2009, the video conferencing service Zoom in 2012, and the communication tool Slack in 2013. Today, cloud- based IT infrastructure and cloud-based applications have become a popular choice for both businesses and individual users and their market share is expected to grow.

Characteristics Of Cloud Computing

- **Multi-Tenant**: Multi-tenancy is an architecture in which a single instance of a software application serves multiple customers. Each customer is called a **tenant**.
- **Self-Sevice**: Self-service cloud computing is a private cloud service where the customer provisions storage and launches applications without an external cloud service provider. With a self-service cloud, users access a web-based portal to request or configure servers and launch applications.
- **Elastic (Scale-Up | Scale-Down)**: Elasticity is the ability to grow or shrink infrastructure resources dynamically as needed to adapt to workload changes in an autonomic

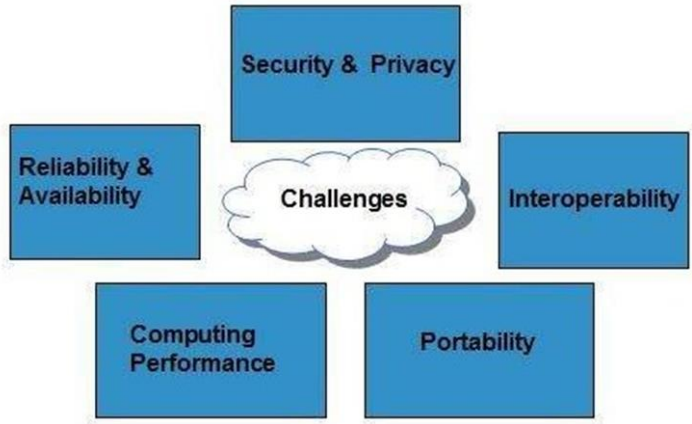
manner, maximizing the use of resources. This can result in savings in infrastructure costs overall.

- **Web-Based**: It means you can access your resources via Web-Based applications.
- **Automated**: Most of the things in the Cloud are automated, and human intervention is less.
- **Pay As You Go Model**: You only have to pay when utilizing cloud resources.
- **Modern Web-Based Integration**: It allows you to configure multiple application programs to share data in the cloud. In a network that incorporates cloud integration, diverse applications communicate either directly or through third-party software.
- **Secure:** Cloud services create a copy of the data that you want to store to prevent any form of data loss. If one server loses the data by any chance, the copy version is restored from the other server.



5.8. Issues in Cloud Computing

Cloud computing, an emergent technology, has placed many challenges in different aspects of data and information handling. Some of these are shown in the following diagram:

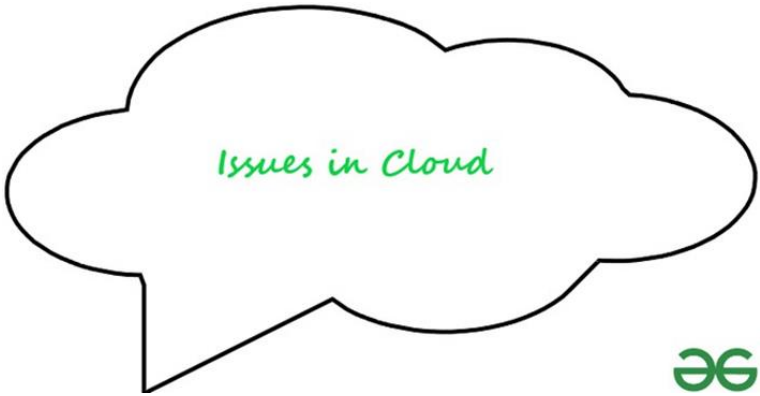


Cloud Computing is a new name for an old concept. The delivery of computing services from a remote location. Cloud Computing is Internet-based computing, where shared resources, software, and information are provided to computers and other devices on demand.

These are major issues in Cloud Computing:

1) **Privacy:** The user data can be accessed by the host company with or without permission. The service provider may access the data that is on the cloud at any point in time. They could accidentally or deliberately alter or even delete information.
2) **Compliance:** There are many regulations in places related to data and hosting. To comply with regulations (Federal Information Security Management Act, Health Insurance Portability and Accountability Act, etc.) the user may have to adopt deployment modes that are expensive.

3) **Security:** Cloud-based services involve third-party for storage and security. Can one assume that a cloud-based company will protect and secure one's data if one is using their services at a very low or for free? They may share users' information with others. Security presents a real threat to the cloud.
4) **Sustainability:** This issue refers to minimizing the effect of cloud computing on the environment. Citing the server's effects on the environmental effects of cloud computing, in areas where climate favors natural cooling and renewable electricity is readily available, the countries with favorable conditions, such as Finland, Sweden, and Switzerland are trying to attract cloud computing data centers. But other than nature's favors, would these countries have enough technical infrastructure to sustain the high-end clouds?
5) **Abuse:** While providing cloud services, it should be ascertained that the client is not purchasing the services of cloud computing for a nefarious purpose. In 2009, a banking Trojan illegally used the popular Amazon service as a command and control channel that issued software updates and malicious instructions to PCs that were infected by the malware So the hosting companies and the servers should have proper measures to address these issues.



1) Higher Cost: If you want to use cloud services uninterruptedly then you need to have a powerful network with higher bandwidth than ordinary internet networks, and also if your organization is broad and large so ordinary cloud service subscription won't suit your organization. Otherwise, you might face hassle in utilizing an ordinary cloud service while working on complex projects and applications. This is a major problem before small organizations, that restricts them from diving into cloud technology for their business.
2) Recovery of lost data in contingency: Before subscribing any cloud service provider goes through all norms and documentations and check whether their services match your requirements and sufficient well- maintained resource infrastructure with proper upkeeping. Once you subscribed to the service you almost hand over your data into the hands of a third party. If you are able to choose proper cloud service then in the future you don't need to worry about the recovery of lost data in any contingency.
3) Upkeeping(management) of Cloud: Maintaining a cloud is a herculin task because a cloud architecture contains a large resources infrastructure and other challenges and risks as well, user satisfaction, etc. As users usually pay for how much they have consumed the resources. So, sometimes it becomes hard to decide how much should be charged in case the user wants scalability and extend the services.
4) Lack of resources/skilled expertise: One of the major issues that companies and enterprises are going through today is the lack of resources and skilled employees. Every second organization is seeming interested or has already been moved to cloud services. That's why the workload in the cloud is increasing so the cloud service hosting companies need continuous rapid advancement. Due to these factors, organizations are having a tough time keeping up to date with the tools. As new tools and technologies are emerging every day so more

skilled/trained employees need to grow. These challenges can only be minimized through additional training of IT and development staff.

5) **Pay-per-use service charges:** Cloud computing services are on-demand services a user can extend or compress the volume of the resource as per needs. so you paid for how much you have consumed the resources. It is difficult to define a certain pre-defined cost for a particular quantity of services. Such types of ups and downs and price variations make the implementation of cloud computing very difficult and intricate. It is not easy for a firm's owner to study consistent demand and fluctuations with the seasons and various events. So it is hard to build a budget for a service that could consume several months of the budget in a few days of heavy use.

Real Life Ghosts of Cloud Computing

Here we have discussed some real life ghosts of cloud computing.

1. Data Security concern

When we talk about the security concern of the cloud technology, then a lot of questions remain unanswered. Multiple serious threats like virus attack and hacking of the client's site are the biggest cloud computing data security issues. Entrepreneurs have to think on these issues before adopting cloud computing technology for their business. Since you are transferring your company's important details to a third party so it is important to ensure yourself about the manageability and security system of the cloud.

2. Selecting the perfect cloud set-up

Choosing the appropriate cloud mechanism as per the needs of your business is very necessary. There are three types of clouds configuration such as public, private, and hybrid. The main secret behind successful implementation of the cloud is picking up the right cloud. If you are not selecting the right cloud then maybe

you have to face some serious hazards. Some companies having vast data so they prefer private clouds while small organisations usually use public clouds. A few companies like to go for a balanced approach with hybrid clouds. Choose a cloud computing consulting service which is aware and clearly disclose the terms and conditions regarding cloud implementation and data security.

3. Real time monitoring requirements

In some agencies, it is required to monitor their system in real time. It is compulsory term for their business that they continuously monitor and maintain their inventory system. Banks and some government agencies need to update their system in real time but cloud service providers are unable to match this requirement. This is really a big challenge for cloud services providers.

4. Resolving the stress

Every organisation wants to have a proper control and access over the data. It is not easy to handover your precious data to a third party. The main tension between enterprise and executives is they desire to control over the new modes of operations while using technology. These tensions are not unsolvable, but they do suggest that providers and clients alike must deliberately address a suite of cloud challenges in the planning, contracting and managing the services.

5. Reliability on new technology

It is a fact of human nature that we trust on the things present in front of our eyes. Normally entrepreneurs feel hesitation in letting out the organisational information to any unknown service provider. They think that information stored in their office premises is more secure and easily accessible. By using cloud computing they have fear of losing control over the data. They think that data is taken from them and handover to an unknown third party. Security threads are increase as they do not know and

where is the information stored and processed. These frights of the unknown service providers must very amicably be dealt with and eliminated form their minds.

6. Dependency on service providers

For uninterrupted services and proper working it is necessary that you acquire a vendor services with proper infrastructural and technical expertise. An authorized vendor who can meet the security standards set by your company's internal policies and government agencies. While selecting the service provider you must carefully read the service level agreement and understand their policies and terms and provision of compensation in case of any outage or lock in clauses.

7. Cultural obstacles

High authority of the company and organisational culture has also become a big obstacle in the proper implementation of the cloud computing. Top authority never wants to store the important data of the company somewhere else where they are not able to control and access the data. They have misconceptions in their minds that it puts the organisation at the risk by seeping out important details. Their mindset is such that the organization on risk averse footing, which makes it more reluctant to migrate to a cloud solution.

8. Cost barrier

For efficient working of this you have to bear the high charges of the bandwidth. Business can cut down the cost on hardware but they have to spend a huge amount on the bandwidth. For smaller application cost is not a big issue but for large and complex applications it is a major concern. For transferring complex and intensive data over the network it is very necessary that you have sufficient bandwidth. This is a major obstacle in front of small organisations, which restrict them for implementing cloud technology in their business.

9. Lack of knowledge and expertise

Every organisation does not have sufficient knowledge about the implementation of the cloud solutions. They have not expertise staff and tools for the proper use of cloud technology. Delivering the information and selection the right cloud is quite difficult without right direction. Teaching your staff about the process and tools of the cloud computing is a very big challenge in itself. Requiring an organisation to shift their business to cloud based technology without having any proper knowledge is like asking for disaster. They would never use this technology for their business functions.

10. Consumption basis services charges

Cloud computing services are on-demand services so it is difficult to define specific cost for a particular quantity of services. These types of fluctuations and price differences make the implementation of cloud computing very difficult and complicated. It is not easy for a normal business owner to study consistent demand and fluctuations with the seasons and various events. So it is hard to budget for a service that could consume several months of budget in a few days of heavy use.

11. Alleviate the threats risk

It is very complicated to certify that the cloud service provider meet the standards for security and threat risk. Every organisation may not have enough mechanism to mitigate these types of threats. Organisations should observe and examine the threats very seriously. There are mainly two types of threat such as internal threats, within the organisations and external threats from the professional hackers who seek out the important information of your business. These threats and security risks put a check on implementing cloud solutions.

12. Unauthorised service providers

It is a new concept for most of the business organisations. A

normal businessman is not able to verify the genuineness of the service provider agency. It's very difficult for them to check the whether the vendors meet the security standards or not. They have not an ICT consultant to evaluate the vendors against the worldwide criteria. It is necessary to verify that the vendor must be operating this business for a sufficient time without having any negative record in past. Vendor continuing business without any data loss complaint and have a number of satisfied clients. Market reputation of the vendor should be unblemished.

13. Hacking of brand

It carries some major risk factors like hacking. Some professional hackers are able to hack the application by breaking the efficient firewalls and steal the sensitive information of the organisations. A cloud provider hosts numerous clients; each can be affected by actions taken against any one of them. When any threat came into the main server it affects all the other clients also. As in distributed denial of service attacks server requests that inundate a provider from widely distributed computers.

14. Recovery of lost data

Cloud services faces issue of data loss. A proper backup policy for the recovery of data must be placed to deal with the loss. Vendors must set proper infrastructures to efficiently handle with server breakdown and outages. All the cloud computing service providers must set up their servers at economically stable locations where they should have proper arrangements for the backup of all the data in at least two different locations. Ideally they should manage a hot backup and a cold backup site.

15. Data portability

Every person wants to leverage of migrating in and out of the cloud. Ensuring data portability is very necessary. Usually, clients complain about being locked in the cloud technology from where they cannot switch without restraints. There should be no lock in

period for switching the cloud. Cloud technology must have capability to integrate efficiently with the on premises. The clients must have a proper contract of data portability with the provider and must have an updated copy of the data to be able to switch service providers, should there be any urgent requirement.

16. Cloud management

Managing a cloud is not an easy task. It consist a lot of technical challenges. A lot of dramatic predictions are famous about the impact of cloud computing. People think that traditional IT department will be outdated and research supports the conclusions that cloud impacts are likely to be more gradual and less linear. Cloud services can easily change and update by the business users. It does not involve any direct involvement of IT department. It is a service provider's responsibility to manage the information and spread it across the organisation. So it is difficult to manage all the complex functionality of cloud computing.

17. Dealing with lock-ins

Cloud providers have an important additional incentive to attempt to exploit lock-ins. A prefixed switching cost is always there for any company receiving external services. Exit strategies and lock-in risks are primary concerns for companies looking to exploit cloud computing.

18. Transparency of service provider

There is no transparency in the service provider's infrastructure and service area. You are not able to see the exact location where your data is stored or being processed. It is a big challenge for an organisation to transfer their business information to such an unknown vendor.

19. Transforming the data into virtual set-up

Transition of business data from a premise set up to a virtual set up is a major issue for various organisations. Data migration and

network configuration are the serious problems behind avoiding cloud computing technology.

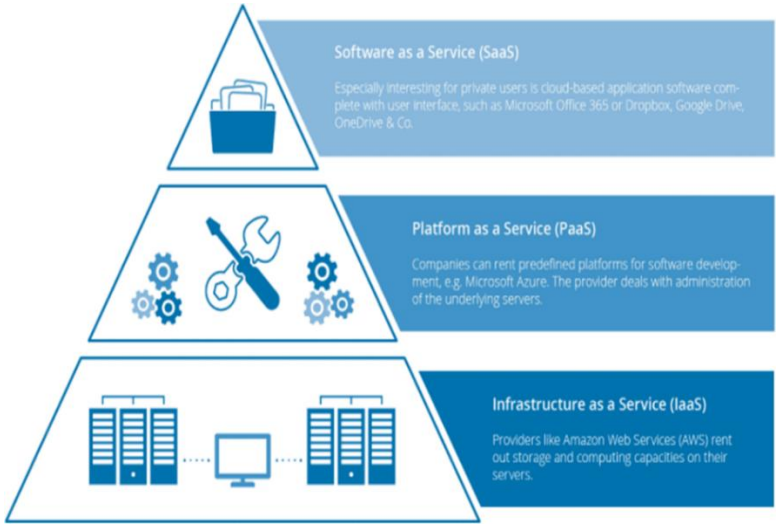
20. Popularization

The idea of cloud has been famous that there is a rush of implementing virtualization amongst CIOs. This has led to more complexities than solutions.

5.9. Service Models

Cloud Computing Service Models

There are three main types of cloud computing services. Each type of cloud computing provides different levels of control, flexibility, and management so that you'll select the proper set of services for your needs. The **three Cloud Service Models** are as follows:



Infrastructure As A Service (IaaS)

- It is the **most flexible** type of cloud service which lets you rent the hardware and contains the basic building blocks for cloud and IT.

- **It gives complete control over the hardware** that runs your application (servers, VMs, storage, networks & operating systems).
- It's an **instant computing** infrastructure, provisioned and managed over the internet.
- IaaS gives you the very best level of flexibility and management control over your IT resources.
- It is most almost like the prevailing IT resources with which many IT departments and developers are familiar.
- **Examples** of IaaS: VM or EC2 (from AWS), Storage or Networking.



Check Out: What is IAM in AWS?

Platform As a Service (PaaS)

- PaaS is a cloud service model that gives a ready-to-use development environment where developers can specialize in writing and executing high-quality code to make customized applications.
- It helps to create an application quickly without

managing the underlying infrastructure. For example, when deploying a web application using PaaS, you don't have to install an operating system, web server, or even system updates. However, you can scale and add new features to your services.

- This cloud service model makes the method of developing and deploying applications simpler and it is more expensive than IaaS but less expensive than SaaS.
- This helps you be more efficient as you don't get to worry about resource procurement, capacity planning, software maintenance, patching, or any of the opposite undifferentiated work involved in running your application.
- Examples of PaaS: Elastic Beanstalk or Lambda from AWS, WebApps, Functions or Azure SQL DB from Azure, Cloud SQL DB from Google Cloud, or Oracle Database Cloud Service from Oracle Cloud.

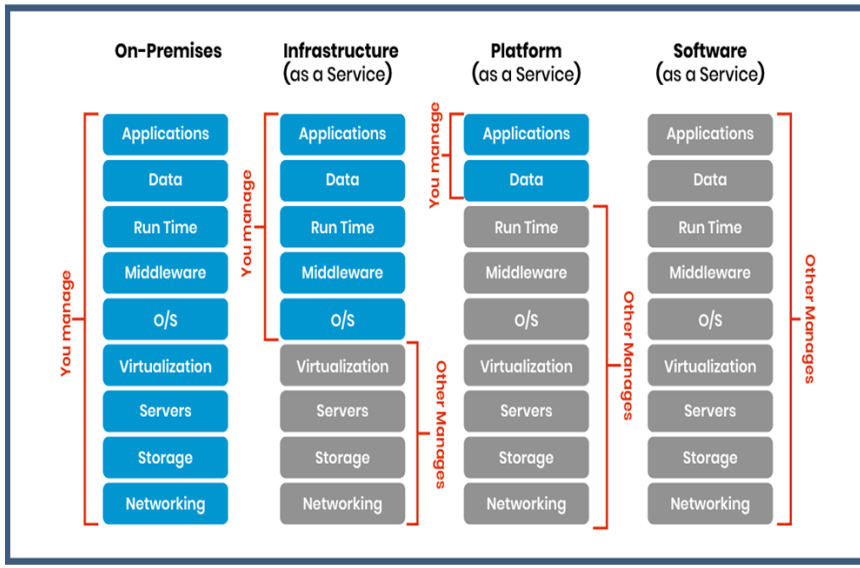


Software As a Service (SaaS)

- SaaS provides you with a complete product that is run and managed by the service provider.
- The software is hosted online and made available to customers on a subscription basis or for purchase in this cloud service model.
- With a SaaS offering, you don't need to worry about how the service is maintained or how the underlying infrastructure is managed. It would help if you believed how you'd use that specific software.
- Examples of SaaS: Microsoft Office 365, Oracle ERP/HCM Cloud, SalesForce, Gmail, or Dropbox.



Graphical Comparison of Cloud Computing Service Models



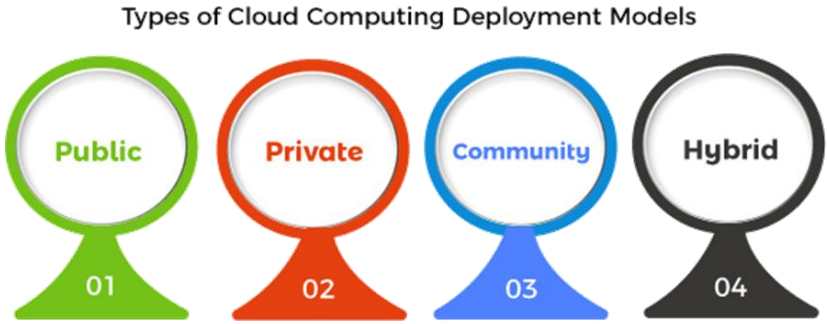
5.10. Cloud Deployment Models

In cloud computing, we have access to a shared pool of computer resources (servers, storage, programs, and so on) in the cloud. You simply need to request additional resources when you require them. Getting resources up and running quickly is a breeze thanks to the clouds. It is possible to release resources that are no longer necessary. This method allows you to just pay for what you use. Your cloud provider is in charge of all upkeep. It functions as a virtual computing environment with a deployment architecture that varies depending on the amount of data you want to store and who has access to the infrastructure.

Deployment Models

The cloud deployment model identifies the specific type of cloud environment based on ownership, scale, and access, as well as the cloud's nature and purpose. The location of the servers you're utilizing and who controls them are defined by a cloud

deployment model. It specifies how your cloud infrastructure will look, what you can change, and whether you will be given services or will have to create everything yourself. Relationships between the infrastructure and your users are also defined by cloud deployment types.



Different types of cloud computing deployment models are:

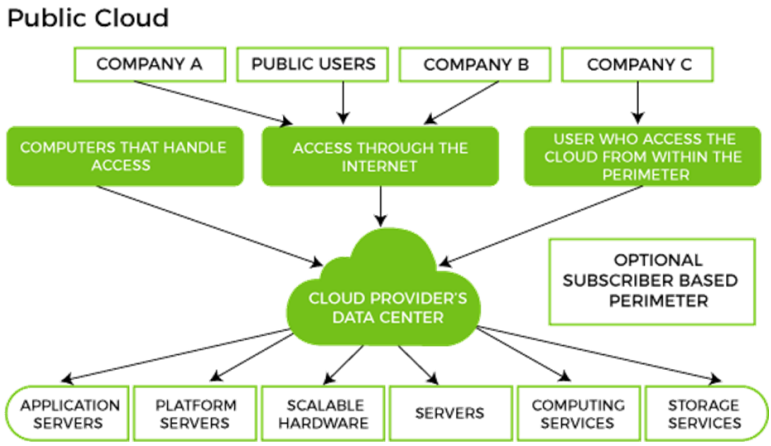
1) Public cloud
2) Private cloud
3) Hybrid cloud
4) Community cloud
5) Multi-cloud

Let us discuss them one by one:

1. **Public Cloud**

The public cloud makes it possible for anybody to access systems and services. The public cloud may be less secure as it is open for everyone. The public cloud is one in which cloud infrastructure services are provided over the internet to the general people or major industry groups. The infrastructure in this cloud model is owned by the entity that delivers the cloud services, not by the consumer. It is a type of cloud hosting that allows customers and users to easily access systems and services. This form of cloud computing is an excellent example of cloud hosting, in which service providers supply services to a variety of customers. In this arrangement, storage backup and retrieval services are given for

free, as a subscription, or on a per-use basis. Example: Google App Engine etc.



Advantages of the public cloud model:

- **Minimal Investment:** Because it is a pay-per-use service, there is no substantial upfront fee, making it excellent for enterprises that require immediate access to resources.
- **No setup cost:** The entire infrastructure is fully subsidized by the cloud service providers, thus there is no need to set up any hardware.
- **Infrastructure Management is not required:** Using the public cloud does not necessitate infrastructure management.
- **No maintenance:** The maintenance work is done by the service provider (Not users).
- **Dynamic Scalability:** To fulfill your company's needs, on-demand resources are accessible.

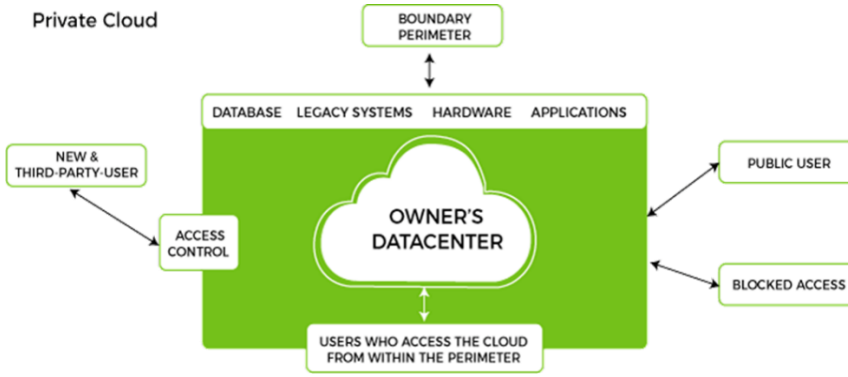
2. **Private Cloud**

The private cloud deployment model is the exact opposite of the public cloud deployment model.

It's a one-on-one environment for a single user (customer). There

is no need to share your hardware with anyone else. The distinction between private and public cloud is in how you handle all of the hardware. It is also called the "internal cloud" & it refers to the ability to access systems and services within a given border or organization. The cloud platform is implemented in a cloud-based secure environment that is protected by powerful firewalls and under the supervision of an organization's IT department.

The private cloud gives the greater flexibility of control over cloud resources.

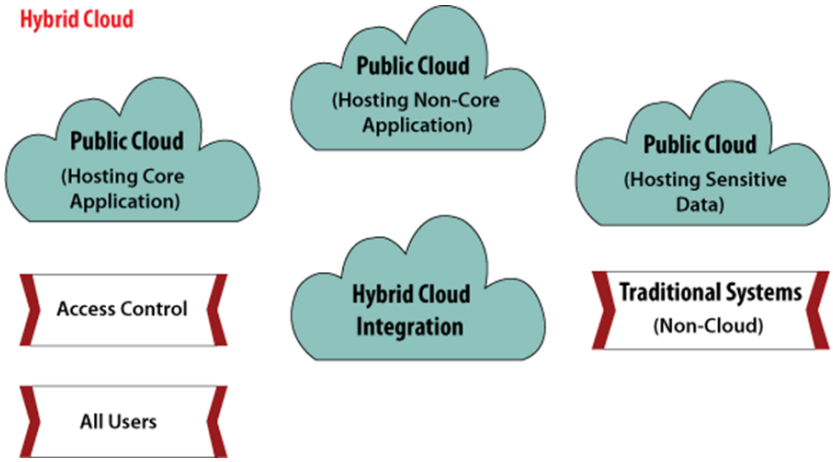


Advantages of the private cloud model:

- **Better Control:** You are the sole owner of the property. You gain complete command over service integration, IT operations, policies, and user behavior.
- **Data Security and Privacy:** It's suitable for storing corporate information to which only authorized staff have access. By segmenting resources within the same infrastructure, improved access and security can be achieved.
- **Supports Legacy Systems:** This approach is designed to work with legacy systems that are unable to access the public cloud.
- **Customization:** Unlike a public cloud deployment, a private cloud allows a company to tailor its solution to meet its specific needs.

3. Hybrid cloud

By bridging the public and private worlds with a layer of proprietary software, hybrid cloud computing gives the best of both worlds. With a hybrid solution, you may host the app in a safe environment while taking advantage of the public cloud's cost savings. Organizations can move data and applications between different clouds using a combination of two or more cloud deployment methods, depending on their needs.



Advantages of the hybrid cloud model:

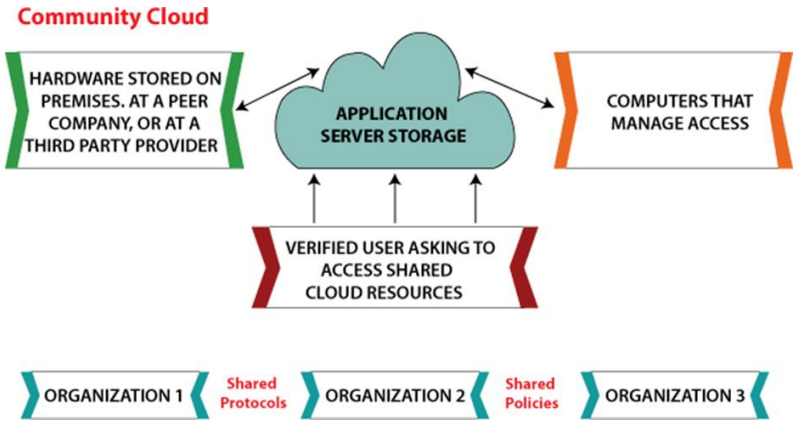
- **Flexibility and control:** Businesses with more flexibility can design personalized solutions that meet their particular needs.
- **Cost:** Because public clouds provide for scalability, you'll only be responsible for paying for the extra capacity if you require it.
- **Security:** Because data is properly separated, the chances of data theft by attackers are considerably reduced.

4. Community cloud

It allows systems and services to be accessible by a group of organizations. It is a distributed system that is created by

integrating the services of different clouds to address the specific needs of a community, industry, or business. The infrastructure of the community could be shared

between the organization which has shared concerns or tasks. It is generally managed by a third party or by the combination of one or more organizations in the community.



Advantages of the community cloud model:

- **Cost Effective:** It is cost-effective because the cloud is shared by multiple organizations or communities.
- **Security:** Community cloud provides better security.
- **Shared resources:** It allows you to share resources, infrastructure, etc. with multiple organizations.
- **Collaboration and data sharing:** It is suitable for both collaboration and data sharing.

5. **Multi-cloud**

We're talking about employing multiple cloud providers at the same time under this paradigm, as

the name implies. It's similar to the hybrid cloud deployment approach, which combines public and private cloud resources. Instead of merging private and public clouds, multi-cloud uses

many public clouds. Although public cloud providers provide numerous tools to improve the reliability of their services, mishaps still occur. It's quite rare that two distinct clouds would have an incident at the same moment. As a result, multi-cloud deployment improves the high availability of your services even more.

Advantages of a multi-cloud model:

- You can mix and match the best features of each cloud provider's services to suit the demands of your apps, workloads, and business by choosing different cloud providers.
- **Reduced Latency:** To reduce latency and improve user experience, you can choose cloud regions and zones that are close to your clients.
- **High availability of service:** It's quite rare that two distinct clouds would have an incident at the same moment. So, the multi-cloud deployment improves the high availability of your services.

5.11. A Comparative Analysis of Cloud Deployment Models

With the below table, we have attempted to analyze the key models with an overview of what each one can do for you:

Important Factors to Consider	**Public**	**Private**	**Community**	**Hybrid**
Setup and ease of use	Easy	Requires professional IT Team	Requires professional IT Team	Requires professional IT Team
Data Security and	Low	High	Very High	High

Privacy				
Scalability and flexibility	High	High	Fixed requirements	High
Cost-Effectiveness	Most affordable	Most expensive	Cost is distributed among members	Cheaper than private but more expensive than public
Reliability	Low	High	Higher	High